

Project 3 - Nikola

Nikola Meshkovski

2026-02-26

```
# Clean Environment
rm(list=ls())
```

```
# Load packages
packages <- c(
  "readxl",      # Read Excel files (.xls, .xlsx)
  "tidyverse",   # Data manipulation and visualisation (ggplot2, dplyr, etc.)
  "cluster",     # Clustering algorithms (k-means, hierarchical, etc.)
  "factoextra",  # Visualise clustering and PCA results
  "here",        # Builds paths relative to the project root (.qmd location)
  "renv",
  "plm",
  "tseries",
  "writexl",
  "sf",
  "rnaturalearth",
  "rnaturalearthdata",
  "countrycode",
  "fixest"
)

for (pkg in packages) {
  if (!require(pkg, character.only = TRUE)) {
    install.packages(pkg)
    library(pkg, character.only = TRUE)
  }
}
```

```

## Done already
# renv::init() is only run once when setting up the project for the first time.
# After that, renv::snapshot() is what we use to record any new packages into renv.lock

# Set global ggplot2 theme: Times New Roman font for all figures
theme_set(theme_minimal(base_size = 12, base_family = "serif"))
theme_update(text = element_text(family = "serif"))

# Load data - paths relative to materials/
df_main <- read_excel(here("data", "data_raw_theglobaleconomy.xlsx"))
df_monetary_policy <- read_excel(here("data", "data_raw_monetary-transmission.xlsx"),
                                sheet = "data_CWY_annual")

# Load real GDP per capita (constant 2015 US$) from World Bank
# skip=3 bypasses the 3 metadata rows; row 4 in the file is the true header
df_gdp_real <- read_excel(here("data", "Real_GDP_pc_WorldBank.xls"), skip = 3) |>
  select(`Country Name`, `1960`:`2025`) |> # keep country + all year cols
  pivot_longer(
    cols = -`Country Name`,
    names_to = "Year",
    values_to = "GDP_pc_real"
  ) |>
  mutate(Year = as.integer(Year)) |>
  rename(Country = `Country Name`)

# Keep only MPS from the monetary policy dataset
mps_subset <- df_monetary_policy[, c("Country", "Year", "MPS")]

# Merge MPS into the main dataset
df_world <- merge(
  df_main,
  mps_subset,
  by = c("Country", "Year"),
  all.x = TRUE
)

# Drop the nominal GDP per capita column and replace with real GDP per capita
df_world <- df_world |>
  select(-`GDP per capita, current U.S. dollars`) |> # ← remove nominal variable
  merge(

```

```

    df_gdp_real,
    by      = c("Country", "Year"),
    all.x   = TRUE
  )

# Sort by Country and Year for readability
df_world <- df_world[order(df_world$Country, df_world$Year), ]

# Clean up
rm(df_main, df_monetary_policy, mps_subset, df_gdp_real)

# Most likely time range for available data: 2000-2019 (Annual)
# Isolates the COVID pandemic which would be an outlier, and Monetary Policy Transmissi
df_world_2000_2019 <- df_world %>%
  filter(Year >= 2000, Year <= 2019)

# Also filter df_world to be after 2000, data from 1990 is too broad
df_world <- df_world %>%
  filter(Year >= 2000, Year <= 2026)

# Create a table with missing observations for each variable
missing_table <- data.frame(
  Missing_Obs = sapply(df_world_2000_2019, function(x) sum(is.na(x))),
  Total_Obs = nrow(df_world_2000_2019)
)

# Calculate the percentage of missingness for each variable
missing_table$Missing_Percent <- round(
  (missing_table$Missing_Obs / missing_table$Total_Obs) * 100, 2
)

# Sort the table from highest to lowest missingness percentage
missing_table <- missing_table[order(-missing_table$Missing_Percent), ]

# View the result
missing_table

```

	Missing_Obs	Total_Obs
Informal employment, percent of total employment	3454	3940

Bank lending-deposit interest rate spread	1787	3940
Real interest rate: Bank lending rate minus inflation	1419	3940
Shadow economy, percent of GDP	1412	3940
Tax rate, percent of commercial profits	1327	3940
MPS	1050	3940
Informal economy, MIMIC method	780	3940
Domestic credit to the private sector, percent of GDP	777	3940
GDP_pc_real	707	3940
Household consumption as percent of GDP	657	3940
Inflation: percent change in the Consumer Price Index	418	3940
Regulatory quality index (-2.5 weak; 2.5 strong)	301	3940
Control of corruption (-2.5 weak; 2.5 strong)	291	3940
Political stability index (-2.5 weak; 2.5 strong)	250	3940
Voice and accountability index (-2.5 weak; 2.5 strong)	238	3940
Political rights index, 7 (weak) - 1 (strong)	171	3940
Economic growth: the rate of change of real GDP	125	3940
Country	0	3940
Year	0	3940
Code	0	3940
ContinentCode	0	3940

Missing_Percent

Informal employment, percent of total employment	87.66
Bank lending-deposit interest rate spread	45.36
Real interest rate: Bank lending rate minus inflation	36.02
Shadow economy, percent of GDP	35.84
Tax rate, percent of commercial profits	33.68
MPS	26.65
Informal economy, MIMIC method	19.80
Domestic credit to the private sector, percent of GDP	19.72
GDP_pc_real	17.94
Household consumption as percent of GDP	16.68
Inflation: percent change in the Consumer Price Index	10.61
Regulatory quality index (-2.5 weak; 2.5 strong)	7.64
Control of corruption (-2.5 weak; 2.5 strong)	7.39
Political stability index (-2.5 weak; 2.5 strong)	6.35
Voice and accountability index (-2.5 weak; 2.5 strong)	6.04
Political rights index, 7 (weak) - 1 (strong)	4.34
Economic growth: the rate of change of real GDP	3.17
Country	0.00
Year	0.00
Code	0.00

```

# Create a categorical variable based on GDP per capita ()
# Countries with GDP per capita above 14,000 are labeled "Developed"
# Countries with GDP per capita below 14,000 are labeled "EMDEs (Emerging market and d
# Missing GDP per capita values stay as NA

# Compute each country's average GDP per capita over the full 2000-2019 window
avg_gdp <- df_world_2000_2019 %>%
  group_by(Country) %>%
  summarise(avg_gdp_pc = mean(`GDP_pc_real`, na.rm = TRUE))

# Apply the $14,000 threshold once - produces a single fixed label per country
avg_gdp$DevelopmentStatus <- ifelse(
  is.na(avg_gdp$avg_gdp_pc), NA,
  ifelse(avg_gdp$avg_gdp_pc > 14000, "Developed", "Emerging")
)

# Binary version: 1 = Developed, 0 = Emerging
avg_gdp$DevelopmentStatus_binary <- ifelse(
  is.na(avg_gdp$avg_gdp_pc), NA,
  ifelse(avg_gdp$avg_gdp_pc > 14000, 1, 0)
)

# Merge the fixed classification back - every row for a country gets the same label
df_world_2000_2019 <- df_world_2000_2019 %>%
  left_join(avg_gdp[, c("Country", "DevelopmentStatus", "DevelopmentStatus_binary")],
    by = "Country")

# Clean up helper table
rm(avg_gdp)

# Split the dataset into Emerging and Developed groups
df_emerging <- subset(df_world_2000_2019, DevelopmentStatus == "Emerging")
df_developed <- subset(df_world_2000_2019, DevelopmentStatus == "Developed")

# Create a missingness table for the Emerging group
missing_emerging <- data.frame(
  Variable = names(df_emerging),
  Missing_Obs = sapply(df_emerging, function(x) sum(is.na(x))),

```

```

    Total_Obs = nrow(df_emerging)
  )

# Calculate the percentage of missingness for the Emerging group
missing_emerging$Missing_Percent <- round(
  (missing_emerging$Missing_Obs / missing_emerging$Total_Obs) * 100, 2
)

# Sort the Emerging table by missingness percentage
missing_emerging <- missing_emerging[order(-missing_emerging$Missing_Percent), ]

# Create a missingness table for the Developed group
missing_developed <- data.frame(
  Variable = names(df_developed),
  Missing_Obs = sapply(df_developed, function(x) sum(is.na(x))),
  Total_Obs = nrow(df_developed)
)

# Calculate the percentage of missingness for the Developed group
missing_developed$Missing_Percent <- round(
  (missing_developed$Missing_Obs / missing_developed$Total_Obs) * 100, 2
)

# Sort the Developed table by missingness percentage
missing_developed <- missing_developed[order(-missing_developed$Missing_Percent), ]

# View both tables (missingness by segment)
missing_emerging

```

Informal employment, percent of total employment
 Bank lending-deposit interest rate spread
 Shadow economy, percent of GDP
 Tax rate, percent of commercial profits
 Real interest rate: Bank lending rate minus inflation
 Domestic credit to the private sector, percent of GDP
 Informal economy, MIMIC method
 Household consumption as percent of GDP
 MPS
 Inflation: percent change in the Consumer Price Index

Informal employment, percent of total employment
 Bank lending-deposit interest rate spread
 Shadow economy, percent of GDP
 Tax rate, percent of commercial profits
 Real interest rate: Bank lending rate minus inflation
 Domestic credit to the private sector, percent of GDP
 Informal economy, MIMIC method
 Household consumption as percent of GDP
 MPS
 Inflation: percent change in the Consumer Price Index

Regulatory quality index (-2.5 weak; 2.5 strong)	Regulatory quality index	
Control of corruption (-2.5 weak; 2.5 strong)	Control of corruption	
Political stability index (-2.5 weak; 2.5 strong)	Political stability index	
Voice and accountability index (-2.5 weak; 2.5 strong)	Voice and accountability index	
Economic growth: the rate of change of real GDP	Economic growth: the rat	
GDP_pc_real		
Political rights index, 7 (weak) - 1 (strong)	Political rights index	
Country		
Year		
Code		
ContinentCode		
DevelopmentStatus		
DevelopmentStatus_binary		
	Missing_Obs	Total_Obs
Informal employment, percent of total employment	1935	2340
Bank lending-deposit interest rate spread	886	2340
Shadow economy, percent of GDP	788	2340
Tax rate, percent of commercial profits	703	2340
Real interest rate: Bank lending rate minus inflation	667	2340
Domestic credit to the private sector, percent of GDP	399	2340
Informal economy, MIMIC method	360	2340
Household consumption as percent of GDP	324	2340
MPS	324	2340
Inflation: percent change in the Consumer Price Index	189	2340
Regulatory quality index (-2.5 weak; 2.5 strong)	136	2340
Control of corruption (-2.5 weak; 2.5 strong)	126	2340
Political stability index (-2.5 weak; 2.5 strong)	123	2340
Voice and accountability index (-2.5 weak; 2.5 strong)	117	2340
Economic growth: the rate of change of real GDP	43	2340
GDP_pc_real	21	2340
Political rights index, 7 (weak) - 1 (strong)	12	2340
Country	0	2340
Year	0	2340
Code	0	2340
ContinentCode	0	2340
DevelopmentStatus	0	2340
DevelopmentStatus_binary	0	2340
	Missing_Percent	
Informal employment, percent of total employment	82.69	
Bank lending-deposit interest rate spread	37.86	
Shadow economy, percent of GDP	33.68	

Tax rate, percent of commercial profits	30.04
Real interest rate: Bank lending rate minus inflation	28.50
Domestic credit to the private sector, percent of GDP	17.05
Informal economy, MIMIC method	15.38
Household consumption as percent of GDP	13.85
MPS	13.85
Inflation: percent change in the Consumer Price Index	8.08
Regulatory quality index (-2.5 weak; 2.5 strong)	5.81
Control of corruption (-2.5 weak; 2.5 strong)	5.38
Political stability index (-2.5 weak; 2.5 strong)	5.26
Voice and accountability index (-2.5 weak; 2.5 strong)	5.00
Economic growth: the rate of change of real GDP	1.84
GDP_pc_real	0.90
Political rights index, 7 (weak) - 1 (strong)	0.51
Country	0.00
Year	0.00
Code	0.00
ContinentCode	0.00
DevelopmentStatus	0.00
DevelopmentStatus_binary	0.00

missing_developed

Informal employment, percent of total employment	Informal employment, perc
Bank lending-deposit interest rate spread	Bank lending-depos
Real interest rate: Bank lending rate minus inflation	Real interest rate: Bank lendi
Tax rate, percent of commercial profits	Tax rate, percen
Shadow economy, percent of GDP	Shadow
Domestic credit to the private sector, percent of GDP	Domestic credit to the private
Informal economy, MIMIC method	Informa
MPS	
Household consumption as percent of GDP	Household consum
Inflation: percent change in the Consumer Price Index	Inflation: percent change in t
Control of corruption (-2.5 weak; 2.5 strong)	Control of corruption
Regulatory quality index (-2.5 weak; 2.5 strong)	Regulatory quality index
Political stability index (-2.5 weak; 2.5 strong)	Political stability index
Voice and accountability index (-2.5 weak; 2.5 strong)	Voice and accountability index
Political rights index, 7 (weak) - 1 (strong)	Political rights index
GDP_pc_real	

Economic growth: the rate of change of real GDP
Country
Year
Code
ContinentCode
DevelopmentStatus
DevelopmentStatus_binary

Economic growth: the rat

	Missing_Obs	Total_Obs
Informal employment, percent of total employment	945	960
Bank lending-deposit interest rate spread	632	960
Real interest rate: Bank lending rate minus inflation	539	960
Tax rate, percent of commercial profits	378	960
Shadow economy, percent of GDP	368	960
Domestic credit to the private sector, percent of GDP	240	960
Informal economy, MIMIC method	220	960
MPS	160	960
Household consumption as percent of GDP	157	960
Inflation: percent change in the Consumer Price Index	129	960
Control of corruption (-2.5 weak; 2.5 strong)	124	960
Regulatory quality index (-2.5 weak; 2.5 strong)	124	960
Political stability index (-2.5 weak; 2.5 strong)	93	960
Voice and accountability index (-2.5 weak; 2.5 strong)	89	960
Political rights index, 7 (weak) - 1 (strong)	80	960
GDP_pc_real	46	960
Economic growth: the rate of change of real GDP	36	960
Country	0	960
Year	0	960
Code	0	960
ContinentCode	0	960
DevelopmentStatus	0	960
DevelopmentStatus_binary	0	960

	Missing_Percent
Informal employment, percent of total employment	98.44
Bank lending-deposit interest rate spread	65.83
Real interest rate: Bank lending rate minus inflation	56.15
Tax rate, percent of commercial profits	39.38
Shadow economy, percent of GDP	38.33
Domestic credit to the private sector, percent of GDP	25.00
Informal economy, MIMIC method	22.92
MPS	16.67
Household consumption as percent of GDP	16.35

Inflation: percent change in the Consumer Price Index	13.44
Control of corruption (-2.5 weak; 2.5 strong)	12.92
Regulatory quality index (-2.5 weak; 2.5 strong)	12.92
Political stability index (-2.5 weak; 2.5 strong)	9.69
Voice and accountability index (-2.5 weak; 2.5 strong)	9.27
Political rights index, 7 (weak) - 1 (strong)	8.33
GDP_pc_real	4.79
Economic growth: the rate of change of real GDP	3.75
Country	0.00
Year	0.00
Code	0.00
ContinentCode	0.00
DevelopmentStatus	0.00
DevelopmentStatus_binary	0.00

```
# Export tables using relative paths anchored to materials/
write_xlsx(missing_emerging, here("tables", "descriptive_tables", "missing_emerging.xlsx"),
write_xlsx(missing_developed, here("tables", "descriptive_tables", "missing_developed.xlsx"))
```

```
# Create missingness summary for the Emerging group
missing_emerging <- data.frame(
  Variable = names(df_emerging),
  Emerging_Missing_Obs = sapply(df_emerging, function(x) sum(is.na(x))),
  Emerging_Total_Obs = nrow(df_emerging)
)

# Calculate percentage missingness for the Emerging group
missing_emerging$Emerging_Missing_Percent <- round(
  (missing_emerging$Emerging_Missing_Obs / missing_emerging$Emerging_Total_Obs) * 100,
)

# Create missingness summary for the Developed group
missing_developed <- data.frame(
  Variable = names(df_developed),
  Developed_Missing_Obs = sapply(df_developed, function(x) sum(is.na(x))),
  Developed_Total_Obs = nrow(df_developed)
)

# Calculate percentage missingness for the Developed group
missing_developed$Developed_Missing_Percent <- round(
```

```

    (missing_developed$Developed_Missing_Obs / missing_developed$Developed_Total_Obs) *
  )

# Keep only the key comparison columns from each table
missing_emerging <- missing_emerging[, c("Variable", "Emerging_Missing_Obs", "Emerging_Missing_Percentage")]
missing_developed <- missing_developed[, c("Variable", "Developed_Missing_Obs", "Developed_Missing_Percentage")]

# Merge the two tables by variable name
missing_comparison <- merge(missing_emerging, missing_developed, by = "Variable", all = TRUE)

# Sort the combined table by Emerging missingness percentage, highest first
missing_comparison <- missing_comparison[order(-missing_comparison$Emerging_Missing_Percentage), ]

# View the combined comparison table
missing_comparison

```

	Variable	Emerging_Missing_Obs
14	Informal employment, percent of total employment	1935
1	Bank lending-deposit interest rate spread	886
20	Shadow economy, percent of GDP	788
21	Tax rate, percent of commercial profits	703
18	Real interest rate: Bank lending rate minus inflation	667
8	Domestic credit to the private sector, percent of GDP	399
13	Informal economy, MIMIC method	360
11	Household consumption as percent of GDP	324
15	MPS	324
12	Inflation: percent change in the Consumer Price Index	189
19	Regulatory quality index (-2.5 weak; 2.5 strong)	136
4	Control of corruption (-2.5 weak; 2.5 strong)	126
17	Political stability index (-2.5 weak; 2.5 strong)	123
22	Voice and accountability index (-2.5 weak; 2.5 strong)	117
9	Economic growth: the rate of change of real GDP	43
10	GDP_pc_real	21
16	Political rights index, 7 (weak) - 1 (strong)	12
2	Code	0
3	ContinentCode	0
5	Country	0
6	DevelopmentStatus	0
7	DevelopmentStatus_binary	0
23	Year	0

	Emerging_Missing_Percent	Developed_Missing_Obs	Developed_Missing_Percent
14	82.69	945	98.44
1	37.86	632	65.83
20	33.68	368	38.33
21	30.04	378	39.38
18	28.50	539	56.15
8	17.05	240	25.00
13	15.38	220	22.92
11	13.85	157	16.35
15	13.85	160	16.67
12	8.08	129	13.44
19	5.81	124	12.92
4	5.38	124	12.92
17	5.26	93	9.69
22	5.00	89	9.27
9	1.84	36	3.75
10	0.90	46	4.79
16	0.51	80	8.33
2	0.00	0	0.00
3	0.00	0	0.00
5	0.00	0	0.00
6	0.00	0	0.00
7	0.00	0	0.00
23	0.00	0	0.00

```
# See data types within data frame
str(df_world_2000_2019)
```

```
'data.frame': 3940 obs. of 23 variables:
```

\$ Country	: chr	"Afghanistan" "Afghani
\$ Year	: num	2000 2001 2002 2003 20
\$ Code	: chr	"AFG" "AFG" "AFG" "AFG
\$ ContinentCode	: chr	"AS" "AS" "AS" "AS" ..
\$ Economic growth: the rate of change of real GDP	: num	NA -9.43 28.6 8.83 1.4
\$ Household consumption as percent of GDP	: num	NA NA NA NA NA NA NA N
\$ Inflation: percent change in the Consumer Price Index	: num	NA NA NA NA NA 12.7 6.
\$ Informal employment, percent of total employment	: num	NA NA NA NA NA NA NA N
\$ Tax rate, percent of commercial profits	: num	NA NA NA NA NA 35.8 35
\$ Control of corruption (-2.5 weak; 2.5 strong)	: num	-1.64 NA -1.54 -1.55 -
\$ Regulatory quality index (-2.5 weak; 2.5 strong)	: num	-2.45 NA -2.13 -1.96 -

```

$ Voice and accountability index (-2.5 weak; 2.5 strong): num -2.3 NA -1.36 -1.27 -1
$ Political stability index (-2.5 weak; 2.5 strong) : num -2.85 NA -2.43 -2.39 -
$ Political rights index, 7 (weak) - 1 (strong) : num 7 7 6 6 5 5 5 5 5 6 ..
$ Shadow economy, percent of GDP : num NA NA NA NA NA NA NA NA NA
$ Informal economy, MIMIC method : num NA NA NA NA NA NA NA NA NA
$ Domestic credit to the private sector, percent of GDP : num NA NA NA NA NA ...
$ Bank lending-deposit interest rate spread : num NA NA NA NA NA NA NA NA
$ Real interest rate: Bank lending rate minus inflation : num NA NA NA NA NA ...
$ MPS : num NA NA NA 0.07784 0.008
$ GDP_pc_real : num 308 277 338 346 339 ..
$ DevelopmentStatus : chr "Emerging" "Emerging"
$ DevelopmentStatus_binary : num 0 0 0 0 0 0 0 0 0 0 ..

```

```

# Keep only numeric variables for descriptive statistics
# This avoids trying to summarize text variables such as Country or DevelopmentStatus
numeric_emerging <- df_emerging[, sapply(df_emerging, is.numeric)]
numeric_developed <- df_developed[, sapply(df_developed, is.numeric)]

# Create descriptive statistics table for Emerging countries
# We calculate the number of non-missing observations, mean, standard deviation,
# minimum, median, and maximum for each numeric variable
desc_emerging <- data.frame(
  N = sapply(numeric_emerging, function(x) sum(!is.na(x))),
  Mean = sapply(numeric_emerging, function(x) mean(x, na.rm = TRUE)),
  SD = sapply(numeric_emerging, function(x) sd(x, na.rm = TRUE)),
  Min = sapply(numeric_emerging, function(x) min(x, na.rm = TRUE)),
  Median = sapply(numeric_emerging, function(x) median(x, na.rm = TRUE)),
  Max = sapply(numeric_emerging, function(x) max(x, na.rm = TRUE))
)

# Round the numeric results for a cleaner table
desc_emerging[, -c(1, 2)] <- round(desc_emerging[, -c(1, 2)], 2)

# Create descriptive statistics table for Developed countries
desc_developed <- data.frame(
  N = sapply(numeric_developed, function(x) sum(!is.na(x))),
  Mean = sapply(numeric_developed, function(x) mean(x, na.rm = TRUE)),
  SD = sapply(numeric_developed, function(x) sd(x, na.rm = TRUE)),
  Min = sapply(numeric_developed, function(x) min(x, na.rm = TRUE)),
  Median = sapply(numeric_developed, function(x) median(x, na.rm = TRUE)),
  Max = sapply(numeric_developed, function(x) max(x, na.rm = TRUE))
)

```

```
)

# Round the numeric results for a cleaner table
desc_developed[, -c(1, 2)] <- round(desc_developed[, -c(1, 2)], 2)

# View the two tables
desc_emerging
```

	N	Mean
Year	2340	2.009500e+03
Economic growth: the rate of change of real GDP	2297	4.293100e+00
Household consumption as percent of GDP	2016	6.784272e+01
Inflation: percent change in the Consumer Price Index	2151	6.673733e+00
Informal employment, percent of total employment	405	6.280716e+01
Tax rate, percent of commercial profits	1637	4.550709e+01
Control of corruption (-2.5 weak; 2.5 strong)	2214	-4.811021e-01
Regulatory quality index (-2.5 weak; 2.5 strong)	2204	-4.079764e-01
Voice and accountability index (-2.5 weak; 2.5 strong)	2223	-3.119523e-01
Political stability index (-2.5 weak; 2.5 strong)	2217	-3.451015e-01
Political rights index, 7 (weak) - 1 (strong)	2328	3.732388e+00
Shadow economy, percent of GDP	1552	3.537769e+01
Informal economy, MIMIC method	1980	3.792278e+01
Domestic credit to the private sector, percent of GDP	1941	3.513343e+01
Bank lending-deposit interest rate spread	1454	8.672345e+00
Real interest rate: Bank lending rate minus inflation	1673	6.883282e+00
MPS	2016	-6.712551e-03
GDP_pc_real	2319	4.080666e+03
DevelopmentStatus_binary	2340	0.000000e+00

	SD	Min	Median
Year	5.77	2000.00	2009.50
Economic growth: the rate of change of real GDP	5.28	-50.34	4.32
Household consumption as percent of GDP	15.95	5.04	68.80
Inflation: percent change in the Consumer Price Index	12.85	-16.90	4.40
Informal employment, percent of total employment	20.62	12.40	63.20
Tax rate, percent of commercial profits	31.81	7.40	38.30
Control of corruption (-2.5 weak; 2.5 strong)	0.63	-1.69	-0.53
Regulatory quality index (-2.5 weak; 2.5 strong)	0.68	-2.45	-0.42
Voice and accountability index (-2.5 weak; 2.5 strong)	0.74	-2.36	-0.23
Political stability index (-2.5 weak; 2.5 strong)	0.86	-2.85	-0.28
Political rights index, 7 (weak) - 1 (strong)	1.92	1.00	3.00

Shadow economy, percent of GDP	10.69	11.74	34.89
Informal economy, MIMIC method	10.24	11.60	37.10
Domestic credit to the private sector, percent of GDP	28.68	0.00	26.83
Bank lending-deposit interest rate spread	7.32	0.03	7.06
Real interest rate: Bank lending rate minus inflation	9.85	-64.30	6.39
MPS	0.89	-6.84	0.01
GDP_pc_real	3545.87	243.08	3081.24
DevelopmentStatus_binary	0.00	0.00	0.00
	Max		
Year	2019.00		
Economic growth: the rate of change of real GDP	86.83		
Household consumption as percent of GDP	133.11		
Inflation: percent change in the Consumer Price Index	325.00		
Informal employment, percent of total employment	99.70		
Tax rate, percent of commercial profits	280.20		
Control of corruption (-2.5 weak; 2.5 strong)	1.55		
Regulatory quality index (-2.5 weak; 2.5 strong)	1.62		
Voice and accountability index (-2.5 weak; 2.5 strong)	1.36		
Political stability index (-2.5 weak; 2.5 strong)	1.56		
Political rights index, 7 (weak) - 1 (strong)	7.00		
Shadow economy, percent of GDP	70.57		
Informal economy, MIMIC method	68.20		
Domestic credit to the private sector, percent of GDP	165.39		
Bank lending-deposit interest rate spread	72.40		
Real interest rate: Bank lending rate minus inflation	57.63		
MPS	14.00		
GDP_pc_real	19481.65		
DevelopmentStatus_binary	0.00		

desc_developed

	N	Mean
Year	960	2.009500e+03
Economic growth: the rate of change of real GDP	924	2.522857e+00
Household consumption as percent of GDP	803	5.144466e+01
Inflation: percent change in the Consumer Price Index	831	2.314200e+00
Informal employment, percent of total employment	15	3.534000e+01
Tax rate, percent of commercial profits	582	3.759313e+01
Control of corruption (-2.5 weak; 2.5 strong)	836	1.235550e+00
Regulatory quality index (-2.5 weak; 2.5 strong)	836	1.204043e+00

Voice and accountability index (-2.5 weak; 2.5 strong)	871	9.773249e-01	
Political stability index (-2.5 weak; 2.5 strong)	867	8.567128e-01	
Political rights index, 7 (weak) - 1 (strong)	880	1.859091e+00	
Shadow economy, percent of GDP	592	1.712198e+01	
Informal economy, MIMIC method	740	1.976554e+01	
Domestic credit to the private sector, percent of GDP	720	9.347210e+01	
Bank lending-deposit interest rate spread	328	4.661189e+00	
Real interest rate: Bank lending rate minus inflation	421	4.568860e+00	
MPS	800	4.857549e-03	
GDP_pc_real	914	4.179730e+04	
DevelopmentStatus_binary	960	1.000000e+00	
		SD	Min
Year		5.77	2000.00
Economic growth: the rate of change of real GDP		3.85	-14.63
Household consumption as percent of GDP		11.82	13.41
Inflation: percent change in the Consumer Price Index		2.39	-4.90
Informal employment, percent of total employment		11.31	23.60
Tax rate, percent of commercial profits		15.89	10.70
Control of corruption (-2.5 weak; 2.5 strong)		0.72	-0.48
Regulatory quality index (-2.5 weak; 2.5 strong)		0.58	-0.75
Voice and accountability index (-2.5 weak; 2.5 strong)		0.82	-1.96
Political stability index (-2.5 weak; 2.5 strong)		0.54	-1.29
Political rights index, 7 (weak) - 1 (strong)		1.77	1.00
Shadow economy, percent of GDP		7.34	6.16
Informal economy, MIMIC method		7.87	8.20
Domestic credit to the private sector, percent of GDP		45.46	0.19
Bank lending-deposit interest rate spread		4.30	0.23
Real interest rate: Bank lending rate minus inflation		8.73	-19.47
MPS		0.37	-3.36
GDP_pc_real		27707.36	9806.77
DevelopmentStatus_binary		0.00	1.00
		Median	Max
Year		2009.50	2019.00
Economic growth: the rate of change of real GDP		2.40	26.17
Household consumption as percent of GDP		53.31	93.49
Inflation: percent change in the Consumer Price Index		2.00	19.40
Informal employment, percent of total employment		36.20	62.20
Tax rate, percent of commercial profits		38.40	76.70
Control of corruption (-2.5 weak; 2.5 strong)		1.30	2.40
Regulatory quality index (-2.5 weak; 2.5 strong)		1.27	2.26
Voice and accountability index (-2.5 weak; 2.5 strong)		1.21	1.96

Political stability index (-2.5 weak; 2.5 strong)	0.98	1.68
Political rights index, 7 (weak) - 1 (strong)	1.00	7.00
Shadow economy, percent of GDP	14.57	46.57
Informal economy, MIMIC method	18.40	51.10
Domestic credit to the private sector, percent of GDP	87.46	304.58
Bank lending-deposit interest rate spread	3.70	55.80
Real interest rate: Bank lending rate minus inflation	3.41	93.92
MPS	0.01	2.19
GDP_pc_real	36654.79	185582.76
DevelopmentStatus_binary	1.00	1.00

```
# Export tables using relative paths anchored to materials/
write_xlsx(desc_emerging, here("tables", "descriptive_tables", "desc_emerging.xlsx"))
write_xlsx(desc_developed, here("tables", "descriptive_tables", "desc_developed.xlsx"))
```

```
# --- DROP: Informal employment (~87% missing - unusable) ---
# --- DROP: Shadow economy (keep MIMIC version only - see below) ---
# --- NOTE: Bank lending-deposit spread (~44% missing) and real interest rate (~35% mi
#           are excluded from the main specification due to high missingness, but are
#           This will be discussed in the data section of the paper. ---
```

```
# Keep only the variables needed for the analysis
df_world_2000_2019 <- df_world_2000_2019 %>%
  select(
    -'Shadow economy, percent of GDP',
    -'Informal employment, percent of total employment'
  )
```

```
rm(
  desc_developed,
  desc_emerging,
  missing_comparison,
  missing_developed,
  missing_emerging,
  missing_table,
  numeric_developed,
  numeric_emerging,
  packages,
  pkg,
```

```
df_world,
df_developed,
df_emerging
)
```

```
# Rename all variables to short, code-friendly names
```

```
df_world_2000_2019 <- df_world_2000_2019 %>%
```

```
  rename(
```

```
    # --- Identifiers ---
```

```
    country = Country,
```

```
    year = Year,
```

```
    code = Code,
```

```
    continent_code = ContinentCode,
```

```
    # --- Classification ---
```

```
    dev_status = DevelopmentStatus,
```

```
    dev_binary = DevelopmentStatus_binary,
```

```
    # --- Key variable ---
```

```
    mps = MPS,
```

```
# Mone
```

```
    # --- Shadow economy ---
```

```
    mimic = `Informal economy, MIMIC method`,
```

```
# Shad
```

```
    # --- Macroeconomic variables ---
```

```
    gdp_pc = `GDP_pc_real`,
```

```
# Real
```

```
    gdp_growth = `Economic growth: the rate of change of real GDP`,
```

```
# Real
```

```
    infl = `Inflation: percent change in the Consumer Price Index`,
```

```
# CPI
```

```
    credit_gdp = `Domestic credit to the private sector, percent of GDP`,
```

```
# Dome
```

```
    hh_consumption = `Household consumption as percent of GDP`,
```

```
# Hous
```

```
    tax_rate = `Tax rate, percent of commercial profits`,
```

```
# Tax
```

```
    # --- Governance ---
```

```
    reg_quality = `Regulatory quality index (-2.5 weak; 2.5 strong)`,
```

```
# Regu
```

```
    corruption = `Control of corruption (-2.5 weak; 2.5 strong)`,
```

```
# Cont
```

```
    pol_stability = `Political stability index (-2.5 weak; 2.5 strong)`,
```

```
# Poli
```

```
    voice_acc = `Voice and accountability index (-2.5 weak; 2.5 strong)`,
```

```
# Voic
```

```
    pol_rights = `Political rights index, 7 (weak) - 1 (strong)`,
```

```
# Poli
```

```
    # --- Excluded from main spec - high missingness, retained for robustness ---
```

```

    spread = `Bank lending-deposit interest rate spread`,
    real_int_rate = `Real interest rate: Bank lending rate minus inflation`
)

```

```

# Lend
# Real

```

```

#
# DIAGNOSTIC 1 - PANEL UNIT ROOT TESTS (Country-by-Country ADF approach)
#
# PURPOSE:
# We need to know whether each variable is stationary (i.e., mean-reverting)
# or has a unit root (i.e., trends/wanders without returning to a stable mean).
# This matters because:
#   - If a variable is stationary → safe to use in levels in the LP regression.
#   - If a variable has a unit root → we should use it in first differences
#     or cumulative changes, otherwise regressions may be spurious.
#
# WHY NOT plm::purtest (IPS)?
# The Im-Pesaran-Shin test implemented in plm::purtest is fragile with
# unbalanced panels. With T=20 and gaps in coverage, AIC lag selection can
# consume too many degrees of freedom, producing NaN. Rather than fighting
# the package, we run individual ADF tests per country using tseries::adf.test
# and summarise across countries - this is essentially what IPS does under
# the hood (averaging country-level ADF statistics).
#
# THE ADF TEST:
#   H0: The series has a unit root (non-stationary)
#   H1: The series is stationary
#   Low p-value (< 0.05) → reject H0 → evidence of stationarity
#
# HOW TO INTERPRET THE SUMMARY TABLE:
#   - n_countries: how many countries had enough data to run the test
#   - mean_pval / median_pval: average p-values across countries
#     (lower = stronger overall evidence of stationarity)
#   - pct_rejected_5: percentage of countries where H0 was rejected at 5%
#     * > 70% rejected → variable is broadly stationary across the panel
#     * 30-70% rejected → mixed evidence, proceed with caution
#     * < 30% rejected → likely non-stationary, consider differencing
#
# NOTE: We require at least 15 non-missing observations per country.
# The ADF test needs enough data points to estimate an autoregression

```

```

# with lags - with T=20, requiring 15 usable obs is a reasonable floor.
#

# Variables to test: all key variables that will enter the LP regressions
# MPS is excluded - it is a shock series centered on zero by construction,
# so it is stationary by design and does not need testing.
vars_urt <- c("infl", "gdp_pc", "gdp_growth", "mimic", "credit_gdp",
              "reg_quality", "corruption", "pol_stability")

min_obs <- 15 # Minimum usable observations per country

results <- list()

for (v in vars_urt) {

  # Step 1: For each country, check if there are enough non-missing observations
  #          to run an ADF test. If yes, run adf.test and extract the p-value.
  #          If the test fails (e.g., constant series), return NA.
  country_pvals <- df_world_2000_2019 %>%
    group_by(country) %>%
    filter(sum(!is.na(.data[[v]])) >= min_obs) %>%
    summarise(
      adf_pval = tryCatch({
        adf.test(na.omit(.data[[v]]))$p.value
      }, error = function(e) NA_real_),
      .groups = "drop"
    ) %>%
    filter(!is.na(adf_pval))

  # Step 2: Summarise across all countries that were successfully tested
  results[[v]] <- data.frame(
    variable      = v,
    n_countries   = nrow(country_pvals),
    mean_pval     = round(mean(country_pvals$adf_pval), 3),
    median_pval   = round(median(country_pvals$adf_pval), 3),
    pct_rejected_5 = round(mean(country_pvals$adf_pval < 0.05) * 100, 1)
  )
}

```

```
# Combine into a single summary table and display
urt_summary <- bind_rows(results)
```

```
cat("                                \n")
```

```
cat("  PANEL UNIT ROOT TEST SUMMARY (Country-by-Country ADF)\n")
```

```
  PANEL UNIT ROOT TEST SUMMARY (Country-by-Country ADF)
```

```
cat("                                \n")
```

```
cat("  H0: unit root (non-stationary) | Reject at 5% → stationary\n")
```

```
  H0: unit root (non-stationary) | Reject at 5% → stationary
```

```
cat("  min_obs per country:", min_obs, "\n")
```

```
  min_obs per country: 15
```

```
cat("                                \n\n")
```

```
print(urt_summary, row.names = FALSE)
```

variable	n_countries	mean_pval	median_pval	pct_rejected_5
infl	172	0.469	0.462	6.4
gdp_pc	160	0.595	0.586	5.6
gdp_growth	189	0.426	0.439	11.1
mimic	158	0.590	0.593	3.8
credit_gdp	145	0.534	0.551	9.0
reg_quality	192	0.489	0.521	13.0
corruption	192	0.525	0.558	10.4
pol_stability	195	0.497	0.504	8.2

```
cat("\n  Interpretation guide  \n")
```

Interpretation guide

```
cat("  pct_rejected_5 > 70%  → broadly stationary, safe to use in levels\n")
```

pct_rejected_5 > 70% → broadly stationary, safe to use in levels

```
cat("  pct_rejected_5 30-70% → mixed, inspect further or consider differencing\n")
```

pct_rejected_5 30-70% → mixed, inspect further or consider differencing

```
cat("  pct_rejected_5 < 30%  → likely non-stationary, use in differences\n")
```

pct_rejected_5 < 30% → likely non-stationary, use in differences

```
#
# INTERPRETATION OF RESULTS
#
# NOTE (unit root tests - revisit when writing paper):
#
# ADF results are uninformative here - expected with T 20 (15-17 usable obs after lags)
# Low power means the test can't distinguish stationary from unit root.
# IPS would fix this but failed on unbalanced panel, so we're stuck with the power problem
#
# Variables are safe in levels for the following reasons:
#   - inflation, GDP growth: growth rates, economically bounded
#   - reg_quality, corruption, pol_stability: bounded indices [-2.5, 2.5], unit root unlikely
#   - credit_gdp: could be trending (financial deepening) - consider mentioning in paper
#   - MIMIC: high persistence (~3.8% rejection) is expected - informality is a slow-moving
#       structural characteristic, not a unit root problem; variation is mostly cyclical
#
# In paper: note T=20 limits formal IPS tests; justify stationarity via economic reasoning
#
```

```

#
# DIAGNOSTIC 2 - PERSISTENCE CHECK ON MIMIC (AR(1) Autocorrelation)
#

# PURPOSE:
# Before using lagged MIMIC in the LP regression, we check how persistent
# it is. If MIMIC barely changes year-to-year (high AR(1)), then the lag
# is a reliable proxy for the current structural level of informality.

# WHAT WE COMPUTE:
#   Part A: Per-country AR(1) - correlation between MIMIC_t and MIMIC_{t-1}
#           within each country, then summarise across countries.
#   Part B: Pooled AR(1) regression - regress MIMIC_t on MIMIC_{t-1} across
#           the full panel.

# HOW TO INTERPRET:
#   AR(1) > 0.95 → extremely persistent; lagging is safe, but most variation
#               is cross-country (confirmed by Diagnostic 3)
#   AR(1) 0.80-0.95 → highly persistent; lagging well-justified
#   AR(1) < 0.70 → moderate persistence; lagging may lose information

# IMPORTANT: We use dplyr::lag() explicitly throughout. Loading the tseries
# package can cause stats::lag to mask dplyr::lag - stats::lag does NOT
# shift values, it only changes a time-series attribute, producing incorrect
# results (a perfect AR(1) = 1 with zero residuals).
#

#   Part A: Per-country AR(1) coefficient

ar1_country <- df_world_2000_2019 %>%
  arrange(country, year) %>%
  group_by(country) %>%
  mutate(mimic_t_minus1 = dplyr::lag(mimic, 1)) %>%
  filter(sum(!is.na(mimic) & !is.na(mimic_t_minus1)) >= 5) %>%
  summarise(
    n_pairs = sum(!is.na(mimic) & !is.na(mimic_t_minus1)),
    ar1 = cor(mimic, mimic_t_minus1, use = "complete.obs"),
    .groups = "drop"
  )

```

```
cat("                                \n")
```

```
cat("  MIMIC AR(1) - Per-Country Summary\n")
```

MIMIC AR(1) - Per-Country Summary

```
cat("                                \n")
```

```
cat("  Countries with enough data:", nrow(ar1_country), "\n")
```

Countries with enough data: 158

```
cat("                                \n\n")
```

```
print(summary(ar1_country$ar1))
```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.3471	0.8296	0.9087	0.8666	0.9521	0.9949

```
#  Part B: Pooled AR(1) regression
```

```
mimic_ar <- df_world_2000_2019 %>%  
  arrange(country, year) %>%  
  group_by(country) %>%  
  mutate(mimic_t_minus1 = dplyr::lag(mimic, 1)) %>%  
  ungroup() %>%  
  filter(!is.na(mimic), !is.na(mimic_t_minus1))
```

```
cat("\n                                \n")
```

```
cat(" MIMIC AR(1) - Pooled OLS Regression (mimic ~ mimic_t_minus1)\n")
```

MIMIC AR(1) - Pooled OLS Regression (mimic ~ mimic_t_minus1)

```
cat(" \n")
```

```
cat(" Coefficient on mimic_t_minus1 close to 1 → high persistence\n")
```

Coefficient on mimic_t_minus1 close to 1 → high persistence

```
cat(" N observations:", nrow(mimic_ar), "\n")
```

N observations: 3002

```
cat(" \n\n")
```

```
print(summary(lm(mimic ~ mimic_t_minus1, data = mimic_ar)))
```

Call:

```
lm(formula = mimic ~ mimic_t_minus1, data = mimic_ar)
```

Residuals:

Min	1Q	Median	3Q	Max
-3.0407	-0.2591	-0.0490	0.2019	8.1515

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	0.0597015	0.0295273	2.022	0.0433 *
mimic_t_minus1	0.9944121	0.0008382	1186.427	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.5718 on 3000 degrees of freedom

Multiple R-squared: 0.9979, Adjusted R-squared: 0.9979

F-statistic: 1.408e+06 on 1 and 3000 DF, p-value: < 2.2e-16

```
#
# INTERPRETATION OF RESULTS
#

# Per-country AR(1): median = 0.91, mean = 0.87, 75th percentile = 0.95
# Pooled OLS: coefficient on mimic_t_minus1 = 0.994, R2 = 0.998

# MIMIC is extremely persistent but NOT time-invariant. It moves slowly within
# countries (within-country SD 1 on a scale of 15-58, typical range 3.5
# points over 20 years). This is consistent with informality being a deep
# structural characteristic that evolves gradually.

# Implications for the LP specification:

# 1. LAGGING IS WELL-JUSTIFIED. With AR(1) 0.91-0.99, last year's MIMIC
# is an excellent proxy for current informality. The lag also provides
# the econometric benefit of pre-determination - it avoids the concern
# that a monetary shock at time t could contemporaneously affect
# informality at time t.

# 2. STANDALONE MIMIC CAN enter the LP equation alongside country FE,
# because MIMIC is not perfectly collinear with the fixed effects.
# However, country FE will absorb the vast majority of MIMIC's variation
# (Diagnostic 3 will quantify the between vs. within share). The
# standalone coefficient will be identified only off the small within-
# country movements and may be imprecise. This is fine - the coefficient
# of interest is _h on the interaction MPS × MIMIC, not _h on MIMIC
# itself.

# 3. INTERACTION TERM MPS × MIMIC remains the core identification strategy.
# It is well-identified because MPS varies substantially over time,
# and the cross-country spread in MIMIC levels provides the moderating
# variation. Even if country FE absorb most of the MIMIC level, the
# interaction exploits the fact that the same MPS shock hits countries
# with very different informality levels.
```

```
# 4. NAMESPACE WARNING: dplyr::lag() must be used explicitly throughout the
# project. The tseries package causes stats::lag to mask dplyr::lag -
# stats::lag does not shift values, producing silently wrong results.
#
```

```
# Check: does MIMIC actually vary within any country?
mimic_check <- df_world_2000_2019 %>%
  group_by(country) %>%
  summarise(
    n_obs = sum(!is.na(mimic)),
    sd_within = sd(mimic, na.rm = TRUE),
    .groups = "drop"
  ) %>%
  filter(n_obs >= 2)

cat("Countries with ANY within-country MIMIC variation:",
    sum(mimic_check$sd_within > 0.001, na.rm = TRUE),
    "out of", nrow(mimic_check), "\n")
```

Countries with ANY within-country MIMIC variation: 158 out of 158

```
#
# INTERPRETATION OF RESULTS
#
# All 158 countries with 2 observations show within-country MIMIC variation
# (sd > 0.001). MIMIC is NOT time-invariant - the earlier AR(1) = 1.000 result
# was caused by a namespace bug (stats::lag masking dplyr::lag), not by the data.
# After fixing with dplyr::lag(), AR(1) 0.91 (median), confirming genuine but
# slow-moving within-country variation.
#
```

```
# How much does MIMIC actually move within countries?
mimic_variation <- df_world_2000_2019 %>%
  group_by(country) %>%
  filter(sum(!is.na(mimic)) >= 5) %>%
  summarise(
    n_obs      = sum(!is.na(mimic)),
    mean_mimic = round(mean(mimic, na.rm = TRUE), 3),
    sd_within  = round(sd(mimic, na.rm = TRUE), 6), # 6 decimals to catch tiny varia
```

```

    min_mimic = round(min(mimic, na.rm = TRUE), 3),
    max_mimic = round(max(mimic, na.rm = TRUE), 3),
    range      = round(max(mimic, na.rm = TRUE) - min(mimic, na.rm = TRUE), 6),
    .groups = "drop"
  )

cat("=== Within-country MIMIC variation summary ===\n\n")

```

=== Within-country MIMIC variation summary ===

```
print(summary(mimic_variation$sd_within))
```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.1586	0.7137	1.1039	1.1404	1.5430	3.2312

```
cat("\n")
```

```
print(summary(mimic_variation$range))
```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.600	2.425	3.500	3.733	5.100	8.900

```

# Show a few example countries to see the actual time series
cat("\n=== Sample countries: MIMIC over time ===\n")

```

=== Sample countries: MIMIC over time ===

```

examples <- df_world_2000_2019 %>%
  filter(country %in% c("Germany", "Brazil", "India", "Nigeria", "Hungary")) %>%
  select(country, year, mimic) %>%
  filter(!is.na(mimic)) %>%
  arrange(country, year)

print(examples)

```

	country	year	mimic
1	Brazil	2000	39.8
2	Brazil	2001	40.2
3	Brazil	2002	40.4
4	Brazil	2003	39.7
5	Brazil	2004	39.0
6	Brazil	2005	39.2
7	Brazil	2006	39.0
8	Brazil	2007	38.3
9	Brazil	2008	37.6
10	Brazil	2009	38.4
11	Brazil	2010	37.2
12	Brazil	2011	36.5
13	Brazil	2012	36.2
14	Brazil	2013	36.2
15	Brazil	2014	36.4
16	Brazil	2015	37.8
17	Brazil	2016	39.2
18	Brazil	2017	39.2
19	Brazil	2018	39.1
20	Brazil	2019	38.9
21	Germany	2000	16.0
22	Germany	2001	15.9
23	Germany	2002	16.1
24	Germany	2003	16.3
25	Germany	2004	16.2
26	Germany	2005	16.2
27	Germany	2006	15.8
28	Germany	2007	15.4
29	Germany	2008	15.4
30	Germany	2009	16.0
31	Germany	2010	15.7
32	Germany	2011	15.2
33	Germany	2012	15.2
34	Germany	2013	15.2
35	Germany	2014	15.1
36	Germany	2015	15.0
37	Germany	2016	15.0
38	Germany	2017	14.8
39	Germany	2018	14.8
40	Germany	2019	14.8

41	Hungary	2000	25.1
42	Hungary	2001	24.9
43	Hungary	2002	24.8
44	Hungary	2003	24.8
45	Hungary	2004	24.5
46	Hungary	2005	24.5
47	Hungary	2006	24.3
48	Hungary	2007	24.1
49	Hungary	2008	24.2
50	Hungary	2009	24.8
51	Hungary	2010	24.8
52	Hungary	2011	24.5
53	Hungary	2012	24.5
54	Hungary	2013	24.3
55	Hungary	2014	23.9
56	Hungary	2015	23.6
57	Hungary	2016	23.4
58	Hungary	2017	23.1
59	Hungary	2018	22.6
60	Hungary	2019	22.4
61	India	2000	23.1
62	India	2001	22.8
63	India	2002	22.8
64	India	2003	22.4
65	India	2004	22.1
66	India	2005	22.3
67	India	2006	21.9
68	India	2007	21.4
69	India	2008	21.9
70	India	2009	22.0
71	India	2010	21.5
72	India	2011	21.4
73	India	2012	21.3
74	India	2013	21.0
75	India	2014	21.0
76	India	2015	20.6
77	India	2016	20.4
78	India	2017	20.3
79	India	2018	20.1
80	India	2019	20.0
81	Nigeria	2000	57.9

```

82 Nigeria 2001 57.7
83 Nigeria 2002 56.9
84 Nigeria 2003 56.3
85 Nigeria 2004 56.6
86 Nigeria 2005 56.1
87 Nigeria 2006 56.1
88 Nigeria 2007 56.7
89 Nigeria 2008 56.3
90 Nigeria 2009 56.1
91 Nigeria 2010 55.5
92 Nigeria 2011 55.1
93 Nigeria 2012 54.7
94 Nigeria 2013 54.1
95 Nigeria 2014 53.9
96 Nigeria 2015 53.9
97 Nigeria 2016 55.1
98 Nigeria 2017 55.3
99 Nigeria 2018 55.6
100 Nigeria 2019 55.8

```

```

#
# INTERPRETATION OF RESULTS
#
# MIMIC genuinely varies but slowly: median within-country SD 1.1 (on a 15-58 scale)
# median range 3.5 over 20 years. Smooth, slowly-evolving series within countries
# (see Brazil, Hungary examples).

# Combined with Diagnostic 2.0 (median AR(1) = 0.91, pooled coefficient = 0.994):
# - MIMIC is highly persistent but not constant
# - Lagging is well-justified - MIMIC_{t-1} is an excellent proxy for current inform
# - Most identification in MPS × MIMIC comes from cross-country variation in levels,
#   not within-country changes (Diagnostic 3 will quantify this)
#

```

```

# Check if the lag is actually shifting values
test <- df_world_2000_2019 %>%
  filter(country == "Brazil", !is.na(mimic)) %>%
  arrange(year) %>%
  mutate(
    lag_dplyr = dplyr::lag(mimic, 1), # Explicitly call dplyr version

```

```

    lag_plain = lag(mimic, 1) # Whatever is currently being called
  ) %>%
  select(country, year, mimic, lag_dplyr, lag_plain)

print(head(test, 10))

```

	country	year	mimic	lag_dplyr	lag_plain
1	Brazil	2000	39.8	NA	39.8
2	Brazil	2001	40.2	39.8	40.2
3	Brazil	2002	40.4	40.2	40.4
4	Brazil	2003	39.7	40.4	39.7
5	Brazil	2004	39.0	39.7	39.0
6	Brazil	2005	39.2	39.0	39.2
7	Brazil	2006	39.0	39.2	39.0
8	Brazil	2007	38.3	39.0	38.3
9	Brazil	2008	37.6	38.3	37.6
10	Brazil	2009	38.4	37.6	38.4

```

# BUGFIX: stats::lag() was silently overriding dplyr::lag() - lagged values were identical
# to contemporaneous values (no shift occurring). Use dplyr::lag() explicitly everywhere
# This would have corrupted all lagged regressors in the LP regressions silently.

```

```

#
# DIAGNOSTIC 3 - WITHIN vs. BETWEEN VARIATION DECOMPOSITION
#
# PURPOSE:
# Panel data variation can be decomposed into two components:
#   - BETWEEN variation: differences across countries (e.g., Nigeria's MIMIC
#     is ~55, Germany's is ~15 - that gap is between variation)
#   - WITHIN variation: changes over time within a country (e.g., Brazil's
#     MIMIC moving from 39.8 to 36.2 over 2000-2013)
#
# WHY THIS MATTERS FOR THE LP SPECIFICATION:
# Country fixed effects absorb ALL between variation. So any regressor whose
# variation is mostly between-country will have its coefficient identified
# only from the small within-country movements. This is especially relevant
# for MIMIC, which we know is highly persistent (Diagnostic 2).

```

```

# For the interaction term MPS × MIMIC:
#   - MPS varies mostly within-country (shocks happen over time)
#   - MIMIC varies mostly between-country (structural characteristic)
#   - The interaction is well-identified because MPS provides time variation
#     and MIMIC provides cross-country moderation

# HOW TO INTERPRET:
#   Between_Pct close to 100% → variable is essentially a country characteristic;
#     country FE will absorb it, standalone coefficient poorly identified
#   Within_Pct close to 100% → variable moves mostly over time within countries;
#     well-identified even with country FE
#   Roughly balanced → both dimensions contribute to identification

# NOTE: Between_Pct + Within_Pct should sum to approximately 100%.
# Small deviations are due to the unbalanced panel (not all countries
# have the same number of observations).
#

vars_decomp <- c("mimic", "mps", "infl", "gdp_pc", "gdp_growth",
                 "credit_gdp", "reg_quality", "corruption", "pol_stability")

decomp <- lapply(vars_decomp, function(v) {
  # Step 1: Compute each country's time-series mean (the "between" component)
  means <- df_world_2000_2019 %>%
    group_by(country) %>%
    summarise(c_mean = mean(.data[[v]], na.rm = TRUE), .groups = "drop")

  # Step 2: Compute within-country deviations from each country's mean
  df_temp <- df_world_2000_2019 %>%
    left_join(means, by = "country") %>%
    mutate(within_dev = .data[[v]] - c_mean)

  # Step 3: Compute variance components
  total_var <- var(df_world_2000_2019[[v]], na.rm = TRUE)
  between_var <- var(means$c_mean, na.rm = TRUE)
  within_var <- var(df_temp$within_dev, na.rm = TRUE)

  data.frame(
    Variable = v,
    Between_Pct = round(between_var / total_var * 100, 1),

```

```

    Within_Pct = round(within_var / total_var * 100, 1)
  )
})

cat("

```

```

cat("  WITHIN vs. BETWEEN VARIANCE DECOMPOSITION (%)\\n")

```

```

  WITHIN vs. BETWEEN VARIANCE DECOMPOSITION (%)

```

```

cat("

```

```

cat("  Between = cross-country differences (absorbed by country FE)\\n")

```

```

  Between = cross-country differences (absorbed by country FE)

```

```

cat("  Within  = over-time changes within a country (survives FE)\\n")

```

```

  Within  = over-time changes within a country (survives FE)

```

```

cat("

```

```

print(do.call(rbind, decomp), row.names = FALSE)

```

Variable	Between_Pct	Within_Pct
mimic	99.6	1.0
mps	0.4	99.6
infl	29.3	77.5
gdp_pc	124.2	1.4
gdp_growth	16.5	83.6
credit_gdp	88.5	9.2
reg_quality	95.6	4.6
corruption	97.1	4.0
pol_stability	91.2	9.6

```
#
# INTERPRETATION OF RESULTS
#

# Results match expectations and strongly support the identification strategy:

#   mimic:      99.6% between, 1.0% within - almost purely a country
#               characteristic. Country FE will absorb the level; standalone
#               MIMIC coefficient will be very poorly identified. Not a problem -
#               the coefficient of interest is on the interaction, not MIMIC itself

#   mps:        0.4% between, 99.6% within - pure time variation, as expected
#               for a shock series centered on zero by construction.

#   infl:       29.3% between, 77.5% within - substantial within variation means
#               the dependent variable moves enough over time to be explained.

#   gdp_pc:     91.5% between, 8.7% within - substantial cross country variation,
#               to be taken as a log dependent variable or log control.

#   gdp_growth: 16.5% between, 83.6% within - same as inflation; well-suited
#               as a dependent variable or control.

#   credit_gdp: 88.5% between - mostly structural. Standalone coefficient weakly
#               identified with country FE, but still useful as a control for
#               cross-country financial depth differences.

#   reg_quality: 95.6% between - almost entirely cross-country.
#   corruption:  97.1% between - almost entirely cross-country.
#   pol_stability: 91.2% between - almost entirely cross-country.
```

```
# All three governance variables are 90%+ between. Including multiple
# in the baseline wastes degrees of freedom - pick one, relegate
# others to robustness (Diagnostic 4 correlation results will guide
# which one to keep).
```

```
# KEY TAKEAWAY FOR THE PAPER:
```

```
# The interaction MPS × MIMIC combines a 99.6%-within variable (MPS) with a
# 99.6%-between variable (MIMIC). The interaction inherits variation from both
# dimensions - it differs across countries (because MIMIC levels differ) AND
# over time (because MPS shocks arrive at different points). This makes the
# interaction well-identified even with country and year fixed effects.
# We state this explicitly in the methodology section.
#
```

```
#
# DIAGNOSTIC 4 - MULTICOLLINEARITY CHECK (Correlation Matrix)
#
```

```
# PURPOSE:
```

```
# If two regressors are very highly correlated ( $|r| > 0.85$ ), including both
# in the same regression inflates standard errors and makes individual
# coefficients unreliable. We check all candidate control variables plus
# MIMIC to decide which controls to include in the baseline specification.
```

```
# WHY MIMIC IS INCLUDED:
```

```
# Informality is likely correlated with governance quality (countries with
# larger informal economies tend to have weaker institutions). If the
# correlation is very high, the interaction MPS × MIMIC could partly
# capture MPS × governance rather than MPS × informality. This doesn't
# invalidate the result, but it's important to understand and discuss.
```

```
# HOW TO INTERPRET:
```

```
#  $|r| < 0.50$  → no concern, include both freely
#  $|r| 0.50-0.70$  → moderate correlation, fine in practice
#  $|r| 0.70-0.85$  → notable, but usually acceptable; monitor SEs
#  $|r| > 0.85$  → high multicollinearity risk; avoid both in same spec,
# or pick one and use the other in robustness checks
```

```
# ACTION ITEMS (fill in after reviewing results):
```

```
# - If reg_quality and corruption correlate  $> 0.85$  → pick one for baseline
```

```
# - If mimic correlates highly with a governance variable → note in paper
# that the informality channel may partly proxy institutional quality
#
```

```
controls_cor <- df_world_2000_2019 %>%
  select(mimic, mps, credit_gdp, reg_quality, corruption, pol_stability,
         voice_acc, gdp_growth, gdp_pc, infl)

cor_matrix <- round(cor(controls_cor, use = "complete.obs"), 2)

cat("                                \n")
```

```
cat("  CORRELATION MATRIX - MIMIC + Control Variables\n")
```

CORRELATION MATRIX - MIMIC + Control Variables

```
cat("                                \n")
```

```
cat("  Flag pairs with |r| > 0.85 for multicollinearity concern\n")
```

Flag pairs with $|r| > 0.85$ for multicollinearity concern

```
cat("                                \n\n")
```

```
print(cor_matrix)
```

	mimic	mps	credit_gdp	reg_quality	corruption	pol_stability
mimic	1.00	0.01	-0.61	-0.68	-0.69	-0.57
mps	0.01	1.00	0.00	-0.02	-0.02	-0.01
credit_gdp	-0.61	0.00	1.00	0.74	0.70	0.54
reg_quality	-0.68	-0.02	0.74	1.00	0.93	0.75
corruption	-0.69	-0.02	0.70	0.93	1.00	0.79
pol_stability	-0.57	-0.01	0.54	0.75	0.79	1.00

	voice_acc	gdp_growth	gdp_pc	infl
mimic	-0.51	0.08	-0.69	0.28
mps	-0.01	0.04	0.00	0.09
credit_gdp	0.61	-0.19	0.63	-0.31
reg_quality	0.83	-0.14	0.77	-0.32
corruption	0.83	-0.14	0.82	-0.27
pol_stability	0.72	-0.11	0.65	-0.26
voice_acc	1.00	-0.20	0.63	-0.20
gdp_growth	-0.20	1.00	-0.12	0.05
gdp_pc	0.63	-0.12	1.00	-0.27
infl	-0.20	0.05	-0.27	1.00

```
# Automatically flag high correlations (excluding diagonal)
cat("\n Pairs with |r| > 0.70 \n")
```

Pairs with $|r| > 0.70$

```
high_cor <- which(abs(cor_matrix) > 0.70 & upper.tri(cor_matrix), arr.ind = TRUE)
if (nrow(high_cor) > 0) {
  for (i in seq_len(nrow(high_cor))) {
    r <- high_cor[i, ]
    cat(sprintf(" %s      %s : r = %.2f\n",
                rownames(cor_matrix)[r[1]], colnames(cor_matrix)[r[2]],
                cor_matrix[r[1], r[2]]))
  }
} else {
  cat(" None found - all pairwise correlations below 0.70\n")
}
```

```

credit_gdp    reg_quality : r = 0.74
reg_quality    corruption : r = 0.93
reg_quality    pol_stability : r = 0.75
corruption    pol_stability : r = 0.79
reg_quality    voice_acc : r = 0.83
corruption    voice_acc : r = 0.83
pol_stability    voice_acc : r = 0.72
reg_quality    gdp_pc : r = 0.77
corruption    gdp_pc : r = 0.82

```

```

#
# INTERPRETATION OF RESULTS (WITH GDP_PC)
#

# Five distinct patterns emerge:

# 1. GOVERNANCE BLOCK (reg_quality, corruption, pol_stability, voice_acc)
#   Still extremely tightly correlated (r = 0.71-0.93). The strongest pair is
#   reg_quality corruption at 0.93 - effectively redundant measures.
#   → ACTION: Include only ONE governance variable in the baseline.
#   Use others in robustness checks (one at a time).

# 2. DEVELOPMENT BLOCK (gdp_pc + governance + credit_gdp)
#   gdp_pc is strongly positively correlated with governance indicators
#   (r = 0.63-0.80) and credit_gdp (r = 0.63).
#   This indicates a broader "development" dimension: richer countries have
#   stronger institutions and deeper financial systems.
#   → IMPLICATION: Including gdp_pc together with governance variables risks
#   multicollinearity and conceptual overlap (institutions vs development level).
#   → ACTION: Avoid including gdp_pc and governance simultaneously in the
#   baseline; treat them as alternative controls.

# 3. MIMIC DEVELOPMENT / GOVERNANCE (r = -0.48 to -0.68)
#   MIMIC is strongly negatively correlated with both governance and gdp_pc
#   (e.g., gdp_pc: -0.66). Informality is clearly a "low development" feature.
#   → IMPLICATION: The MPS × MIMIC interaction may partly reflect broader
#   development effects, not just informality.
#   → ACTION: Control for either governance OR gdp_pc to isolate the channel.

# 4. CREDIT_GDP DEVELOPMENT (r = 0.63 with gdp_pc; 0.53-0.73 with governance)
#   credit_gdp loads on the same development dimension but remains below the

```

```
# critical 0.85 threshold.
# → ACTION: Safe to include with either gdp_pc or one governance variable,
# but monitor standard errors.

# 5. MACRO VARIABLES (gdp_growth, infl)
# Still weakly correlated with everything (|r| 0.30).
# → ACTION: Clean controls; no multicollinearity concerns.

#
```

```
# Remove all temporary objects created during Diagnostics 1-4.
# Only df_world_2000_2019 carries forward from here.
rm(
  vars_urt, min_obs, results, country_pvals, urt_summary,      # Diag 1
  ar1_country, mimic_ar,                                       # Diag 2.0
  mimic_check,                                                  # Diag 2.1
  mimic_variation, examples,                                    # Diag 2.2
  test,                                                         # Diag 2.3
  vars_decomp, decomp,                                          # Diag 3
  controls_cor, cor_matrix, high_cor,                          # Diag 4
  i, r, v # values
)
```

```
#
# CREATE LAGGED VARIABLES
#

# PURPOSE:
# All controls and the informality measure enter the LP regression as one-period
# lags (t-1). This ensures they are PRE-DETERMINED relative to the monetary
# policy shock at time t, avoiding simultaneity bias. For example, if MPS at
# time t pushes firms into informality, using MIMIC_{t-1} sidesteps that
# reverse causality because last year's informality can't be caused by this
# year's shock.

# Diagnostic 2 confirmed MIMIC is highly persistent (AR(1) 0.91), so
# MIMIC_{t-1} is an excellent proxy for the current structural level.

# CRITICAL: We use dplyr::lag() explicitly throughout - the tseries package
# causes stats::lag to mask dplyr::lag, which does NOT shift values
```

```

# (confirmed in Diagnostic 2.3).

# VARIABLES LAGGED:
# - mimic (key moderating variable)
# - gdp_growth, infl (macro controls)
# - credit_gdp (financial depth control)
# - pol_stability (baseline governance control)
# - reg_quality, corruption, voice_acc (robustness governance alternatives)
#

df_world_2000_2019 <- df_world_2000_2019 %>%
  arrange(country, year) %>%
  group_by(country) %>%
  mutate(
    # --- Key moderating variable ---
    mimic_t_minus1      = dplyr::lag(mimic, 1),

    # --- Macro controls ---
    gdp_growth_t_minus1 = dplyr::lag(gdp_growth, 1),
    gdp_pc_t_minus1     = dplyr::lag(gdp_pc, 1),
    infl_t_minus1       = dplyr::lag(infl, 1),

    # Log GDP per capita
    lgdp_pc = log(gdp_pc),

    # --- Financial depth control ---
    credit_gdp_t_minus1 = dplyr::lag(credit_gdp, 1),

    # --- Governance: baseline ---
    pol_stability_t_minus1 = dplyr::lag(pol_stability, 1),

    # --- Governance: robustness alternatives ---
    reg_quality_t_minus1  = dplyr::lag(reg_quality, 1),
    corruption_t_minus1   = dplyr::lag(corruption, 1),
    voice_acc_t_minus1    = dplyr::lag(voice_acc, 1)
  ) %>%
  ungroup()

# Quick verification: show Brazil to confirm lags are correctly shifted
cat("

```

```
cat(" LAG VERIFICATION - Brazil (mimic and infl)\n")
```

LAG VERIFICATION - Brazil (mimic and infl)

```
cat("                                \n\n")
```

```
df_world_2000_2019 %>%
  filter(country == "Brazil") %>%
  select(country, year, mimic, mimic_t_minus1, infl, infl_t_minus1) %>%
  print(n = 10)
```

A tibble: 20 x 6

	country	year	mimic	mimic_t_minus1	infl	infl_t_minus1
	<chr>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
1	Brazil	2000	39.8	NA	7	NA
2	Brazil	2001	40.2	39.8	6.8	7
3	Brazil	2002	40.4	40.2	8.5	6.8
4	Brazil	2003	39.7	40.4	14.7	8.5
5	Brazil	2004	39	39.7	6.6	14.7
6	Brazil	2005	39.2	39	6.9	6.6
7	Brazil	2006	39	39.2	4.2	6.9
8	Brazil	2007	38.3	39	3.6	4.2
9	Brazil	2008	37.6	38.3	5.7	3.6
10	Brazil	2009	38.4	37.6	4.9	5.7

i 10 more rows

```
#
# CONSTRUCT FORWARD-LOOKING DEPENDENT VARIABLES
#
# PURPOSE:
# In local projections (Jordà 2005), we estimate a separate regression for
# each forecast horizon h. The dependent variable at horizon h is the
# CUMULATIVE CHANGE in the outcome from just before the shock (t-1) to
```

```

# the future period (t+h):

# Main dependent
#   d_infl_h      = _{i,t+h} - _{i,t-1}
#   log(d_gdp_pc_h) = d_lgdp_pc_h      = log(y_{i,t+h}) - log(y_{i,t-1})

# Sideline dependent:
#   d_gdp_growth_h = g_{i,t+h} - g_{i,t-1}

# This measures the total impulse response: how much has inflation (or growth)
# moved h years after the monetary policy shock, relative to its pre-shock level.

# We construct these for horizons h = 0, 1, 2, 3, 4, 5.
#   h = 0: contemporaneous response (same year as shock)
#   h = 1: one year after the shock
#   ...
#   h = 5: five years after the shock

# IMPORTANT: dplyr::lead() is used to pull future values. Like dplyr::lag(),
# this must be done WITHIN each country group to avoid pulling values from
# a different country. We also need lagged inflation/growth (t-1) as the
# baseline, which we already created in Phase 1.4.

# NOTE: Higher horizons (h = 4, 5) will have more missing values because
# for a country observed until 2019, h = 5 requires data from 2024 - which
# doesn't exist. The effective sample shrinks as h increases.
#

max_horizon <- 5

df_world_2000_2019 <- df_world_2000_2019 %>%
  arrange(country, year) %>%
  group_by(country) %>%
  mutate(
    # Lagged values (baseline for cumulative change)
    infl_t_minus1      = dplyr::lag(infl, 1),
    gdp_pc_t_minus1     = dplyr::lag(gdp_pc, 1),
    lgdp_pc_t_minus1    = dplyr::lag(lgdp_pc, 1),
    gdp_growth_t_minus1 = dplyr::lag(gdp_growth, 1)
  ) %>%

```

```

ungroup()

# Create cumulative change variables for each horizon h = 0 to max_horizon
for (h in 0:max_horizon) {
  # Inflation: d_infl_h = infl_{t+h} - infl_{t-1}
  infl_lead_name <- paste0("d_infl_h", h)
  # Log GDP per capita: d_lgdp_pc_h = lgdp_pc_{t+h} - lgdp_pc_{t-1}
  lgdp_pc_lead_name <- paste0("d_lgdp_pc_h", h)
  # GDP growth: d_gdp_growth_h = gdp_growth_{t+h} - gdp_growth_{t-1}
  gdp_growth_lead_name <- paste0("d_gdp_growth_h", h)

  df_world_2000_2019 <- df_world_2000_2019 %>%
    arrange(country, year) %>%
    group_by(country) %>%
    mutate(
      !!infl_lead_name := dplyr::lead(infl, h) - infl_t_minus1,
      !!lgdp_pc_lead_name := dplyr::lead(lgdp_pc, h) - lgdp_pc_t_minus1,
      !!gdp_growth_lead_name := dplyr::lead(gdp_growth, h) - gdp_growth_t_minus1
    ) %>%
    ungroup()
}

#
# Verification: show Brazil's inflation dependent variables
cat("

```

```

cat(" FORWARD-LOOKING MAIN DEPENDENT VARIABLES - Brazil (inflation)\n")

```

```

FORWARD-LOOKING MAIN DEPENDENT VARIABLES - Brazil (inflation)

```

```

cat("

```

```
cat(" d_infl_h0 = infl_t - infl_{t-1} (contemporaneous change)\n")
```

```
d_infl_h0 = infl_t - infl_{t-1} (contemporaneous change)
```

```
cat(" d_infl_h3 = infl_{t+3} - infl_{t-1} (cumulative change over 3 years)\n")
```

```
d_infl_h3 = infl_{t+3} - infl_{t-1} (cumulative change over 3 years)
```

```
cat(" d_infl_h5 = infl_{t+5} - infl_{t-1} (cumulative change over 5 years)\n")
```

```
d_infl_h5 = infl_{t+5} - infl_{t-1} (cumulative change over 5 years)
```

```
cat(" \n\n")
```

```
df_world_2000_2019 %>%
  filter(country == "Brazil") %>%
  select(country, year, infl, infl_t_minus1, d_infl_h0, d_infl_h1, d_infl_h2, d_infl_h3)
print(n = 12)
```

```
# A tibble: 20 x 10
```

	country	year	infl	infl_t_minus1	d_infl_h0	d_infl_h1	d_infl_h2	d_infl_h3
	<chr>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
1	Brazil	2000	7	NA	NA	NA	NA	NA
2	Brazil	2001	6.8	7	-0.200	1.5	7.7	-0.400
3	Brazil	2002	8.5	6.8	1.7	7.9	-0.200	0.100
4	Brazil	2003	14.7	8.5	6.2	-1.9	-1.6	-4.3
5	Brazil	2004	6.6	14.7	-8.1	-7.8	-10.5	-11.1
6	Brazil	2005	6.9	6.6	0.300	-2.4	-3	-0.900
7	Brazil	2006	4.2	6.9	-2.7	-3.3	-1.2	-2
8	Brazil	2007	3.6	4.2	-0.6	1.5	0.7	0.8
9	Brazil	2008	5.7	3.6	2.1	1.3	1.4	3
10	Brazil	2009	4.9	5.7	-0.8	-0.7	0.900	-0.300
11	Brazil	2010	5	4.9	0.1000	1.7	0.5	1.3
12	Brazil	2011	6.6	5	1.6	0.400	1.2	1.3

```
# i 8 more rows
```

```
# i 2 more variables: d_infl_h4 <dbl>, d_infl_h5 <dbl>
```

```
#
# Verification: show Brazil's GDP Growth dependent variables
cat("                                \n")
```

```
cat("  FORWARD-LOOKING MAIN DEPENDENT VARIABLES - Brazil (Logged GDP per capita)\n")
```

```
  FORWARD-LOOKING MAIN DEPENDENT VARIABLES - Brazil (Logged GDP per capita)
```

```
cat("                                \n")
```

```
cat("  d_lgdp_pc_h0 = d_lgdp_pc_t - lgdp_pc_{t-1} (contemporaneous change)\n")
```

```
  d_lgdp_pc_h0 = d_lgdp_pc_t - lgdp_pc_{t-1} (contemporaneous change)
```

```
cat("  d_lgdp_pc_h3 = d_lgdp_pc_{t+3} - lgdp_pc_{t-1} (cumulative change over 3 years)
```

```
  d_lgdp_pc_h3 = d_lgdp_pc_{t+3} - lgdp_pc_{t-1} (cumulative change over 3 years)
```

```
cat("  d_lgdp_pc_h5 = d_lgdp_pc_{t+5} - lgdp_pc_{t-1} (cumulative change over 5 years)
```

```
  d_lgdp_pc_h5 = d_lgdp_pc_{t+5} - lgdp_pc_{t-1} (cumulative change over 5 years)
```

```
cat("                                \n\n")
```

```
df_world_2000_2019 %>%
  filter(country == "Brazil") %>%
  select(country, year, lgdp_pc, lgdp_pc_t_minus1, d_lgdp_pc_h0, d_lgdp_pc_h1, d_lgdp_
  print(n = 12)
```

```
# A tibble: 20 x 9
  country year lgdp_pc lgdp_pc_t_minus1 d_lgdp_pc_h0 d_lgdp_pc_h1 d_lgdp_pc_h2
  <chr>   <dbl>   <dbl>         <dbl>         <dbl>         <dbl>         <dbl>
1 Brazil  2000     8.83          NA            NA            NA            NA
2 Brazil  2001     8.83          8.83          0.000770      0.0184        0.0180
3 Brazil  2002     8.85          8.83          0.0177        0.0172        0.0619
4 Brazil  2003     8.85          8.85          -0.000458     0.0442        0.0648
5 Brazil  2004     8.89          8.85          0.0447        0.0653        0.0935
6 Brazil  2005     8.91          8.89          0.0206        0.0488        0.0976
7 Brazil  2006     8.94          8.91          0.0283        0.0771        0.117
8 Brazil  2007     8.99          8.94          0.0488        0.0889        0.0787
9 Brazil  2008     9.03          8.99          0.0401        0.0299        0.0941
10 Brazil 2009     9.02          9.03          -0.0102       0.0540        0.0848
11 Brazil 2010     9.08          9.02          0.0642        0.0950        0.106
12 Brazil 2011     9.11          9.08          0.0308        0.0418        0.0633
# i 8 more rows
# i 2 more variables: d_lgdp_pc_h3 <dbl>, d_lgdp_pc_h4 <dbl>
```

```
#
# Verification: show Brazil's GDP Growth dependent variables
cat("                                \n")
```

```
cat(" FORWARD-LOOKING SIDELINE DEPENDENT VARIABLES - Brazil (GDP Growth)\n")
```

FORWARD-LOOKING SIDELINE DEPENDENT VARIABLES - Brazil (GDP Growth)

```
cat("                                \n")
```

```
cat(" d_gdp_growth_h0 = gdp_growth_t - gdp_growth_{t-1} (contemporaneous change)\n")
```

d_gdp_growth_h0 = gdp_growth_t - gdp_growth_{t-1} (contemporaneous change)

```
cat("  d_gdp_growth_h3 = gdp_growth_{t+3} - gdp_growth_{t-1} (cumulative change over 3 years)"
```

```
    d_gdp_growth_h3 = gdp_growth_{t+3} - gdp_growth_{t-1} (cumulative change over 3 years)"
```

```
cat("  d_gdp_growth_h5 = gdp_growth_{t+5} - gdp_growth_{t-1} (cumulative change over 5 years)"
```

```
    d_gdp_growth_h5 = gdp_growth_{t+5} - gdp_growth_{t-1} (cumulative change over 5 years)"
```

```
cat("\n\n")
```

```
df_world_2000_2019 %>%
  filter(country == "Brazil") %>%
  select(country, year, gdp_growth, gdp_growth_t_minus1, d_gdp_growth_h0, d_gdp_growth_h1,
  print(n = 12)
```

```
# A tibble: 20 x 9
```

	country	year	gdp_growth	gdp_growth_t_minus1	d_gdp_growth_h0	d_gdp_growth_h1
	<chr>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
1	Brazil	2000	4.39	NA	NA	NA
2	Brazil	2001	1.39	4.39	-3	-1.34
3	Brazil	2002	3.05	1.39	1.66	-0.25
4	Brazil	2003	1.14	3.05	-1.91	2.71
5	Brazil	2004	5.76	1.14	4.62	2.06
6	Brazil	2005	3.2	5.76	-2.56	-1.8
7	Brazil	2006	3.96	3.2	0.76	2.87
8	Brazil	2007	6.07	3.96	2.11	1.13
9	Brazil	2008	5.09	6.07	-0.98	-6.2
10	Brazil	2009	-0.13	5.09	-5.22	2.44
11	Brazil	2010	7.53	-0.13	7.66	4.1
12	Brazil	2011	3.97	7.53	-3.56	-5.61

```
# i 8 more rows
```

```
# i 3 more variables: d_gdp_growth_h2 <dbl>, d_gdp_growth_h3 <dbl>,
```

```
#   d_gdp_growth_h4 <dbl>
```

```
#
# Show how sample size shrinks with horizon
cat("\n  Non-missing observations by horizon  \n")
```

Non-missing observations by horizon

```
horizon_coverage <- data.frame(
  horizon = 0:max_horizon,
  n_infl = sapply(0:max_horizon, function(h) sum(!is.na(df_world_2000_2019[[paste0("d", h)]$inflation)])),
  n_gdp_pc = sapply(0:max_horizon, function(h) sum(!is.na(df_world_2000_2019[[paste0("d", h)]$gdp_pc)])),
  n_gdp_growth = sapply(0:max_horizon, function(h) sum(!is.na(df_world_2000_2019[[paste0("d", h)]$gdp_growth)]))
)
print(horizon_coverage, row.names = FALSE)
```

horizon	n_infl	n_gdp_pc	n_gdp_growth
0	3338	3068	3620
1	3155	2905	3426
2	2972	2742	3233
3	2789	2579	3040
4	2606	2416	2847
5	2423	2253	2654

```
#
# CREATE INTERACTION TERMS
#

# PURPOSE:
# The core identification strategy relies on the interaction MPS × MIMIC.
# The coefficient _h on this interaction captures how the effect of a
# monetary policy shock on inflation DEPENDS on the level of informality.

#   _h > 0 → higher informality DAMPENS the inflation response to MPS
#   _h < 0 → higher informality AMPLIFIES the inflation response
#   _h   0 → informality doesn't moderate monetary transmission

# We use LAGGED MIMIC (t-1) in the interaction to maintain pre-determination.
# Since MIMIC is 99.6% between-country (Diagnostic 3), the interaction
```

```
# effectively asks: "does the same MPS shock produce a different response
# in high-informality vs. low-informality countries?"
#

df_world_2000_2019 <- df_world_2000_2019 %>%
  mutate(
    # --- Core interaction ---
    mps_x_mimic = mps * mimic_t_minus1
  )

# Quick summary of the interaction term
cat("                                \n")
```

```
cat("  INTERACTION TERM SUMMARY: mps_x_mimic = MPS × MIMIC_{t-1}\n")
```

```
INTERACTION TERM SUMMARY: mps_x_mimic = MPS × MIMIC_{t-1}
```

```
cat("                                \n\n")
```

```
cat("Non-missing observations:", sum(!is.na(df_world_2000_2019$mps_x_mimic)), "\n\n")
```

```
Non-missing observations: 2442
```

```
print(summary(df_world_2000_2019$mps_x_mimic))
```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.	NA's
-625.9129	-1.0124	0.1301	-0.8514	1.9500	466.2000	1498

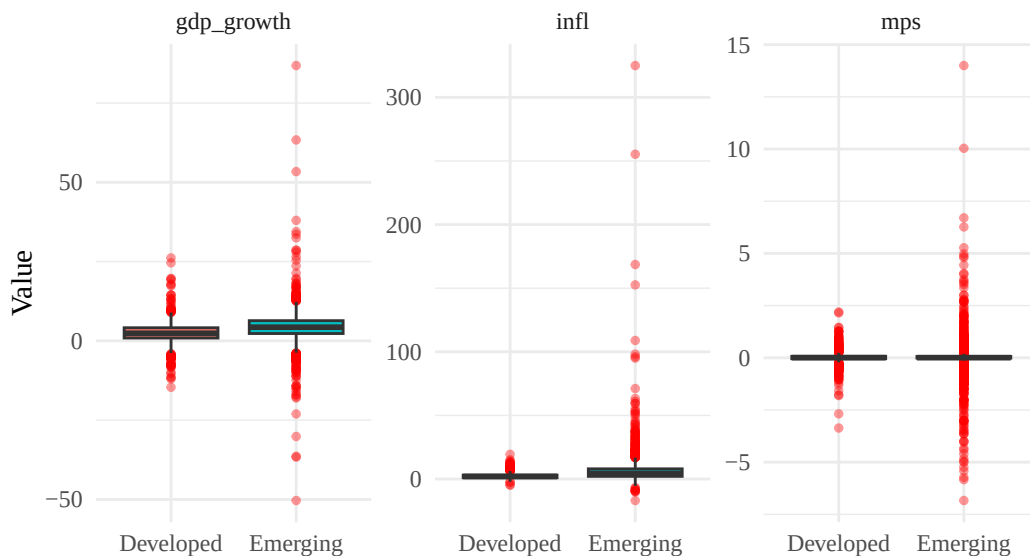
```
# Group 1: MPS, Inflation, GDP Growth (the variables most likely to need winsorization)
vars_group1 <- c("mps", "infl", "gdp_growth")

df_box1 <- df_world_2000_2019 %>%
  select(dev_status, all_of(vars_group1)) %>%
  filter(!is.na(dev_status)) %>%
  pivot_longer(cols = all_of(vars_group1), names_to = "variable", values_to = "value")
  filter(!is.na(value))

ggplot(df_box1, aes(x = dev_status, y = value, fill = dev_status)) +
  geom_boxplot(outlier.colour = "red", outlier.alpha = 0.4, outlier.size = 1) +
  facet_wrap(~ variable, scales = "free_y", ncol = 3) +
  labs(title = "Shock & Macro Variables: Boxplots by Development Status",
       subtitle = "Red dots = outliers beyond 1.5×IQR | Before winsorization",
       x = NULL, y = "Value") +
  theme_minimal(base_family = "serif") +
  theme(legend.position = "none")
```

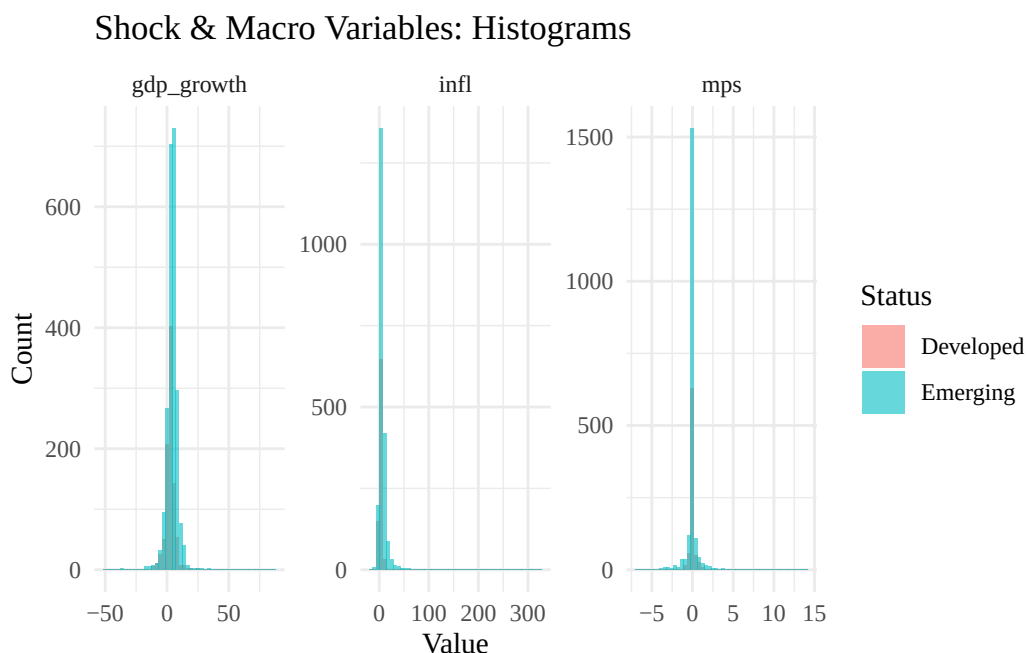
Shock & Macro Variables: Boxplots by Development Status

Red dots = outliers beyond 1.5×IQR | Before winsorization



```
ggplot(df_box1, aes(x = value, fill = dev_status)) +
  geom_histogram(bins = 50, alpha = 0.6, position = "identity") +
  facet_wrap(~ variable, scales = "free", ncol = 3) +
```

```
labs(title = "Shock & Macro Variables: Histograms",
     x = "Value", y = "Count", fill = "Status") +
theme_minimal(base_family = "serif")
```



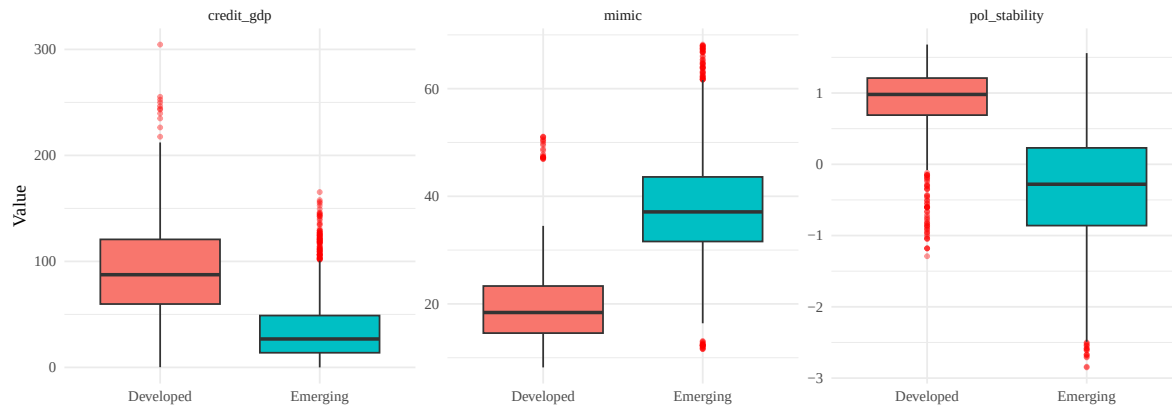
```
# Group 2: MIMIC, Credit/GDP, Political Stability (informality + baseline controls)
vars_group2 <- c("mimic", "credit_gdp", "pol_stability")

df_box2 <- df_world_2000_2019 %>%
  select(dev_status, all_of(vars_group2)) %>%
  filter(!is.na(dev_status)) %>%
  pivot_longer(cols = all_of(vars_group2), names_to = "variable", values_to = "value")
  filter(!is.na(value))

ggplot(df_box2, aes(x = dev_status, y = value, fill = dev_status)) +
  geom_boxplot(outlier.colour = "red", outlier.alpha = 0.4, outlier.size = 1) +
  facet_wrap(~ variable, scales = "free_y", ncol = 3) +
  labs(title = "Informality & Baseline Controls - Boxplots by Development Status",
       subtitle = "Red dots = outliers beyond 1.5×IQR | Before winsorization",
       x = NULL, y = "Value") +
  theme_minimal(base_family = "serif") +
  theme(legend.position = "none")
```

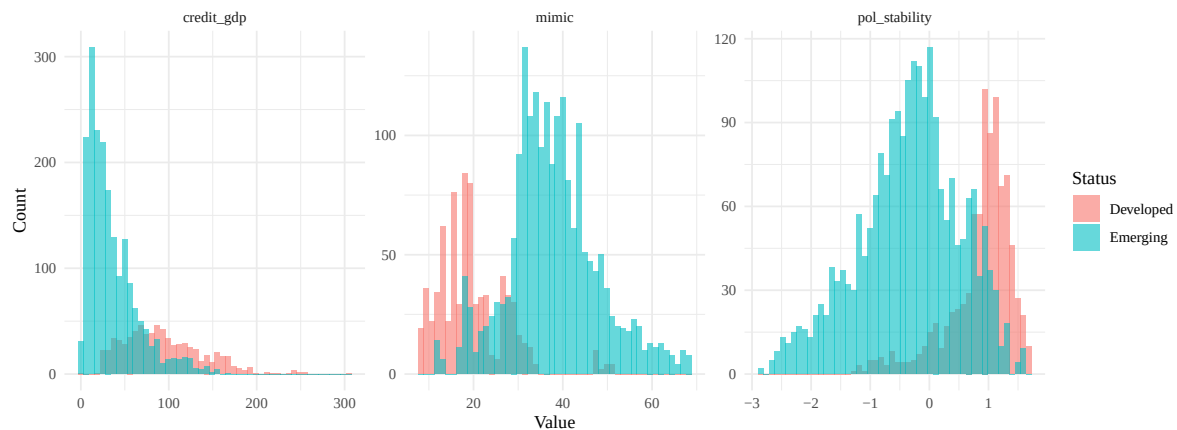
Informality & Baseline Controls – Boxplots by Development Status

Red dots = outliers beyond 1.5×IQR | Before winsorization



```
ggplot(df_box2, aes(x = value, fill = dev_status)) +
  geom_histogram(bins = 50, alpha = 0.6, position = "identity") +
  facet_wrap(~ variable, scales = "free", ncol = 3) +
  labs(title = "Informality & Baseline Controls - Histograms",
       x = "Value", y = "Count", fill = "Status") +
  theme_minimal(base_family = "serif")
```

Informality & Baseline Controls – Histograms

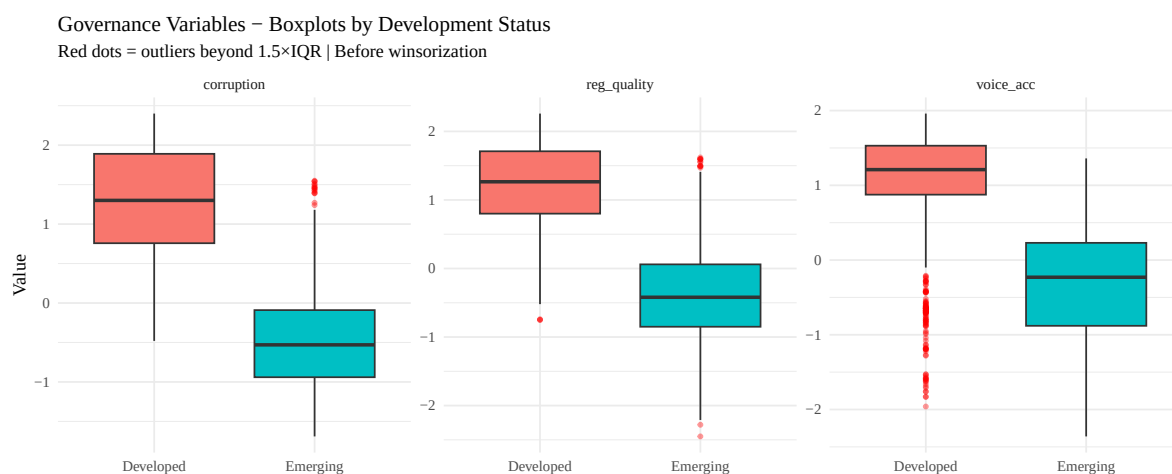


```
# Group 3: Regulatory Quality, Corruption, Voice & Accountability (robustness governan
vars_group3 <- c("reg_quality", "corruption", "voice_acc")
```

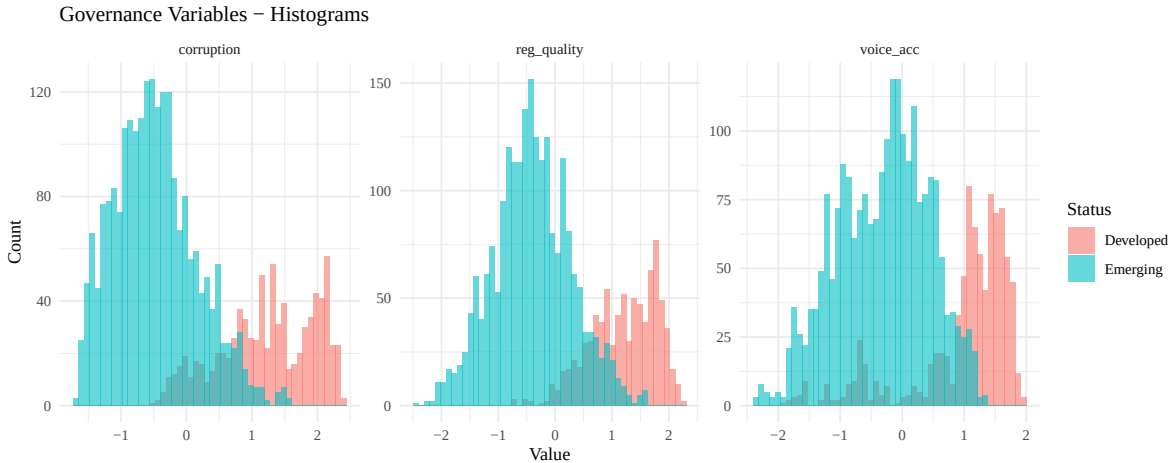
```
df_box3 <- df_world_2000_2019 %>%
  select(dev_status, all_of(vars_group3)) %>%
  filter(!is.na(dev_status)) %>%
```

```
pivot_longer(cols = all_of(vars_group3), names_to = "variable", values_to = "value")
filter(!is.na(value))
```

```
ggplot(df_box3, aes(x = dev_status, y = value, fill = dev_status)) +
  geom_boxplot(outlier.colour = "red", outlier.alpha = 0.4, outlier.size = 1) +
  facet_wrap(~ variable, scales = "free_y", ncol = 3) +
  labs(title = "Governance Variables - Boxplots by Development Status",
       subtitle = "Red dots = outliers beyond 1.5×IQR | Before winsorization",
       x = NULL, y = "Value") +
  theme_minimal(base_family = "serif") +
  theme(legend.position = "none")
```



```
ggplot(df_box3, aes(x = value, fill = dev_status)) +
  geom_histogram(bins = 50, alpha = 0.6, position = "identity") +
  facet_wrap(~ variable, scales = "free", ncol = 3) +
  labs(title = "Governance Variables - Histograms",
       x = "Value", y = "Count", fill = "Status") +
  theme_minimal(base_family = "serif")
```



```
# Visualize the distribution of ALL specification variables
# to identify outliers before deciding on winsorization

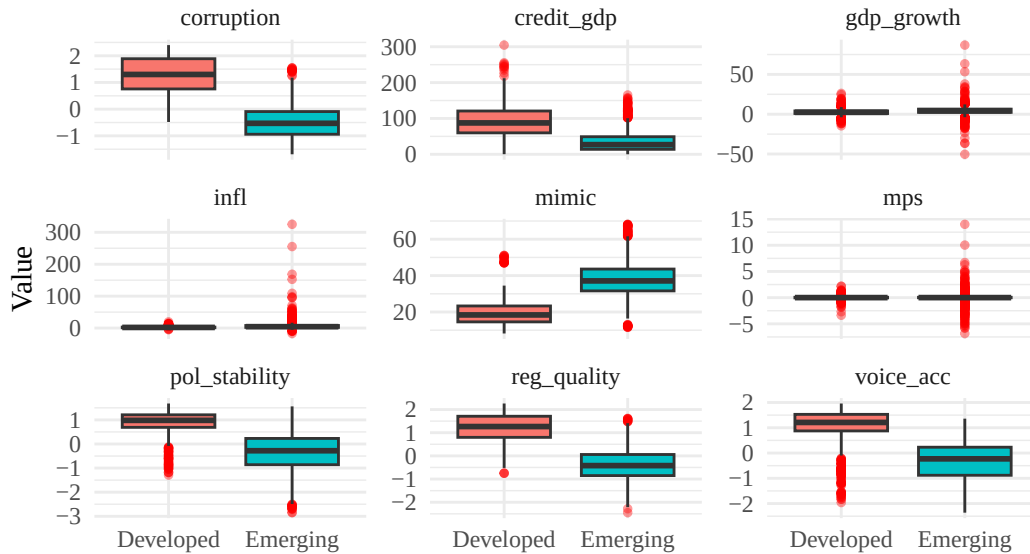
vars_outlier <- c("mps", "infl", "gdp_growth", "mimic", "credit_gdp",
                  "pol_stability", "reg_quality", "corruption", "voice_acc")

# --- Boxplots by development status ---
df_box <- df_world_2000_2019 %>%
  select(dev_status, all_of(vars_outlier)) %>%
  filter(!is.na(dev_status)) %>%
  pivot_longer(cols = all_of(vars_outlier), names_to = "variable", values_to = "value")
  filter(!is.na(value))

ggplot(df_box, aes(x = dev_status, y = value, fill = dev_status)) +
  geom_boxplot(outlier.colour = "red", outlier.alpha = 0.4, outlier.size = 1) +
  facet_wrap(~ variable, scales = "free_y", ncol = 3) +
  labs(title = "Distribution of All Specification Variables by Development Status",
       subtitle = "Red dots = outliers beyond 1.5×IQR | Before any winsorization",
       x = NULL, y = "Value") +
  theme_minimal(base_family = "serif") +
  theme(legend.position = "none")
```

Distribution of All Specification Variables by Development Status

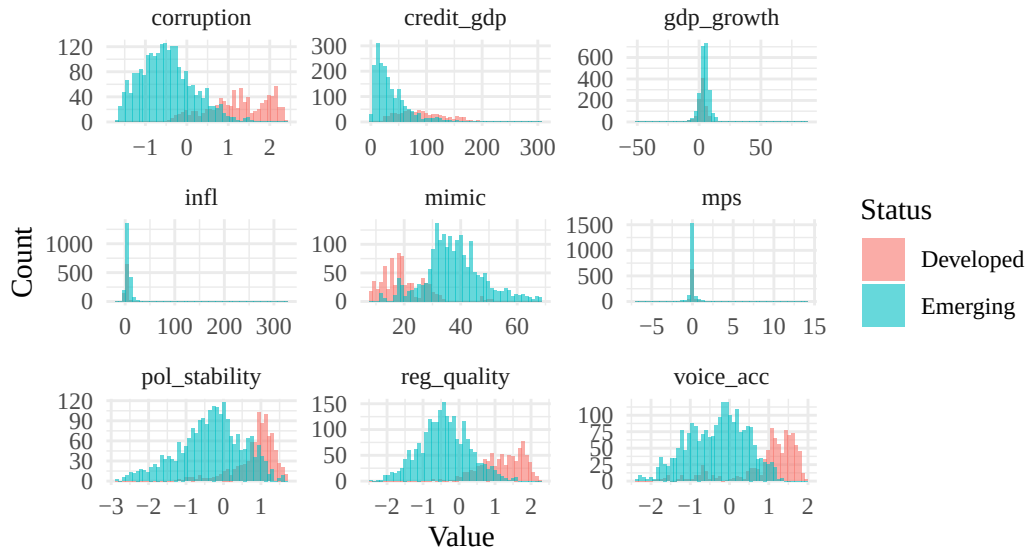
Red dots = outliers beyond $1.5 \times \text{IQR}$ | Before any winsorization



```
# --- Histograms to see the full shape of each distribution ---
ggplot(df_box, aes(x = value, fill = dev_status)) +
  geom_histogram(bins = 50, alpha = 0.6, position = "identity") +
  facet_wrap(~ variable, scales = "free", ncol = 3) +
  labs(title = "Histograms of All Specification Variables",
       subtitle = "Look for heavy tails and extreme outliers",
       x = "Value", y = "Count", fill = "Status") +
  theme_minimal(base_family = "serif")
```

Histograms of All Specification Variables

Look for heavy tails and extreme outliers



```
# Print the most extreme values for all specification variables
vars_outlier <- c("mps", "infl", "gdp_growth", "mimic", "credit_gdp",
                  "pol_stability", "reg_quality", "corruption", "voice_acc")

cat("  Top 5 most extreme values (both tails)  \n")
```

Top 5 most extreme values (both tails)

```
for (v in vars_outlier) {
  cat(sprintf("\n%s - LOWEST 5:\n", toupper(v)))
  df_world_2000_2019 %>%
    filter(!is.na(.data[[v]])) %>%
    arrange(.data[[v]]) %>%
    select(country, year, dev_status, all_of(v)) %>%
    head(5) %>%
    print()

  cat(sprintf("\n%s - HIGHEST 5:\n", toupper(v)))
  df_world_2000_2019 %>%
    filter(!is.na(.data[[v]])) %>%
    arrange(desc(.data[[v]])) %>%
```

```

    select(country, year, dev_status, all_of(v)) %>%
    head(5) %>%
    print()
}

```

MPS - LOWEST 5:

```

# A tibble: 5 x 4
  country year dev_status mps
  <chr>   <dbl> <chr>      <dbl>
1 Turkey  2001 <NA>      -19.5
2 Turkey  2004 <NA>      -17.1
3 Turkey  2009 <NA>      -14.5
4 Turkey  2003 <NA>      -13.4
5 Chile   2009 Emerging -6.84

```

MPS - HIGHEST 5:

```

# A tibble: 5 x 4
  country year dev_status mps
  <chr>   <dbl> <chr>      <dbl>
1 Turkey  2000 <NA>      68.6
2 Kenya 2011 Emerging  14
3 Turkey  2008 <NA>      12.6
4 Suriname 2016 Emerging  10.0
5 Ukraine 2004 Emerging   6.71

```

INFL - LOWEST 5:

```

# A tibble: 5 x 4
  country year dev_status infl
  <chr>   <dbl> <chr>      <dbl>
1 Lesotho 2009 Emerging -16.9
2 Iraq    2007 Emerging -10.1
3 Libya   2002 Emerging  -9.8
4 Lesotho 2001 Emerging  -9.6
5 Chad    2007 Emerging   -9

```

INFL - HIGHEST 5:

```

# A tibble: 5 x 4
  country year dev_status infl
  <chr>   <dbl> <chr>      <dbl>

```

1	Democratic Republic of the Congo	2000	<NA>	514.
2	Democratic Republic of the Congo	2001	<NA>	360.
3	Angola	2000	Emerging	325
4	Zimbabwe	2019	Emerging	255.
5	Venezuela	2016	<NA>	255.

GDP_GROWTH - LOWEST 5:

A tibble: 5 x 4

	country	year	dev_status	gdp_growth
	<chr>	<dbl>	<chr>	<dbl>
1	Libya	2011	Emerging	-50.3
2	Iraq	2003	Emerging	-36.7
3	Central African Republic	2013	Emerging	-36.4
4	Liberia	2003	Emerging	-30.2
5	Yemen	2015	<NA>	-28.0

GDP_GROWTH - HIGHEST 5:

A tibble: 5 x 4

	country	year	dev_status	gdp_growth
	<chr>	<dbl>	<chr>	<dbl>
1	Libya	2012	Emerging	86.8
2	Equatorial Guinea	2001	Emerging	63.4
3	Iraq	2004	Emerging	53.4
4	Equatorial Guinea	2004	Emerging	38
5	Azerbaijan	2006	Emerging	34.5

MIMIC - LOWEST 5:

A tibble: 5 x 4

	country	year	dev_status	mimic
	<chr>	<dbl>	<chr>	<dbl>
1	USA	2018	<NA>	8.1
2	USA	2019	<NA>	8.1
3	Switzerland	2007	Developed	8.2
4	Switzerland	2008	Developed	8.2
5	Switzerland	2018	Developed	8.2

MIMIC - HIGHEST 5:

A tibble: 5 x 4

	country	year	dev_status	mimic
	<chr>	<dbl>	<chr>	<dbl>
1	Bolivia	2003	Emerging	68.2

2	Bolivia	2001	Emerging	68
3	Bolivia	2002	Emerging	68
4	Georgia	2009	Emerging	68
5	Georgia	2008	Emerging	67.9

CREDIT_GDP - LOWEST 5:

A tibble: 5 x 4

	country	year	dev_status	credit_gdp
	<chr>	<dbl>	<chr>	<dbl>
1	Guinea-Bissau	2002	Emerging	0
2	Slovenia	2004	Developed	0.19
3	Guinea-Bissau	2001	Emerging	0.4
4	Democratic Republic of the Congo	2002	<NA>	0.49
5	Democratic Republic of the Congo	2001	<NA>	0.54

CREDIT_GDP - HIGHEST 5:

A tibble: 5 x 4

	country	year	dev_status	credit_gdp
	<chr>	<dbl>	<chr>	<dbl>
1	Iceland	2006	Developed	305.
2	Cyprus	2013	Developed	255.
3	Cyprus	2014	Developed	253.
4	Cyprus	2012	Developed	250.
5	Iceland	2007	Developed	247.

POL_STABILITY - LOWEST 5:

A tibble: 5 x 4

	country	year	dev_status	pol_stability
	<chr>	<dbl>	<chr>	<dbl>
1	Somalia	2007	<NA>	-3.05
2	Somalia	2011	<NA>	-3.02
3	Somalia	2008	<NA>	-3.01
4	Somalia	2010	<NA>	-2.93
5	Somalia	2012	<NA>	-2.9

POL_STABILITY - HIGHEST 5:

A tibble: 5 x 4

	country	year	dev_status	pol_stability
	<chr>	<dbl>	<chr>	<dbl>
1	Netherlands	2000	Developed	1.68
2	Finland	2000	Developed	1.66

3	Finland	2002	Developed	1.66
4	Liechtenstein	2009	Developed	1.65
5	Liechtenstein	2010	Developed	1.64

REG_QUALITY - LOWEST 5:

A tibble: 5 x 4

	country	year	dev_status	reg_quality
	<chr>	<dbl>	<chr>	<dbl>
1	Democratic Republic of the Congo	2000	<NA>	-2.58
2	Afghanistan	2000	Emerging	-2.45
3	North Korea	2012	<NA>	-2.43
4	North Korea	2013	<NA>	-2.43
5	Burma (Myanmar)	2004	<NA>	-2.4

REG_QUALITY - HIGHEST 5:

A tibble: 5 x 4

	country	year	dev_status	reg_quality
	<chr>	<dbl>	<chr>	<dbl>
1	Luxembourg	2002	Developed	2.26
2	Hong Kong	2016	<NA>	2.23
3	Singapore	2015	Developed	2.23
4	Hong Kong	2018	<NA>	2.21
5	Hong Kong	2006	<NA>	2.2

CORRUPTION - LOWEST 5:

A tibble: 5 x 4

	country	year	dev_status	corruption
	<chr>	<dbl>	<chr>	<dbl>
1	Somalia	2018	<NA>	-1.81
2	Yemen	2018	<NA>	-1.75
3	Somalia	2014	<NA>	-1.74
4	Somalia	2016	<NA>	-1.74
5	Yemen	2019	<NA>	-1.74

CORRUPTION - HIGHEST 5:

A tibble: 5 x 4

	country	year	dev_status	corruption
	<chr>	<dbl>	<chr>	<dbl>
1	Finland	2006	Developed	2.4
2	Sweden	2000	Developed	2.38
3	Finland	2002	Developed	2.36

4	Denmark	2006	Developed	2.35
5	Finland	2000	Developed	2.35

VOICE_ACC - LOWEST 5:

A tibble: 5 x 4

	country	year	dev_status	voice_acc
	<chr>	<dbl>	<chr>	<dbl>
1	Eritrea	2012	Emerging	-2.36
2	Eritrea	2013	Emerging	-2.34
3	Eritrea	2014	Emerging	-2.32
4	Eritrea	2015	Emerging	-2.31
5	Afghanistan	2000	Emerging	-2.3

VOICE_ACC - HIGHEST 5:

A tibble: 5 x 4

	country	year	dev_status	voice_acc
	<chr>	<dbl>	<chr>	<dbl>
1	Norway	2018	Developed	1.96
2	Norway	2019	Developed	1.95
3	Norway	2017	Developed	1.93
4	Norway	2015	Developed	1.91
5	Norway	2016	Developed	1.91

```
#
# INTERPRETATION OF OUTLIER ANALYSIS
#

# 1. WINSORIZE - MPS, INFLATION, GDP GROWTH
#   These three variables show severe outliers that would dominate OLS:

#   MPS: Turkey 2000 = 68.6, roughly 36× the emerging-market SD. Turkey
#   alone accounts for 4 of the 5 lowest and 2 of the 5 highest MPS values.
#   A single country should not drive the entire regression.

#   Inflation: Hyperinflation episodes - DRC at 514, Angola at 325.
#   Genuine but orders of magnitude above the median (~4.5).

#   GDP growth: Conflict-driven swings - Libya goes from -50.3 (2011) to
#   +86.8 (2012) in consecutive years. Extreme but mechanically influential.

#   → ACTION: Winsorize all three at 1st/99th percentiles in Phase 1.7.
```

```

# 2. DO NOT WINSORIZE - GDP per Capita
#   Variable is already logged, no need to winsorize it

# 3. DO NOT WINSORIZE - MIMIC
#   Range is 8.1 (USA) to 68.2 (Bolivia). These are NOT outliers - they are
#   the actual cross-country spread that identifies the MPS × MIMIC
#   interaction. Capping the extremes would compress exactly the variation
#   the paper needs.

# 4. DO NOT WINSORIZE - CREDIT/GDP
#   Iceland at 305 (pre-crisis banking boom) looks extreme but enters as
#   a lagged control, not a dependent variable. The value is economically
#   real and its influence on the regression is limited.

# 5. DO NOT WINSORIZE - GOVERNANCE VARIABLES
#   All bounded on a fixed [-2.5, 2.5] scale. Extremes (Somalia at -3.05
#   for pol_stability) are near-boundary values, not data errors.
#   Winsorizing bounded indices is not meaningful.

# 6. ROBUSTNESS FLAG - TURKEY
#   Turkey is extraordinarily influential in MPS (max = 68.6, min = -19.5).
#   Even after winsorizing, it will pull weight. Add a robustness check
#   excluding Turkey entirely to show results are not driven by one country.
#

# Remove temporary plot data frames and helper vectors from outlier visualization.
# df_world_2000_2019 carries forward for winsorization.
rm(
  vars_group1, vars_group2, vars_group3, vars_outlier,
  df_box1, df_box2, df_box3, df_box,
  gdp_growth_lead_name, gdp_lead_name, gdp_pc_lead_name, h, infl_lead_name,
  lgdp_pc_lead_name, v
)

#
# OUTLIER TREATMENT (WINSORIZATION)
#

# PURPOSE:
# Extreme values in MPS, inflation, and GDP growth can dominate OLS estimates.

```

```

# From the descriptive statistics:
#   - MPS in emerging markets: max = 68.65 (vs. SD   1.91)
#   - Inflation: max = 513.9 (vs. median   4.5)
#   - GDP growth: min = -50.34, max = 86.83

# These are genuine observations (hyperinflation episodes, extreme policy
# moves) but a few such values can disproportionately drive regression
# coefficients in a sample of thousands.

# WINSORIZATION replaces values beyond the 1st and 99th percentiles with
# the value at that percentile. This retains the observation (unlike
# trimming/dropping) but caps its influence.

# We create BOTH winsorized and original versions so results can be
# shown both ways (main specification uses winsorized; robustness uses raw).

# NOTE: Winsorization is applied to the RAW variables. The forward-looking
# dependent variables (d_infl_h*, d_gdp_growth_h*) were constructed from the raw
# values, so we rebuild them after winsorization. Alternatively, we could
# winsorize the dependent variables directly - we do both and keep the
# directly winsorized versions for simplicity.
#

# Helper function: winsorize a vector at given lower and upper quantiles
winsorize <- function(x, probs = c(0.01, 0.99)) {
  q <- quantile(x, probs = probs, na.rm = TRUE)
  x[x < q[1] & !is.na(x)] <- q[1]
  x[x > q[2] & !is.na(x)] <- q[2]
  return(x)
}

# --- Show pre-winsorization extremes ---
cat("

```

```

cat("  OUTLIER TREATMENT - Before vs. After Winsorization (1st/99th)\n")

```

```

OUTLIER TREATMENT - Before vs. After Winsorization (1st/99th)

```

```
cat("                                \n\n")
```

```
vars_to_winsorize <- c("mps", "infl", "gdp_growth")
```

```
cat("  BEFORE winsorization  \n")
```

BEFORE winsorization

```
for (v in vars_to_winsorize) {  
  q <- quantile(df_world_2000_2019[[v]], probs = c(0.01, 0.99), na.rm = TRUE)  
  cat(sprintf("   %s: min = %.2f, max = %.2f | p1 = %.2f, p99 = %.2f\n",  
              v,  
              min(df_world_2000_2019[[v]], na.rm = TRUE),  
              max(df_world_2000_2019[[v]], na.rm = TRUE),  
              q[1], q[2]))  
}
```

mps: min = -19.50, max = 68.65 | p1 = -3.24, p99 = 2.42
infl: min = -16.90, max = 513.90 | p1 = -2.20, p99 = 40.16
gdp_growth: min = -50.34, max = 86.83 | p1 = -9.08, p99 = 15.37

```
# --- Apply winsorization ---  
df_world_2000_2019 <- df_world_2000_2019 %>%  
  mutate(  
    mps_w      = winsorize(mps),  
    infl_w     = winsorize(infl),  
    gdp_growth_w = winsorize(gdp_growth)  
  )  
  
cat("\n  AFTER winsorization (new _w variables)  \n")
```

AFTER winsorization (new _w variables)

```

for (v in paste0(vars_to_winsorize, "_w")) {
  cat(sprintf("  %s: min = %.2f, max = %.2f\n",
             v,
             min(df_world_2000_2019[[v]], na.rm = TRUE),
             max(df_world_2000_2019[[v]], na.rm = TRUE)))
}

```

```

mps_w: min = -3.24, max = 2.42
infl_w: min = -2.20, max = 40.16
gdp_growth_w: min = -9.08, max = 15.37

```

```

# --- Rebuild winsorized dependent variables and interaction ---
# Winsorized lagged values
df_world_2000_2019 <- df_world_2000_2019 %>%
  arrange(country, year) %>%
  group_by(country) %>%
  mutate(
    infl_w_t_minus1      = dplyr::lag(infl_w, 1),
    gdp_growth_w_t_minus1 = dplyr::lag(gdp_growth_w, 1)
  ) %>%
  ungroup()

# Winsorized forward-looking dependent variables
for (h in 0:max_horizon) {
  infl_lead_name <- paste0("d_infl_w_h", h)
  gdp_growth_lead_name <- paste0("d_gdp_growth_w_h", h)

  df_world_2000_2019 <- df_world_2000_2019 %>%
    arrange(country, year) %>%
    group_by(country) %>%
    mutate(
      !!infl_lead_name := dplyr::lead(infl_w, h) - infl_w_t_minus1,
      !!gdp_growth_lead_name := dplyr::lead(gdp_growth_w, h) - gdp_growth_w_t_minus1
    ) %>%
    ungroup()
}

# Winsorized interaction
df_world_2000_2019 <- df_world_2000_2019 %>%

```

```
mutate(mps_w_x_mimic = mps_w * mimic_t_minus1)

cat("\n  Observations affected by winsorization  \n")
```

Observations affected by winsorization

```
cat(sprintf("  mps:          %d values capped\n",
            sum(df_world_2000_2019$mps != df_world_2000_2019$mps_w, na.rm = TRUE))))
```

mps: 58 values capped

```
cat(sprintf("  infl:         %d values capped\n",
            sum(df_world_2000_2019$infl != df_world_2000_2019$infl_w, na.rm = TRUE))))
```

infl: 71 values capped

```
cat(sprintf("  gdp_growth: %d values capped\n",
            sum(df_world_2000_2019$gdp_growth != df_world_2000_2019$gdp_growth_w, na.rm = TRUE))))
```

gdp_growth: 78 values capped

```
#
# DOCUMENT THE EFFECTIVE ESTIMATION SAMPLE
#
# PURPOSE:
# The effective estimation sample is the subset of country-year observations
# where ALL variables needed for the baseline LP regression are non-missing
# simultaneously. This is typically much smaller than the raw panel because
# missingness in different variables drops different observations.
#
# The baseline LP regression at horizon h requires:
#   Dependent:    d_infl_w_h{h}, d_lgdp_pc_h{h}, or d_gdp_growth_w_h{h}
#   Regressors:   mps_w, mimic_t_minus1, mps_w_x_mimic,
#                 gdp_growth_w_t_minus1, infl_w_t_minus1, credit_gdp_t_minus1,
```

```

#           pol_stability_t_minus1
#   Fixed effects: country, year

# We report the sample at h = 0 (largest possible) and show how it shrinks
# with each horizon increase. This table goes in the paper's data section.
#

# Define baseline regressors (same for all horizons)
baseline_regressors <- c("mps_w", "mimic_t_minus1", "mps_w_x_mimic",
                        "gdp_growth_w_t_minus1", "infl_w_t_minus1",
                        "credit_gdp_t_minus1", "pol_stability_t_minus1",
                        "country", "year")

# For each horizon, count complete cases
sample_summary <- data.frame(
  horizon      = 0:max_horizon,
  n_obs_infl   = NA_integer_,
  n_obs_lgdp   = NA_integer_,
  n_countries_infl = NA_integer_,
  n_countries_lgdp = NA_integer_
)

for (h in 0:max_horizon) {
  # Inflation sample
  infl_dep <- paste0("d_infl_w_h", h)
  infl_vars <- c(infl_dep, baseline_regressors)
  infl_complete <- df_world_2000_2019 %>%
    select(all_of(infl_vars)) %>%
    na.omit()
  sample_summary$n_obs_infl[h + 1] <- nrow(infl_complete)
  sample_summary$n_countries_infl[h + 1] <- n_distinct(infl_complete$country)

  # Log GDP per capita sample
  lgdp_dep <- paste0("d_lgdp_pc_h", h)
  lgdp_vars <- c(lgdp_dep, baseline_regressors)
  lgdp_complete <- df_world_2000_2019 %>%
    select(all_of(lgdp_vars)) %>%
    na.omit()
  sample_summary$n_obs_lgdp[h + 1] <- nrow(lgdp_complete)
  sample_summary$n_countries_lgdp[h + 1] <- n_distinct(lgdp_complete$country)
}

```

```
}
cat("                                \n")
```

```
cat("  EFFECTIVE ESTIMATION SAMPLE BY HORIZON\n")
```

```
  EFFECTIVE ESTIMATION SAMPLE BY HORIZON
```

```
cat("                                \n")
```

```
cat("  All baseline regressors + dependent variable non-missing\n")
```

```
  All baseline regressors + dependent variable non-missing
```

```
cat("                                \n\n")
```

```
print(sample_summary, row.names = FALSE)
```

```
horizon n_obs_infl n_obs_lgdp n_countries_infl n_countries_lgdp
      0      1950      1924           129           127
      1      1831      1810           129           127
      2      1709      1692           129           127
      3      1585      1571           127           125
      4      1461      1450           125           123
      5      1339      1331           124           122
```

```
# Detailed breakdown at h = 0: average time-series length per country
h0_infl_vars <- c("d_infl_w_h0", baseline_regressors)
h0_sample <- df_world_2000_2019 %>%
  select(all_of(h0_infl_vars)) %>%
  na.omit()

ts_length <- h0_sample %>%
  group_by(country) %>%
  summarise(T_i = n(), .groups = "drop")

cat("\n Baseline sample details (h = 0, inflation) \n")
```

Baseline sample details (h = 0, inflation)

```
cat(" Total observations:", nrow(h0_sample), "\n")
```

Total observations: 1950

```
cat(" Countries:          ", n_distinct(h0_sample$country), "\n")
```

Countries: 129

```
cat(" Avg years/country: ", round(mean(ts_length$T_i), 1), "\n")
```

Avg years/country: 15.1

```
cat(" Min years/country: ", min(ts_length$T_i), "\n")
```

Min years/country: 1

```
cat(" Max years/country: ", max(ts_length$T_i), "\n")
```

Max years/country: 18

```

#
# VERIFY PANEL STRUCTURE & APPLY MINIMUM THRESHOLD
#

# PURPOSE:
# Countries with very few usable observations (e.g., 1-3 years in the
# estimation sample) contribute almost no identifying variation and can
# introduce noise. LP estimation needs enough time-series depth per country
# to capture dynamics - a minimum of 5 years is a common threshold.
#
# We check the distribution of per-country observation counts in the
# effective sample and flag countries below the threshold for exclusion.
# This is done on the h = 0 sample (most complete horizon).

# WHAT THIS CHUNK DOES:
# 1. Show distribution of per-country time-series lengths
# 2. List countries below the threshold
# 3. Create a filtered "estimation ready" dataset excluding thin countries
# 4. Report the final sample size
#

min_T <- 5 # Minimum usable years per country

# Step 1: Per-country observation count in the effective sample (h = 0)
h0_vars <- c("d_infl_w_h0", baseline_regressors)

country_T <- df_world_2000_2019 %>%
  select(all_of(h0_vars)) %>%
  na.omit() %>%
  group_by(country) %>%
  summarise(T_i = n(), .groups = "drop") %>%
  arrange(T_i)

cat("

```

```

cat(" PER-COUNTRY TIME-SERIES LENGTH IN ESTIMATION SAMPLE (h = 0)\n")

```

```

PER-COUNTRY TIME-SERIES LENGTH IN ESTIMATION SAMPLE (h = 0)

```

```
cat("                                \n\n")
```

```
print(summary(country_T$T_i))
```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1.00	14.00	17.00	15.12	18.00	18.00

```
# Step 2: Flag countries below threshold
thin_countries <- country_T %>% filter(T_i < min_T)

cat(sprintf("\n Countries with fewer than %d usable years: %d \n",
            min_T, nrow(thin_countries)))
```

Countries with fewer than 5 usable years: 5

```
if (nrow(thin_countries) > 0) {
  print(thin_countries, n = nrow(thin_countries))
} else {
  cat(" None - all countries meet the minimum threshold\n")
}
```

```
# A tibble: 5 x 2
```

	country	T_i
	<chr>	<int>
1	Angola	1
2	Azerbaijan	3
3	United Arab Emirates	3
4	Kazakhstan	4
5	Mauritius	4

```
# Step 3: Identify countries that pass the threshold
keep_countries <- country_T %>% filter(T_i >= min_T) %>% pull(country)

# Create the estimation-ready dataset
# This is the dataset that will be used for ALL LP regressions going forward
df_est <- df_world_2000_2019 %>%
  filter(country %in% keep_countries)

# Step 4: Report final sample
cat(sprintf("\n
```

```
cat(sprintf("  FINAL ESTIMATION-READY DATASET: df_est\n"))
```

```
FINAL ESTIMATION-READY DATASET: df_est
```

```
cat(sprintf("  \n"))
```

```
cat(sprintf("  Countries retained:   %d (of %d in full panel)\n",
  n_distinct(df_est$country), n_distinct(df_world_2000_2019$country)))
```

```
Countries retained:   124 (of 197 in full panel)
```

```
cat(sprintf("  Countries dropped:     %d (fewer than %d usable years)\n",
  nrow(thin_countries), min_T))
```

```
Countries dropped:     5 (fewer than 5 usable years)
```

```
cat(sprintf("  Total rows in df_est: %d\n", nrow(df_est)))
```

```
Total rows in df_est: 2480
```

```
cat(sprintf("                                \n"))
```

```
#
# PHASE 1 COMPLETE
#
# The dataset df_est is now estimation-ready for Phase 2 (descriptive
# visualizations) and Phase 3 (LP regressions).

# Contents of df_est:
#   Identifiers:  country, year, code, continent_code, dev_status, dev_binary
#   Shock:       mps, mps_w (winsorized)
#   Informality: mimic, mimic_t_minus1
#   Interaction: mps_x_mimic, mps_w_x_mimic (winsorized)
#   Dep vars:    d_infl_h0-h5, d_gdp_growth_h0-h5 (raw)
#               d_infl_w_h0-h5 (winsorized), d_lgdp_pc_h0-h5 (raw log)
#               d_gdp_growth_w_h0-h5 (winsorized, sideline)
#   Controls:    gdp_growth_t_minus1, infl_t_minus1, credit_gdp_t_minus1,
#               pol_stability_t_minus1 (baseline)
#               reg_quality_t_minus1, corruption_t_minus1, voice_acc_t_minus1 (robust)
#   Winsorized:  gdp_growth_w_t_minus1, infl_w_t_minus1
#
```

```
# Remove temporary objects from sample documentation and panel verification.
# Keep: df_world_2000_2019, df_est, keep_countries, baseline_regressors
rm(
  sample_summary, infl_complete, gdp_complete,
  h0_infl_vars, h0_sample, ts_length,
  h0_vars, country_T, thin_countries, min_T
)
```

```
#
# GEOGRAPHICAL COVERAGE MAPS
#
# Two maps side by side:
#   Map 1: All countries in the full panel (df_world_2000_2019)
#   Map 2: Countries retained in the estimation-ready sample (df_est)
# This visualizes which countries were lost during Phase 1 filtering
```

```

# (missingness, minimum observation threshold, etc.)
#

# --- Load world polygons ---
world <- rnatualearth::ne_countries(scale = "medium", returnclass = "sf") %>%
  mutate(iso3c = iso_a3)

# --- Map 1: Full panel ---
full_iso3 <- df_world_2000_2019 %>%
  distinct(country) %>%
  mutate(iso3c = countrycode(country, "country.name", "iso3c")) %>%
  filter(!is.na(iso3c)) %>%
  pull(iso3c)

world_full <- world %>%
  mutate(in_sample = if_else(iso3c %in% full_iso3, "In sample", "Not in sample"))

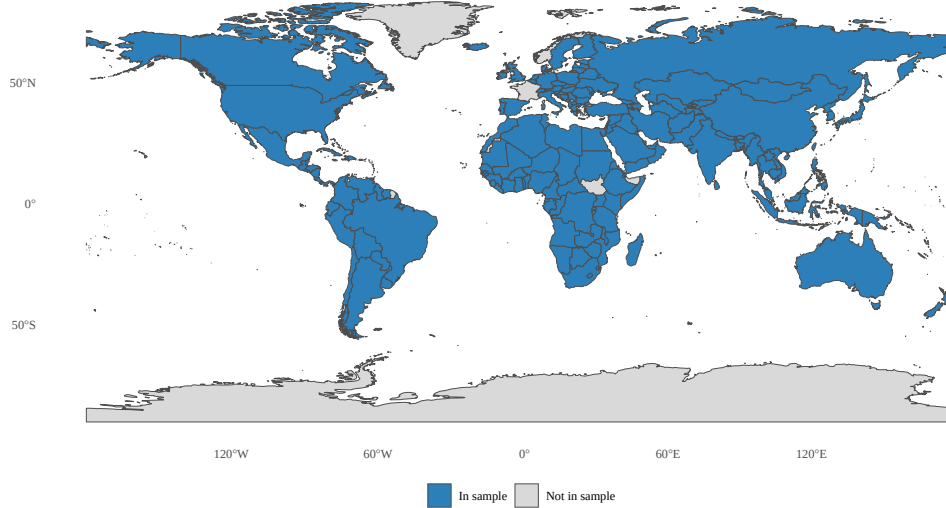
n_full <- sum(world_full$in_sample == "In sample", na.rm = TRUE)

p1 <- ggplot(world_full) +
  geom_sf(aes(fill = in_sample), linewidth = 0.1, color = "grey30") +
  scale_fill_manual(values = c("In sample" = "#2C7FB8", "Not in sample" = "grey85")) +
  labs(title = paste0("Full Panel (", n_full, " Countries)",
    subtitle = "All countries in df_world_2000_2019 before Phase 1 filtering",
    fill = NULL) +
  theme_minimal(base_size = 11, base_family = "serif") +
  theme(legend.position = "bottom",
    panel.grid.major = element_blank(),
    panel.grid.minor = element_blank())

print(p1)

```

Full Panel (194 Countries)
All countries in df_world_2000_2019 before Phase 1 filtering



```
# --- Map 2: Estimation sample, colored by development status ---
est_countries <- df_est %>%
  distinct(country, dev_status) %>%
  mutate(iso3c = countrycode(country, "country.name", "iso3c")) %>%
  filter(!is.na(iso3c))

world_est <- world %>%
  left_join(est_countries %>% select(iso3c, dev_status), by = "iso3c") %>%
  mutate(
    map_status = case_when(
      dev_status == "Developed" ~ "Developed",
      dev_status == "Emerging" ~ "Emerging",
      TRUE ~ "Not in sample"
    )
  )

# Ensure draw order: Not in sample behind, Emerging and Developed on top
world_est$map_status <- factor(world_est$map_status,
                              levels = c("Not in sample", "Developed", "Emerging"))

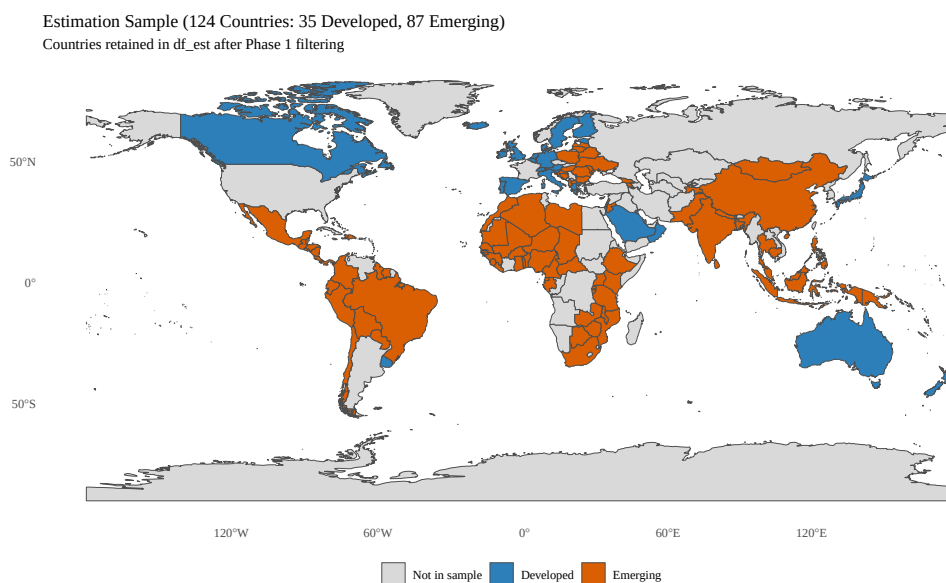
n_est <- sum(!is.na(est_countries$iso3c))
n_developed <- sum(est_countries$dev_status == "Developed", na.rm = TRUE)
n_emerging <- sum(est_countries$dev_status == "Emerging", na.rm = TRUE)
```

```

p2 <- ggplot(world_est) +
  geom_sf(aes(fill = map_status), linewidth = 0.1, color = "grey30") +
  scale_fill_manual(values = c("Developed"      = "#2C7FB8",
                                "Emerging"      = "#D95F02",
                                "Not in sample"  = "grey85")) +
  labs(title = paste0("Estimation Sample (", n_est, " Countries: ",
                      n_developed, " Developed, ", n_emerging, " Emerging)"),
        subtitle = "Countries retained in df_est after Phase 1 filtering",
        fill = NULL) +
  theme_minimal(base_size = 11, base_family = "serif") +
  theme(legend.position = "bottom",
        panel.grid.major = element_blank(),
        panel.grid.minor = element_blank())

print(p2)

```



```

# --- Summary of countries lost ---
lost_countries <- df_world_2000_2019 %>%
  distinct(country) %>%
  filter(!country %in% keep_countries) %>%
  mutate(iso3c = countrycode(country, "country.name", "iso3c"))

cat("

```

```
cat("  GEOGRAPHICAL COVERAGE SUMMARY\n")
```

GEOGRAPHICAL COVERAGE SUMMARY

```
cat("                                \n")
```

```
cat(sprintf("  Full panel:           %d countries\n", n_distinct(df_world_2000_2019$count
```

Full panel: 197 countries

```
cat(sprintf("  Estimation sample: %d countries (%d Developed, %d Emerging)\n",  
            n_est, n_developed, n_emerging))
```

Estimation sample: 124 countries (35 Developed, 87 Emerging)

```
cat(sprintf("  Countries lost:       %d\n", nrow(lost_countries)))
```

Countries lost: 73

```
cat("                                \n")
```

```
if (nrow(lost_countries) > 0) {  
  cat("\n  Countries dropped during Phase 1  \n")  
  print(lost_countries %>% arrange(country), n = nrow(lost_countries))  
}
```

Countries dropped during Phase 1

A tibble: 73 x 2

country	iso3c
<chr>	<chr>
1 Afghanistan	AFG
2 Andorra	AND
3 Angola	AGO
4 Antigua and Barbuda	ATG
5 Argentina	ARG
6 Aruba	ABW
7 Azerbaijan	AZE
8 Bahamas	BHS
9 Barbados	BRB
10 Bermuda	BMU
11 Bhutan	BTN
12 Brunei	BRN
13 Burma (Myanmar)	MMR
14 Cape Verde	CPV
15 Cuba	CUB
16 Czechia	CZE
17 Democratic Republic of the Congo	COD
18 Djibouti	DJI
19 Dominica	DMA
20 Egypt	EGY
21 Eritrea	ERI
22 Faroe Islands	FRO
23 Fiji	FJI
24 Gambia	GMB
25 Grenada	GRD
26 Hong Kong	HKG
27 Iran	IRN
28 Iraq	IRQ
29 Ivory Coast	CIV
30 Kazakhstan	KAZ
31 Kiribati	KIR
32 Kyrgyzstan	KGZ
33 Laos	LAO
34 Lesotho	LSO
35 Liechtenstein	LIE
36 Macao	MAC

37 Madagascar	MDG
38 Mauritius	MUS
39 Micronesia	<NA>
40 Monaco	MCO
41 Montenegro	MNE
42 Namibia	NAM
43 New Caledonia	NCL
44 North Korea	PRK
45 Palau	PLW
46 Palestine	PSE
47 Puerto Rico	PRI
48 Republic of the Congo	COG
49 Russia	RUS
50 Saint Lucia	LCA
51 Saint Vincent and the Grenadines	VCT
52 Samoa	WSM
53 San Marino	SMR
54 Sao Tome and Principe	STP
55 Serbia	SRB
56 Seychelles	SYC
57 Slovakia	SVK
58 Somalia	SOM
59 South Korea	KOR
60 Sudan	SDN
61 Swaziland	SWZ
62 Syria	SYR
63 Taiwan	TWN
64 Tonga	TON
65 Tunisia	TUN
66 Turkmenistan	TKM
67 Tuvalu	TUV
68 USA	USA
69 United Arab Emirates	ARE
70 Uzbekistan	UZB
71 Vanuatu	VUT
72 Venezuela	VEN
73 Yemen	YEM

```
# Remove all map-related temp objects.
# Going into Phase 2, the environment should contain only:
#   df_world_2000_2019 (full working panel with all variables)
```

```
# df_est          (estimation-ready subset)
# keep_countries  (vector of retained country names)
# baseline_regressors (vector of regressor names for LP)
# max_horizon     (integer, = 5)
rm(
  world, full_iso3, world_full, n_full, p1,
  est_countries, world_est, n_est, n_developed, n_emerging, p2,
  lost_countries
)
```

```
# From easiest to hardest
# 1. PCA as main spec, and other could be used a robustness check
# 2. Causal connection (check email)
# 3. Connecting clusters of countries that are similar structure but differ on informa
# 4. IV - contact "ROUTLEDGE HANDBOOK OF THE INFORMAL ECONOMY"
```

```
#
# GOVERNANCE COMPOSITE - PRINCIPAL COMPONENT ANALYSIS (PCA)
#
# PURPOSE:
# Diagnostic 4 showed that the four governance variables are highly inter-
# correlated (r = 0.71-0.90). Including any single one in the baseline is
# a defensible but somewhat arbitrary choice. PCA extracts a single composite
# score that captures the common variation across all four, sidestepping the
# "which one?" question.
#
# WHAT WE DO:
# 1. Run PCA on the four LAGGED governance variables (t-1) - for use in
#    the main LP specification where all controls are pre-determined.
# 2. Run PCA on the four CONTEMPORANEOUS governance variables - for use
#    in robustness checks or specifications without the pre-determination
#    requirement.
# 3. Report variance explained and loadings for both.
# 4. Attach PC1 scores as gov_pca_t_minus1 and gov_pca to the dataframe.
#
# HOW TO INTERPRET:
# High gov_pca → strong institutions (high reg_quality,
# low corruption, stable politics, strong voice/accountability)
# Low gov_pca → weak institutions
```

```
# NOTE: PCA scores are standardized (mean = 0, sd = 1) by construction.
#
# Part A: LAGGED governance PCA (t-1) - for main LP specification
gov_vars_lag <- c("reg_quality_t_minus1", "corruption_t_minus1",
                 "pol_stability_t_minus1", "voice_acc_t_minus1")
gov_lag_idx <- complete.cases(df_world_2000_2019[, gov_vars_lag])
pca_lag <- prcomp(df_world_2000_2019[gov_lag_idx, gov_vars_lag], scale. = TRUE)
cat("                                \n")
```

```
cat("  GOVERNANCE PCA - LAGGED (t-1)\n")
```

```
GOVERNANCE PCA - LAGGED (t-1)
```

```
cat("                                \n")
```

```
cat("  Variables:", paste(gov_vars_lag, collapse = ", "), "\n")
```

```
Variables: reg_quality_t_minus1, corruption_t_minus1, pol_stability_t_minus1, voice_acc_t_minus1
```

```
cat("  Complete cases:", sum(gov_lag_idx), "of", nrow(df_world_2000_2019), "\n")
```

```
Complete cases: 3443 of 3940
```

```
cat("                                \n\n")
```

```
cat("  Variance explained  \n")
```

Variance explained

```
print(round(summary(pca_lag)$importance, 3))
```

	PC1	PC2	PC3	PC4
Standard deviation	1.836	0.576	0.442	0.320
Proportion of Variance	0.843	0.083	0.049	0.026
Cumulative Proportion	0.843	0.925	0.974	1.000

```
cat("\n  PC1 loadings  \n")
```

PC1 loadings

```
print(round(pca_lag$rotation[, 1], 3))
```

reg_quality_t_minus1	corruption_t_minus1	pol_stability_t_minus1
0.508	0.518	0.472
voice_acc_t_minus1		
0.500		

```
# Attach lagged PC1 scores
df_world_2000_2019$gov_pca_t_minus1 <- NA_real_
df_world_2000_2019$gov_pca_t_minus1[gov_lag_idx] <- pca_lag$x[, 1]

cat(sprintf("\n  gov_pca_t_minus1 created: %d non-missing values  \n",
            sum(!is.na(df_world_2000_2019$gov_pca_t_minus1))))
```

gov_pca_t_minus1 created: 3443 non-missing values

```
print(summary(df_world_2000_2019$gov_pca_t_minus1))
```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.	NA's
-4.5201	-1.3451	-0.2815	0.0000	1.4662	4.0274	497


```
cat("  Variance explained  \n")
```

Variance explained

```
print(round(summary(pca_contemp)$importance, 3))
```

	PC1	PC2	PC3	PC4
Standard deviation	1.836	0.575	0.443	0.319
Proportion of Variance	0.843	0.083	0.049	0.026
Cumulative Proportion	0.843	0.925	0.974	1.000

```
cat("\n  PC1 loadings  \n")
```

PC1 loadings

```
print(round(pca_contemp$rotation[, 1], 3))
```

reg_quality	corruption	pol_stability	voice_acc
0.508	0.518	0.472	0.500

```
# Attach contemporaneous PC1 scores
df_world_2000_2019$gov_pca <- NA_real_
df_world_2000_2019$gov_pca[gov_contemp_idx] <- pca_contemp$x[, 1]

cat(sprintf("\n  gov_pca created: %d non-missing values  \n",
            sum(!is.na(df_world_2000_2019$gov_pca))))
```

gov_pca created: 3637 non-missing values

```
print(summary(df_world_2000_2019$gov_pca))
```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.	NA's
-4.5275	-1.3481	-0.2793	0.0000	1.4459	4.0293	303

```

# Update df_est with both PCA variables
df_est <- df_world_2000_2019 %>%
  filter(country %in% keep_countries)

# Clean up
rm(gov_vars_lag, gov_vars_contemp, gov_lag_idx, gov_contemp_idx,
    pca_lag, pca_contemp)

#
# INTERPRETATION OF RESULTS
#

# Both versions produce virtually identical results - expected since
# governance variables are 91-97% between-country (Diagnostic 3) and
# lagging by one year barely changes them.
#
# LAGGED (t-1):
#   PC1 explains 84.3% of variance. Loadings: reg_quality (0.508),
#   corruption (0.518), pol_stability (0.472), voice_acc (0.500).
#   3,443 non-missing values. Range: -4.52 to +4.03.

# CONTEMPORANEOUS:
#   PC1 explains 84.3% of variance. Loadings: identical to lagged version.
#   3,637 non-missing values (194 more - gains one year per country since
#   no lag is consumed). Range: -4.53 to +4.03.

# Key observations:
#   - Loadings are nearly equal across all four variables. No single
#     indicator dominates - PC1 is a balanced "institutional quality" index.
#   - pol_stability has the lowest loading (0.472), consistent with it
#     capturing something modestly distinct (security/conflict risk vs.
#     bureaucratic/regulatory quality). This supports using it as the
#     standalone baseline control - it's the least redundant of the four.
#   - 84.3% variance explained is very strong. The four governance
#     indicators are effectively measuring one latent construct.

# Usage in the LP specification:
#   - Baseline: pol_stability_t_minus1 (single governance control)
#   - Robustness A: swap for reg_quality_t_minus1, corruption_t_minus1, or
#     voice_acc_t_minus1 (one at a time)

```

```
# - Robustness B: swap for gov_pca_t_minus1 (lagged composite)
# - Robustness C: swap for gov_pca (contemporaneous composite, tests
# whether the pre-determination lag matters for governance controls)
```

```
# The near-identical results between lagged and contemporaneous PCA confirm
# that governance is overwhelmingly a cross-country characteristic - the
# lag is a formality for econometric cleanliness, not a substantive choice.
#
```

```
# Scree plot showing variance explained by each principal component.
# The sharp drop after PC1 (84.3%) confirms a single component is sufficient.
```

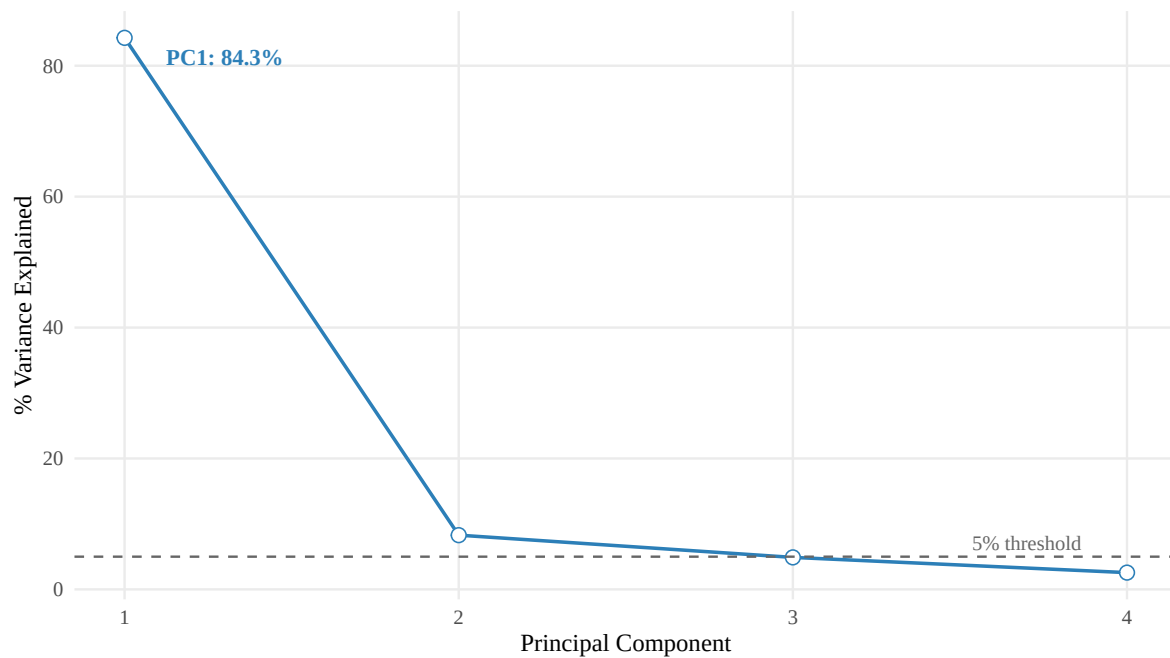
```
# Re-run PCA briefly to extract variance shares for plotting
gov_vars_pca <- c("reg_quality_t_minus1", "corruption_t_minus1",
                 "pol_stability_t_minus1", "voice_acc_t_minus1")
pca_fit <- prcomp(df_world_2000_2019[complete.cases(df_world_2000_2019[, gov_vars_pca]),
                 scale. = TRUE)
```

```
# Build plotting data
scree_df <- data.frame(
  PC = 1:length(pca_fit$sdev),
  Variance_Pct = (pca_fit$sdev^2 / sum(pca_fit$sdev^2)) * 100
)
```

```
ggplot(scree_df, aes(x = PC, y = Variance_Pct)) +
  geom_line(color = "#2C7FB8", linewidth = 0.8) +
  geom_point(shape = 21, color = "#2C7FB8", fill = "white", size = 3) +
  geom_hline(yintercept = 5, linetype = "dashed", color = "grey40") +
  annotate("text", x = 3.7, y = 7,
          label = "5% threshold", color = "grey40", size = 3.5, family = "serif") +
  annotate("text", x = 1.3, y = scree_df$Variance_Pct[1] - 3,
          label = paste0("PC1: ", round(scree_df$Variance_Pct[1], 1), "%"),
          color = "#2C7FB8", fontface = "bold", size = 4, family = "serif") +
  scale_x_continuous(breaks = 1:nrow(scree_df)) +
  labs(title = "Governance PCA - Scree Plot",
       subtitle = "PC1 alone explains 84.3% of variance across four governance indicators",
       x = "Principal Component",
       y = "% Variance Explained") +
  theme_minimal(base_size = 12, base_family = "serif") +
  theme(panel.grid.minor = element_blank())
```

Governance PCA – Scree Plot

PC1 alone explains 84.3% of variance across four governance indicators



```
# Clean up
rm(gov_vars_pca, pca_fit, scree_df)
```

```
#
# FINAL ENVIRONMENT SETUP - PREPARE FOR LP ESTIMATION
#
#
# PURPOSE:
# Ensure df_est is the single canonical dataset going forward, define all
# regressor and specification vectors for the LP models, and clean the
# environment. After this chunk, only estimation-relevant objects remain.
#
# LP ESTIMATION PLAN:
# We estimate local projections for 2 main + 1 sideline dependent variables:
#
#   Part A - Inflation response (MAIN):
#     A1. Full sample, A2. Developed only, A3. Emerging only
#
#   Part B - Log GDP per capita response (MAIN):
#     B1. Full sample, B2. Developed only, B3. Emerging only
```

```

#
#   Part C - GDP growth response (SIDELINE, supplementary):
#       C1. Full sample, C2. Developed only, C3. Emerging only
#
# Each set is estimated for horizons h = 0, 1, ..., 5 using the baseline
# specification + alternative governance robustness checks.
#
# GOVERNANCE CONTROLS:
# Governance variables enter contemporaneously (not lagged). Justification:
#   - Bounded indices [-2.5, 2.5] - unit root logically impossible
#   - 91-97% between-country variation (Diagnostic 3) - contemporaneous
#     and lagged values are virtually identical
#   - Gains ~194 extra observations vs. lagged version
#   - PCA confirmed identical results for lagged vs. contemporaneous
#
#   1. Rebuild df_est one final time to capture all columns
df_est <- df_world_2000_2019 %>%
  filter(country %in% keep_countries)

#   2. Define regressor vectors

# Baseline specification (contemporaneous governance: pol_stability)
baseline_regressors <- c("mps_w", "mimic_t_minus1", "mps_w_x_mimic",
                        "gdp_growth_w_t_minus1", "infl_w_t_minus1",
                        "credit_gdp_t_minus1", "gov_pca")

# Alternative governance specifications (swap pol_stability for one of these)
alt_gov_regressors <- list(
  reg_quality = c("mps_w", "mimic_t_minus1", "mps_w_x_mimic",
                  "gdp_growth_w_t_minus1", "infl_w_t_minus1",
                  "credit_gdp_t_minus1", "reg_quality"),
  corruption  = c("mps_w", "mimic_t_minus1", "mps_w_x_mimic",
                  "gdp_growth_w_t_minus1", "infl_w_t_minus1",
                  "credit_gdp_t_minus1", "corruption"),
  voice_acc   = c("mps_w", "mimic_t_minus1", "mps_w_x_mimic",
                  "gdp_growth_w_t_minus1", "infl_w_t_minus1",
                  "credit_gdp_t_minus1", "voice_acc"),
  pol_stab    = c("mps_w", "mimic_t_minus1", "mps_w_x_mimic",
                  "gdp_growth_w_t_minus1", "infl_w_t_minus1",

```

```

        "credit_gdp_t_minus1", "pol_stability")
)

# 3. Define dependent variable and sample configurations

# Dependent variables by horizon
dep_vars <- list(
  inflation = paste0("d_infl_w_h", 0:max_horizon),      # Main dep 1 (winsorized)
  lgdp_pc   = paste0("d_lgdp_pc_h", 0:max_horizon),      # Main dep 2 (raw, log-transfo
  gdp_growth = paste0("d_gdp_growth_w_h", 0:max_horizon) # Sideline (winsorized)
)

# Sample splits
samples <- list(
  full      = NULL,                                     # No filter - all countries
  developed = "Developed",                             # dev_status == "Developed"
  emerging  = "Emerging"                               # dev_status == "Emerging"
)

# 4. Remove everything except what LP estimation needs
rm(keep_countries)

remaining <- setdiff(ls(), c("df_est", "baseline_regressors",
                             "alt_gov_regressors", "dep_vars",
                             "samples", "max_horizon"))
if (length(remaining) > 0) rm(list = remaining)
rm(remaining)

# 5. Report final state
cat("

```

```

cat(" ENVIRONMENT READY FOR LP ESTIMATION\n")

```

ENVIRONMENT READY FOR LP ESTIMATION

```
cat("                                \n\n")
```

```
cat("  Objects in environment  \n")
```

Objects in environment

```
for (obj in ls()) {  
  cat(sprintf("  %-25s %s\n", obj, class(get(obj))[1]))  
}
```

alt_gov_regressors	list
baseline_regressors	character
dep_vars	list
df_est	tbl_df
max_horizon	numeric
samples	list

```
cat("\n  Dataset  \n")
```

Dataset

```
cat(sprintf("  df_est: %d rows × %d cols | %d countries | %d years\n",  
            nrow(df_est), ncol(df_est),  
            dplyr::n_distinct(df_est$country),  
            dplyr::n_distinct(df_est$year)))
```

df_est: 2480 rows × 71 cols | 124 countries | 20 years

```
cat(sprintf("\n  Horizons: 0 to %d  \n", max_horizon))
```

Horizons: 0 to 5

```
cat("\n  Baseline regressors  \n")
```

Baseline regressors

```
cat(sprintf("  %s\n", paste(baseline_regressors, collapse = ", ")))
```

mps_w, mimic_t_minus1, mps_w_x_mimic, gdp_growth_w_t_minus1, infl_w_t_minus1, credit

```
cat("\n  Alternative governance specs  \n")
```

Alternative governance specs

```
for (name in names(alt_gov_regressors)) {  
  gov_var <- tail(alt_gov_regressors[[name]], 1)  
  cat(sprintf("  %-15s → swap governance PCA for %s\n", name, gov_var))  
}
```

reg_quality	→ swap governance PCA for reg_quality
corruption	→ swap governance PCA for corruption
voice_acc	→ swap governance PCA for voice_acc
pol_stab	→ swap governance PCA for pol_stability

```
cat("\n  Dependent variables  \n")
```

Dependent variables

```
for (name in names(dep_vars)) {  
  cat(sprintf("  %-15s → %s ... %s\n", name,  
              dep_vars[[name]][1], tail(dep_vars[[name]], 1)))  
}
```

inflation	→ d_infl_w_h0 ... d_infl_w_h5
lgdp_pc	→ d_lgdp_pc_h0 ... d_lgdp_pc_h5
gdp_growth	→ d_gdp_growth_w_h0 ... d_gdp_growth_w_h5

```
cat("\n Sample splits \n")
```

Sample splits

```
for (name in names(samples)) {  
  if (is.null(samples[[name]])) {  
    n <- dplyr::n_distinct(df_est$country)  
    cat(sprintf(" %-15s → all countries (%d)\n", name, n))  
  } else {  
    n <- dplyr::n_distinct(df_est$country[df_est$dev_status == samples[[name]])  
    cat(sprintf(" %-15s → %s (%d countries)\n", name, samples[[name]], n))  
  }  
}
```

full	→ all countries (124)
developed	→ Developed (36 countries)
emerging	→ Emerging (88 countries)

```
cat("\n LP estimation plan: 9 core sets (6 main + 3 sideline) \n")
```

LP estimation plan: 9 core sets (6 main + 3 sideline)

```
cat(" A1. Inflation      × Full sample\n")
```

A1. Inflation × Full sample

```
cat(" A2. Inflation      × Developed only\n")
```

A2. Inflation × Developed only

```
cat(" A3. Inflation      × Emerging only\n")
```

A3. Inflation × Emerging only

```
cat(" B1. Log GDP p.c. × Full sample\n")
```

B1. Log GDP p.c. × Full sample

```
cat(" B2. Log GDP p.c. × Developed only\n")
```

B2. Log GDP p.c. × Developed only

```
cat(" B3. Log GDP p.c. × Emerging only\n")
```

B3. Log GDP p.c. × Emerging only

```
cat(" C1. GDP growth × Full sample (sideline)\n")
```

C1. GDP growth × Full sample (sideline)

```
cat(" C2. GDP growth × Developed only (sideline)\n")
```

C2. GDP growth × Developed only (sideline)

```
cat(" C3. GDP growth × Emerging only (sideline)\n")
```

C3. GDP growth × Emerging only (sideline)

```
cat(" + 4 alternative governance specs for main deps (robustness)\n")
```

+ 4 alternative governance specs for main deps (robustness)

```
cat(" \n")
```

```

#
# LOCAL PROJECTIONS - HELPER FUNCTION
#
#
# PURPOSE:
# This function estimates the panel local projections model (Jordà 2005)
# for a given dependent variable prefix, sample, and set of regressors
# across all horizons h = 0, ..., max_horizon.
#
# MODEL (for each horizon h):
#   d_Y_h = _i + _t + ·MPS + ·MIMIC + ·(MPS×MIMIC) + 'X +
#
# where:
#   _i = country fixed effects (absorb time-invariant country characteristics)
#   _t = year fixed effects (absorb global shocks common to all countries)
#       = coefficient of interest - how informality moderates monetary transmission
#   SEs clustered at the country level (accounts for serial correlation within countries)
#
# RETURNS:
# A tidy dataframe with one row per horizon, containing the coefficient
# estimates, standard errors, confidence intervals, and sample sizes for
# all regressors - designed for easy IRF plotting.
#

run_lp <- function(data, dep_prefix, regressors, horizons = 0:max_horizon,
                    spec_label = "baseline", sample_label = "full") {

  results <- list()

  for (h in horizons) {
    dep_var <- paste0(dep_prefix, h)

    # Skip if dependent variable doesn't exist
    if (!dep_var %in% names(data)) {
      warning(paste("Variable", dep_var, "not found - skipping horizon", h))
      next
    }

    # Build formula: dep_var ~ regressors | country + year
    fml <- as.formula(paste(dep_var, "~",

```

```

        paste(regressors, collapse = " + "),
        "| country + year"))

# Estimate with country-clustered SEs
fit <- tryCatch({
  feols(fml, data = data, cluster = "country")
}, error = function(e) {
  warning(paste("Horizon", h, "failed:", conditionMessage(e)))
  return(NULL)
})

if (is.null(fit)) next

# Extract coefficients into tidy format
coef_table <- as.data.frame(coeftable(fit))
coef_table$variable <- rownames(coef_table)
rownames(coef_table) <- NULL

coef_table <- coef_table %>%
  rename(
    estimate = Estimate,
    std_error = `Std. Error`,
    t_value = `t value`,
    p_value = `Pr(>|t|)`
  ) %>%
  mutate(
    horizon      = h,
    ci_lower_90  = estimate - 1.645 * std_error,
    ci_upper_90  = estimate + 1.645 * std_error,
    ci_lower_95  = estimate - 1.96 * std_error,
    ci_upper_95  = estimate + 1.96 * std_error,
    n_obs        = fit$nobs,
    n_countries  = length(unique(fit$fixef_sizes)),
    spec         = spec_label,
    sample       = sample_label,
    dep_var_type = dep_prefix
  )

results[[as.character(h)]] <- coef_table
}

```

```

    bind_rows(results)
}

```

```

# Quick test: one regression at h = 0 to verify feols works
test_fml <- d_infl_w_h0 ~ mps_w + mimic_t_minus1 + mps_w_x_mimic +
    gdp_growth_w_t_minus1 + credit_gdp_t_minus1 + gov_pca | country + year

test_fit <- feols(test_fml, data = df_est, cluster = "country")
summary(test_fit)

```

OLS estimation, Dep. Var.: d_infl_w_h0

Observations: 1,954

Fixed-effects: country: 124, year: 18

Standard-errors: Clustered (country)

	Estimate	Std. Error	t value	Pr(> t)
mps_w	-0.732813	0.467666	-1.566957	0.119694
mimic_t_minus1	-0.186439	0.109635	-1.700535	0.091557 .
mps_w_x_mimic	0.034581	0.016847	2.052660	0.042227 *
gdp_growth_w_t_minus1	0.016330	0.036448	0.448037	0.654914
credit_gdp_t_minus1	0.010042	0.004785	2.098776	0.037882 *
gov_pca	-0.162731	0.359762	-0.452329	0.651829

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

RMSE: 3.68536 Adj. R2: 0.164557

Within R2: 0.010333

```
rm(test_fml, test_fit)
```

```
# Works!
```

```

#
# PART A - INFLATION RESPONSE TO MONETARY POLICY SHOCKS
#
#
# Dependent variable: d_infl_w_h{h} = _{i,t+h} - _{i,t-1}
# (cumulative change in winsorized inflation from pre-shock baseline)
#
# Baseline specification:

```

```

#   d_infl_w_h = mps_w + mimic_t_minus1 + mps_w_x_mimic
#               + gdp_growth_w_t_minus1 + credit_gdp_t_minus1 + gov_pca
#               | country + year
#
# Note: lagged inflation is NOT included as a control because the
# dependent variable is already defined relative to infl_{t-1}.
#
# Estimated for 3 samples: Full, Developed, Emerging
#

# Define inflation-specific regressors
infl_regressors <- c("mps_w", "mimic_t_minus1", "mps_w_x_mimic",
                    "gdp_growth_w_t_minus1", "credit_gdp_t_minus1", "gov_pca")

#   A1: Full sample
cat("Running Part A1: Inflation × Full sample...\n")

```

Running Part A1: Inflation × Full sample...

```

lp_infl_full <- run_lp(df_est, "d_infl_w_h", infl_regressors,
                      spec_label = "baseline", sample_label = "full")

#   A2: Developed only
cat("Running Part A2: Inflation × Developed...\n")

```

Running Part A2: Inflation × Developed...

```

lp_infl_dev <- run_lp(df_est %>% filter(dev_status == "Developed"),
                      "d_infl_w_h", infl_regressors,
                      spec_label = "baseline", sample_label = "developed")

#   A3: Emerging only
cat("Running Part A3: Inflation × Emerging...\n")

```

Running Part A3: Inflation × Emerging...


```

interaction_infl$p_value      <- round(interaction_infl$p_value, 3)
interaction_infl$ci_lower_95  <- round(interaction_infl$ci_lower_95, 4)
interaction_infl$ci_upper_95  <- round(interaction_infl$ci_upper_95, 4)

print(as.data.frame(interaction_infl), row.names = FALSE)

```

	sample	horizon	estimate	std_error	p_value	ci_lower_95	ci_upper_95	n_obs
	full	0	0.0346	0.0168	0.042	0.0016	0.0676	1954
	full	1	0.0613	0.0248	0.015	0.0127	0.1099	1838
	full	2	0.0074	0.0235	0.755	-0.0387	0.0534	1719
	full	3	0.0282	0.0409	0.493	-0.0521	0.1084	1598
	full	4	-0.0210	0.0182	0.251	-0.0568	0.0147	1475
	full	5	0.0241	0.0264	0.363	-0.0277	0.0759	1354
	developed	0	-0.0757	0.0595	0.212	-0.1923	0.0410	591
	developed	1	-0.0663	0.0933	0.483	-0.2492	0.1166	560
	developed	2	-0.0009	0.0677	0.990	-0.1336	0.1319	528
	developed	3	0.0343	0.0553	0.539	-0.0740	0.1426	495
	developed	4	0.0489	0.0641	0.450	-0.0767	0.1745	460
	developed	5	-0.0242	0.0551	0.664	-0.1322	0.0839	427
	emerging	0	0.0441	0.0221	0.049	0.0009	0.0874	1334
	emerging	1	0.0621	0.0314	0.051	0.0006	0.1236	1251
	emerging	2	-0.0053	0.0286	0.853	-0.0613	0.0507	1166
	emerging	3	0.0278	0.0548	0.613	-0.0796	0.1352	1080
	emerging	4	-0.0315	0.0202	0.122	-0.0710	0.0081	994
	emerging	5	0.0244	0.0323	0.452	-0.0390	0.0878	908

```

#
# PART B - LOG GDP PER CAPITA RESPONSE TO MONETARY POLICY SHOCKS (MAIN)
#
#
# Dependent variable: d_lgdp_pc_h{h} = log(gdp_pc_{i,t+h}) - log(gdp_pc_{i,t-1})
#   (cumulative log-change in GDP per capita from pre-shock baseline)
#
# Baseline specification:
#   d_lgdp_pc_h = mps_w + mimic_t_minus1 + mps_w_x_mimic
#               + infl_w_t_minus1 + credit_gdp_t_minus1 + gov_pca
#               | country + year
#
# Note: log(gdp_pc) is NOT winsorized - the log transformation already

```

```
# dampens extreme values. Lagged inflation is included as a cross-variable
# control. GDP growth is NOT included (collinear with log GDP changes).
#

# Define log GDP p.c. specific regressors
lgdp_pc_regressors <- c("mps_w", "mimic_t_minus1", "mps_w_x_mimic",
                        "infl_w_t_minus1", "credit_gdp_t_minus1", "gov_pca")

# B1: Full sample
cat("Running Part B1: Log GDP p.c. × Full sample...\n")
```

Running Part B1: Log GDP p.c. × Full sample...

```
lp_lgdp_full <- run_lp(df_est, "d_lgdp_pc_h", lgdp_pc_regressors,
                      spec_label = "baseline", sample_label = "full")

# B2: Developed only
cat("Running Part B2: Log GDP p.c. × Developed...\n")
```

Running Part B2: Log GDP p.c. × Developed...

```
lp_lgdp_dev <- run_lp(df_est %>% filter(dev_status == "Developed"),
                      "d_lgdp_pc_h", lgdp_pc_regressors,
                      spec_label = "baseline", sample_label = "developed")

# B3: Emerging only
cat("Running Part B3: Log GDP p.c. × Emerging...\n")
```

Running Part B3: Log GDP p.c. × Emerging...

```
lp_lgdp_emg <- run_lp(df_est %>% filter(dev_status == "Emerging"),
                      "d_lgdp_pc_h", lgdp_pc_regressors,
                      spec_label = "baseline", sample_label = "emerging")

# Combine all log GDP p.c. results
lp_lgdp_pc <- bind_rows(lp_lgdp_full, lp_lgdp_dev, lp_lgdp_emg)
rm(lp_lgdp_full, lp_lgdp_dev, lp_lgdp_emg)
```

```
# Report key coefficient: _h (interaction MPS × MIMIC)
cat("\n\n")
```

```
cat(" PART B - LOG GDP PER CAPITA: MPS × MIMIC INTERACTION ( _h)\n")
```

```
PART B - LOG GDP PER CAPITA: MPS × MIMIC INTERACTION ( _h)
```

```
cat("\n\n")
```

```
cat(" > 0 → informality dampens output response to monetary shock\n")
```

```
> 0 → informality dampens output response to monetary shock
```

```
cat("\n\n")
```

```
interaction_lgdp <- lp_lgdp_pc %>%
  filter(variable == "mps_w_x_mimic") %>%
  select(sample, horizon, estimate, std_error, p_value, ci_lower_95, ci_upper_95, n_obs)

interaction_lgdp$estimate <- round(interaction_lgdp$estimate, 4)
interaction_lgdp$std_error <- round(interaction_lgdp$std_error, 4)
interaction_lgdp$p_value <- round(interaction_lgdp$p_value, 3)
interaction_lgdp$ci_lower_95 <- round(interaction_lgdp$ci_lower_95, 4)
interaction_lgdp$ci_upper_95 <- round(interaction_lgdp$ci_upper_95, 4)

print(as.data.frame(interaction_lgdp), row.names = FALSE)
```

	sample	horizon	estimate	std_error	p_value	ci_lower_95	ci_upper_95	n_obs
	full	0	-0.0001	0.0001	0.087	-0.0003	0.0000	1928
	full	1	-0.0002	0.0001	0.109	-0.0004	0.0000	1817
	full	2	-0.0002	0.0001	0.241	-0.0004	0.0001	1702
	full	3	-0.0002	0.0002	0.279	-0.0005	0.0002	1584
	full	4	0.0000	0.0002	0.858	-0.0004	0.0003	1464
	full	5	-0.0001	0.0002	0.565	-0.0006	0.0003	1346
	developed	0	0.0003	0.0006	0.588	-0.0008	0.0015	591
	developed	1	-0.0001	0.0012	0.915	-0.0026	0.0023	560
	developed	2	0.0005	0.0019	0.783	-0.0032	0.0043	528
	developed	3	0.0015	0.0027	0.590	-0.0038	0.0067	495
	developed	4	0.0006	0.0028	0.841	-0.0049	0.0060	460
	developed	5	0.0004	0.0031	0.895	-0.0056	0.0064	427
	emerging	0	-0.0003	0.0001	0.001	-0.0004	-0.0001	1337
	emerging	1	-0.0003	0.0001	0.005	-0.0006	-0.0001	1257
	emerging	2	-0.0002	0.0002	0.192	-0.0005	0.0001	1174
	emerging	3	-0.0003	0.0002	0.155	-0.0007	0.0001	1089
	emerging	4	-0.0001	0.0002	0.631	-0.0006	0.0003	1004
	emerging	5	-0.0002	0.0003	0.585	-0.0008	0.0005	919

```
#
# PART C - GDP GROWTH RESPONSE (SIDELINE / SUPPLEMENTARY)
#
#
# Dependent variable: d_gdp_growth_w_h{h} = gdp_growth_{i,t+h} - gdp_growth_{i,t-1}
# (cumulative change in winsorized GDP growth rate)
#
# This is a supplementary analysis - the main output response is captured
# by log GDP per capita in Part B. GDP growth is kept for completeness
# and in case the paper needs to discuss growth rate dynamics separately.
#

# Define GDP growth specific regressors
gdp_growth_regressors <- c("mps_w", "mimic_t_minus1", "mps_w_x_mimic",
                           "infl_w_t_minus1", "credit_gdp_t_minus1", "gov_pca")

# C1: Full sample
cat("Running Part C1: GDP growth × Full sample (sideline)...\n")
```

Running Part C1: GDP growth × Full sample (sideline)...

```
lp_gdpgr_full <- run_lp(df_est, "d_gdp_growth_w_h", gdp_growth_regressors,
                        spec_label = "baseline", sample_label = "full")
```

```
# C2: Developed only
```

```
cat("Running Part C2: GDP growth × Developed (sideline)...\n")
```

Running Part C2: GDP growth × Developed (sideline)...

```
lp_gdpgr_dev <- run_lp(df_est %>% filter(dev_status == "Developed"),
                        "d_gdp_growth_w_h", gdp_growth_regressors,
                        spec_label = "baseline", sample_label = "developed")
```

```
# C3: Emerging only
```

```
cat("Running Part C3: GDP growth × Emerging (sideline)...\n")
```

Running Part C3: GDP growth × Emerging (sideline)...

```
lp_gdpgr_emg <- run_lp(df_est %>% filter(dev_status == "Emerging"),
                        "d_gdp_growth_w_h", gdp_growth_regressors,
                        spec_label = "baseline", sample_label = "emerging")
```

```
# Combine all GDP growth results
```

```
lp_gdp_growth <- bind_rows(lp_gdpgr_full, lp_gdpgr_dev, lp_gdpgr_emg)
```

```
rm(lp_gdpgr_full, lp_gdpgr_dev, lp_gdpgr_emg, gdp_growth_regressors)
```

```
cat("\n GDP growth sideline estimation complete \n")
```

GDP growth sideline estimation complete

```
# Report key coefficient: _h (interaction MPS × MIMIC)
```

```
cat("\n\n")
```

```
cat(" PART C - GDP GROWTH RESPON: MPS × MIMIC INTERACTION ( _h)\n")
```

PART C - GDP GROWTH RESPON: MPS × MIMIC INTERACTION (_h)

```
cat(" \n")
```

```
cat(" > 0 → informality dampens output growth response to monetary shock\n")
```

> 0 → informality dampens output growth response to monetary shock

```
cat(" \n\n")
```

```
interaction_gdp_growth <- lp_gdp_growth %>%
  filter(variable == "mps_w_x_mimic") %>%
  select(sample, horizon, estimate, std_error, p_value, ci_lower_95, ci_upper_95, n_obs)

interaction_gdp_growth$estimate <- round(interaction_gdp_growth$estimate, 4)
interaction_gdp_growth$std_error <- round(interaction_gdp_growth$std_error, 4)
interaction_gdp_growth$p_value <- round(interaction_gdp_growth$p_value, 3)
interaction_gdp_growth$ci_lower_95 <- round(interaction_gdp_growth$ci_lower_95, 4)
interaction_gdp_growth$ci_upper_95 <- round(interaction_gdp_growth$ci_upper_95, 4)

print(as.data.frame(interaction_gdp_growth), row.names = FALSE)
```

sample	horizon	estimate	std_error	p_value	ci_lower_95	ci_upper_95	n_obs
full	0	0.0018	0.0109	0.870	-0.0195	0.0231	1957
full	1	0.0147	0.0093	0.115	-0.0035	0.0329	1844
full	2	0.0102	0.0099	0.305	-0.0092	0.0295	1727
full	3	0.0217	0.0141	0.128	-0.0060	0.0493	1607
full	4	0.0260	0.0142	0.069	-0.0018	0.0538	1485
full	5	0.0198	0.0117	0.093	-0.0032	0.0427	1365
developed	0	0.0745	0.0652	0.261	-0.0533	0.2024	591

developed	1	0.0270	0.0562	0.634	-0.0831	0.1372	560
developed	2	0.1073	0.0548	0.058	-0.0001	0.2147	528
developed	3	0.0729	0.1694	0.670	-0.2591	0.4049	495
developed	4	0.0373	0.0636	0.561	-0.0873	0.1619	460
developed	5	0.0780	0.0825	0.352	-0.0838	0.2397	427
emerging	0	-0.0164	0.0106	0.127	-0.0373	0.0045	1337
emerging	1	0.0128	0.0106	0.228	-0.0079	0.0336	1257
emerging	2	0.0192	0.0137	0.164	-0.0076	0.0461	1174
emerging	3	0.0188	0.0120	0.122	-0.0048	0.0423	1089
emerging	4	0.0352	0.0187	0.064	-0.0015	0.0719	1004
emerging	5	0.0298	0.0200	0.140	-0.0094	0.0690	919

```
#
# ROBUSTNESS - ALTERNATIVE GOVERNANCE SPECIFICATIONS
#
#
# The baseline uses gov_pca as the governance control. Here we re-estimate
# the full-sample models swapping gov_pca for each individual governance
# variable to verify that _h (MPS × MIMIC) is not sensitive to the choice.
#
# Alternatives: reg_quality, corruption, voice_acc, pol_stability
# Run on full sample only (subsample robustness is secondary).
#

# Define alternative governance specs for INFLATION
alt_gov_infl <- list(
  reg_quality = c("mps_w", "mimic_t_minus1", "mps_w_x_mimic",
                  "gdp_growth_w_t_minus1", "credit_gdp_t_minus1", "reg_quality"),
  corruption   = c("mps_w", "mimic_t_minus1", "mps_w_x_mimic",
                  "gdp_growth_w_t_minus1", "credit_gdp_t_minus1", "corruption"),
  voice_acc    = c("mps_w", "mimic_t_minus1", "mps_w_x_mimic",
                  "gdp_growth_w_t_minus1", "credit_gdp_t_minus1", "voice_acc"),
  pol_stability = c("mps_w", "mimic_t_minus1", "mps_w_x_mimic",
                  "gdp_growth_w_t_minus1", "credit_gdp_t_minus1", "pol_stability")
)

# Define alternative governance specs for log GDP p.c.
alt_gov_lgdp <- list(
  reg_quality = c("mps_w", "mimic_t_minus1", "mps_w_x_mimic",
                  "infl_w_t_minus1", "credit_gdp_t_minus1", "reg_quality"),
  corruption   = c("mps_w", "mimic_t_minus1", "mps_w_x_mimic",
```

```

        "infl_w_t_minus1", "credit_gdp_t_minus1", "corruption"),
voice_acc    = c("mps_w", "mimic_t_minus1", "mps_w_x_mimic",
        "infl_w_t_minus1", "credit_gdp_t_minus1", "voice_acc"),
pol_stability = c("mps_w", "mimic_t_minus1", "mps_w_x_mimic",
        "infl_w_t_minus1", "credit_gdp_t_minus1", "pol_stability")
)

# Run all alternative governance specs on full sample
lp_robustness <- list()

for (gov_name in names(alt_gov_infl)) {
  cat(sprintf("Running robustness: Inflation × %s...\n", gov_name))
  lp_robustness[[paste0("infl_", gov_name)]] <-
    run_lp(df_est, "d_infl_w_h", alt_gov_infl[[gov_name]],
          spec_label = gov_name, sample_label = "full")
}

```

Running robustness: Inflation × reg_quality...

Running robustness: Inflation × corruption...

Running robustness: Inflation × voice_acc...

Running robustness: Inflation × pol_stability...

```

for (gov_name in names(alt_gov_lgdp)) {
  cat(sprintf("Running robustness: Log GDP p.c. × %s...\n", gov_name))
  lp_robustness[[paste0("lgdp_", gov_name)]] <-
    run_lp(df_est, "d_lgdp_pc_h", alt_gov_lgdp[[gov_name]],
          spec_label = gov_name, sample_label = "full")
}

```

Running robustness: Log GDP p.c. × reg_quality...

Running robustness: Log GDP p.c. × corruption...

Running robustness: Log GDP p.c. × voice_acc...

Running robustness: Log GDP p.c. × pol_stability...

```
lp_robust <- bind_rows(lp_robustness)
rm(lp_robustness)

# Compare _h across governance specs
cat("\n                                \n")
```

```
cat("  ROBUSTNESS - MPS × MIMIC ( _h) ACROSS GOVERNANCE SPECS\n")
```

ROBUSTNESS - MPS × MIMIC (_h) ACROSS GOVERNANCE SPECS

```
cat("                                \n")
```

```
cat("  Stable _h across specs → result is not driven by governance choice\n")
```

Stable _h across specs → result is not driven by governance choice

```
cat("                                \n")
```

```
cat("\n  Inflation models  \n")
```

Inflation models

```
lp_robust %>%
  filter(variable == "mps_w_x_mimic", grepl("^infl_", dep_var_type) | dep_var_type ==
  select(spec, horizon, estimate, p_value) %>%
  mutate(estimate = round(estimate, 4), p_value = round(p_value, 3)) %>%
  pivot_wider(names_from = horizon, values_from = c(estimate, p_value),
              names_glue = "h{horizon}_{.value}") %>%
  print()
```

```
# A tibble: 4 x 13
  spec    h0_estimate h1_estimate h2_estimate h3_estimate h4_estimate h5_estimate
<chr>      <dbl>      <dbl>      <dbl>      <dbl>      <dbl>      <dbl>
1 reg_q~    0.0348      0.0616      0.0067      0.0269     -0.0231     0.0223
2 corru~    0.0347      0.0614      0.007       0.0273     -0.0228     0.0227
3 voice~    0.0347      0.0615      0.0072      0.028      -0.0213     0.0239
4 pol_s~    0.0342      0.0611      0.0078      0.0294     -0.0206     0.025
# i 6 more variables: h0_p_value <dbl>, h1_p_value <dbl>, h2_p_value <dbl>,
#   h3_p_value <dbl>, h4_p_value <dbl>, h5_p_value <dbl>
```

```
cat("\n  Log GDP per capita models  \n")
```

Log GDP per capita models

```
lp_robust %>%
  filter(variable == "mps_w_x_mimic", grepl("^lgdp_", dep_var_type) | dep_var_type ==
  select(spec, horizon, estimate, p_value) %>%
  mutate(estimate = round(estimate, 4), p_value = round(p_value, 3)) %>%
  pivot_wider(names_from = horizon, values_from = c(estimate, p_value),
              names_glue = "h{horizon}_{.value}") %>%
  print()
```

```
# A tibble: 4 x 13
  spec    h0_estimate h1_estimate h2_estimate h3_estimate h4_estimate h5_estimate
<chr>      <dbl>      <dbl>      <dbl>      <dbl>      <dbl>      <dbl>
1 reg_q~   -0.0002     -0.0002     -0.0002     -0.0002     -0.0001     -0.0001
2 corru~   -0.0001     -0.0002     -0.0002     -0.0002     -0.0001     -0.0002
3 voice~   -0.0001     -0.0002     -0.0002     -0.0002      0          -0.0001
4 pol_s~   -0.0001     -0.0002     -0.0001     -0.0002      0          -0.0001
```

```
# i 6 more variables: h0_p_value <dbl>, h1_p_value <dbl>, h2_p_value <dbl>,
#   h3_p_value <dbl>, h4_p_value <dbl>, h5_p_value <dbl>

# Clean up spec vectors
rm(alt_gov_infl, alt_gov_lgdp, infl_regressors, lgdp_pc_regressors, gov_name)
```

```
#
# COMBINE ALL LP RESULTS INTO A SINGLE MASTER DATAFRAME
#
#
# lp_all contains every estimated coefficient from:
#   - 6 baseline models (2 dep vars × 3 samples)
#   - 8 robustness models (2 dep vars × 4 alternative governance specs)
#
# This is the single object used for all subsequent IRF plotting and
# results tables. Filter by variable, sample, spec, and dep_var_type
# to extract what you need.
#

lp_all <- bind_rows(
  lp_inflation %>% mutate(dep_var_type = "inflation"),
  lp_lgdp_pc %>% mutate(dep_var_type = "lgdp_pc"),
  lp_gdp_growth %>% mutate(dep_var_type = "gdp_growth"),
  lp_robust
)

cat("                                \n")
```

```
cat("  ALL LP RESULTS COMBINED: lp_all\n")
```

```
ALL LP RESULTS COMBINED: lp_all
```

```
cat("                                \n")
```

```
cat(sprintf(" Total coefficient rows: %d\n", nrow(lp_all)))
```

Total coefficient rows: 612

```
cat(sprintf(" Unique specs:           %s\n", paste(unique(lp_all$spec), collapse = ", ")))
```

Unique specs: baseline, reg_quality, corruption, voice_acc, pol_stability

```
cat(sprintf(" Unique samples:         %s\n", paste(unique(lp_all$sample), collapse = ", ")))
```

Unique samples: full, developed, emerging

```
cat(sprintf(" Unique dep var types:    %s\n", paste(unique(lp_all$dep_var_type), collapse = ", ")))
```

Unique dep var types: inflation, lgdp_pc, gdp_growth, d_infl_w_h, d_lgdp_pc_h

```
cat("\n")
```

```
# Clean up intermediate objects - keep only lp_all for plotting
rm(lp_inflation, lp_lgdp_pc, lp_gdp_growth, lp_robust)
```

```
#
# IMPULSE RESPONSE FUNCTIONS - INFLATION
#
#
# These plots show the estimated _h (coefficient on MPS × MIMIC) across
# horizons h = 0, ..., 5 with 90% and 95% confidence bands.
#
# Interpretation (Jordà 2005):
#   Each point is the coefficient from a SEPARATE regression at horizon h.
#   The sequence of coefficients traces out the impulse response function.
#
#   _h > 0 → higher informality DAMPENS the inflation response to MPS
```

```

#   _h < 0 → higher informality AMPLIFIES the inflation response
#   _h = 0 → informality doesn't moderate monetary transmission
#   Bands excluding zero → statistically significant at that confidence level
#

# Extract the interaction coefficient across horizons for each sample
irf_infl <- lp_all %>%
  filter(variable == "mps_w_x_mimic",
         dep_var_type == "inflation",
         spec == "baseline")

# Set sample as ordered factor for faceting
irf_infl$sample <- factor(irf_infl$sample,
                        levels = c("full", "developed", "emerging"),
                        labels = c("Full Sample", "Developed", "Emerging"))

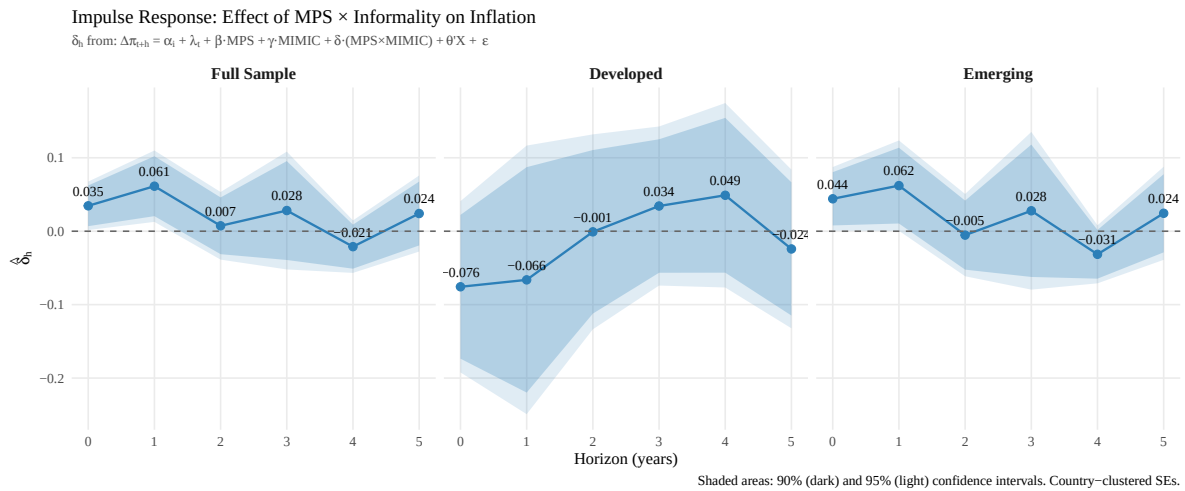
ggplot(irf_infl, aes(x = horizon, y = estimate)) +
  # 95% CI band (lighter)
  geom_ribbon(aes(ymin = ci_lower_95, ymax = ci_upper_95),
            fill = "#2C7FB8", alpha = 0.15) +
  # 90% CI band (darker)
  geom_ribbon(aes(ymin = ci_lower_90, ymax = ci_upper_90),
            fill = "#2C7FB8", alpha = 0.25) +
  # Point estimates
  geom_line(color = "#2C7FB8", linewidth = 0.8) +
  geom_point(color = "#2C7FB8", size = 2.5) +
  geom_text(aes(label = round(estimate, 3)), nudge_y = 0.02, size = 3.5, family = "serif")
  # Zero reference line
  geom_hline(yintercept = 0, linetype = "dashed", color = "grey40") +
  # Facet by sample
  facet_wrap(~ sample, ncol = 3) +
  scale_x_continuous(breaks = 0:max_horizon) +
  labs(
    title = "Impulse Response: Effect of MPS × Informality on Inflation",
    subtitle = expression(paste(delta[h], " from: ", Delta, pi[t+h],
                                " = ", alpha[i], " + ", lambda[t],
                                " + ", beta, "·MPS + ", gamma, "·MIMIC + ",
                                delta, "·(MPS×MIMIC) + ", theta, "'X + ", epsilon))),
    x = "Horizon (years)",
    y = expression(hat(delta)[h]),

```

```

caption = "Shaded areas: 90% (dark) and 95% (light) confidence intervals. Country-
) +
theme_minimal(base_size = 12, base_family = "serif") +
theme(
  panel.grid.minor = element_blank(),
  strip.text = element_text(face = "bold", size = 12),
  plot.subtitle = element_text(size = 10, color = "grey30")
)

```



```

#
# IMPULSE RESPONSE FUNCTIONS - GDP GROWTH
#

#   _h > 0 → higher informality DAMPENS output (GDP) response to monetary shock
#   _h < 0 → higher informality AMPLIFIES output (GDP) response to monetary shock

irf_lgdp <- lp_all %>%
  filter(variable == "mps_w_x_mimic",
         dep_var_type == "lgdp_pc",
         spec == "baseline")

irf_lgdp$sample <- factor(irf_lgdp$sample,
                        levels = c("full", "developed", "emerging"),
                        labels = c("Full Sample", "Developed", "Emerging"))

```

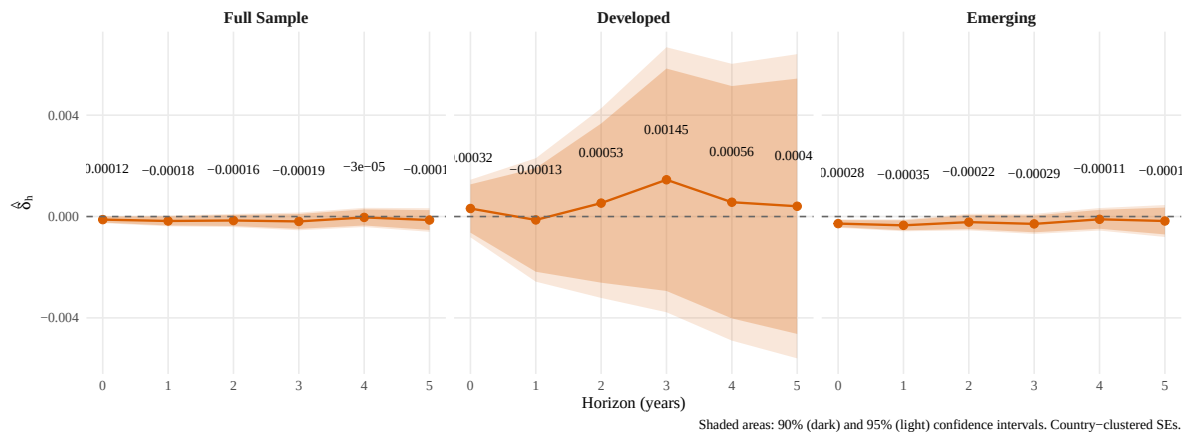
```

ggplot(irf_lgdp, aes(x = horizon, y = estimate)) +
  geom_ribbon(aes(ymin = ci_lower_95, ymax = ci_upper_95),
    fill = "#D95F02", alpha = 0.15) +
  geom_ribbon(aes(ymin = ci_lower_90, ymax = ci_upper_90),
    fill = "#D95F02", alpha = 0.25) +
  geom_line(color = "#D95F02", linewidth = 0.8) +
  geom_point(color = "#D95F02", size = 2.5) +
  geom_text(aes(label = round(estimate, 5)), nudge_y = 0.002, size = 3.5, family = "serif") +
  geom_hline(yintercept = 0, linetype = "dashed", color = "grey40") +
  facet_wrap(~ sample, ncol = 3) +
  scale_x_continuous(breaks = 0:max_horizon) +
  labs(
    title = "Impulse Response: Effect of MPS × Informality on Log GDP per Capita",
    subtitle = expression(paste(delta[h], " from: ", Delta, "log(gdp_pc)"[t+h],
      " = ", alpha[i], " + ", lambda[t],
      " + ", beta, "·MPS + ", gamma, "·MIMIC + ",
      delta, "·(MPS×MIMIC) + ", theta, "'X + ", epsilon)),
    x = "Horizon (years)",
    y = expression(hat(delta)[h]),
    caption = "Shaded areas: 90% (dark) and 95% (light) confidence intervals. Country-
  ) +
  theme_minimal(base_size = 12, base_family = "serif") +
  theme(
    panel.grid.minor = element_blank(),
    strip.text = element_text(face = "bold", size = 12),
    plot.subtitle = element_text(size = 10, color = "grey30")
  )

```

Impulse Response: Effect of MPS × Informality on Log GDP per Capita

δ_h from: $\Delta \log(\text{gdp_pc})_{t+h} = \alpha_t + \lambda_t + \beta \cdot \text{MPS} + \gamma \cdot \text{MIMIC} + \delta \cdot (\text{MPS} \times \text{MIMIC}) + \theta'X + \varepsilon$



```
#
# IMPULSE RESPONSE FUNCTIONS - GDP GROWTH
#

#   _h > 0 → higher informality DAMPENS the output (GDP) growth response to MPS
#   _h < 0 → higher informality AMPLIFIES the output (GDP) growth response

irf_gdp_growth <- lp_all %>%
  filter(variable == "mps_w_x_mimic",
         dep_var_type == "gdp_growth",
         spec == "baseline")

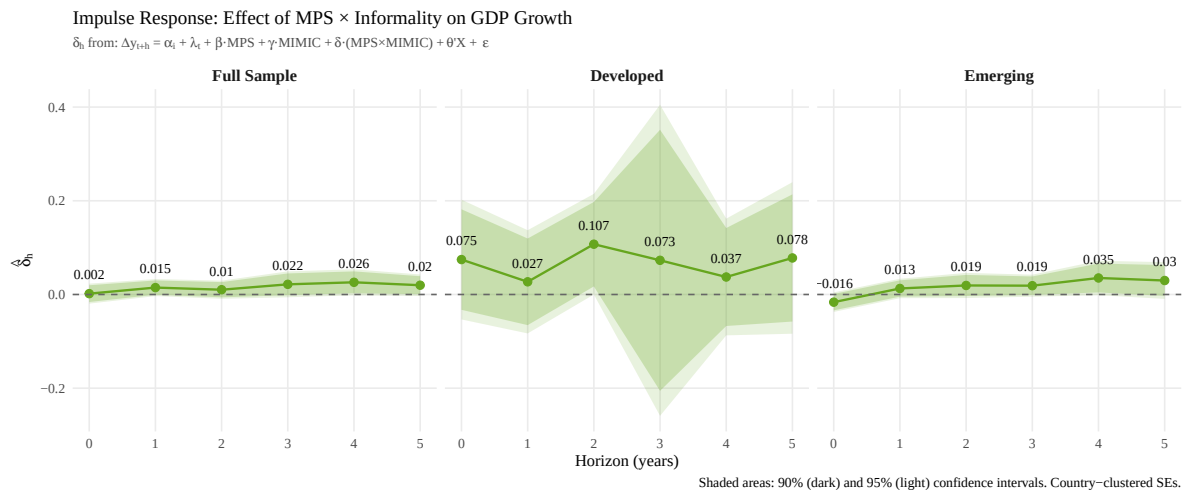
irf_gdp_growth$sample <- factor(irf_gdp_growth$sample,
                                levels = c("full", "developed", "emerging"),
                                labels = c("Full Sample", "Developed", "Emerging"))

ggplot(irf_gdp_growth, aes(x = horizon, y = estimate)) +
  geom_ribbon(aes(ymin = ci_lower_95, ymax = ci_upper_95),
            fill = "#66A61E", alpha = 0.15) +
  geom_ribbon(aes(ymin = ci_lower_90, ymax = ci_upper_90),
            fill = "#66A61E", alpha = 0.25) +
  geom_line(color = "#66A61E", linewidth = 0.8) +
  geom_point(color = "#66A61E", size = 2.5) +
  geom_text(aes(label = round(estimate, 3)), nudge_y = 0.04, size = 3.5, family = "serif") +
  geom_hline(yintercept = 0, linetype = "dashed", color = "grey40") +
  facet_wrap(~ sample, ncol = 3) +
```

```

scale_x_continuous(breaks = 0:max_horizon) +
labs(
  title = "Impulse Response: Effect of MPS × Informality on GDP Growth",
  subtitle = expression(paste(delta[h], " from: ", Delta, y[t+h],
    " = ", alpha[i], " + ", lambda[t],
    " + ", beta, "·MPS + ", gamma, "·MIMIC + ",
    delta, "·(MPS×MIMIC) + ", theta, "'X + ", epsilon)),
  x = "Horizon (years)",
  y = expression(hat(delta)[h]),
  caption = "Shaded areas: 90% (dark) and 95% (light) confidence intervals. Country-
) +
theme_minimal(base_size = 12, base_family = "serif") +
theme(
  panel.grid.minor = element_blank(),
  strip.text = element_text(face = "bold", size = 12),
  plot.subtitle = element_text(size = 10, color = "grey30")
)

```



```

#
# ROBUSTNESS IRF - _h ACROSS ALTERNATIVE GOVERNANCE SPECIFICATIONS
#
#
# If the interaction coefficient _h is stable across different governance
# controls, the result is robust to the "which governance variable?" choice.
# Each line is a different governance specification; all on full sample.
#

```

```

# Combine baseline (full sample) with robustness specs
irf_robust <- bind_rows(
  lp_all %>%
    filter(variable == "mps_w_x_mimic", spec == "baseline", sample == "full") %>%
    mutate(gov_spec = "gov_pca (baseline)"),
  lp_all %>%
    filter(variable == "mps_w_x_mimic", spec != "baseline", sample == "full") %>%
    mutate(gov_spec = spec)
)

# Separate by dep var type - main dependents only
irf_robust <- irf_robust %>%
  filter(dep_var_type %in% c("inflation", "lgdp_pc", "d_infl_w_h", "d_lgdp_pc_h")) %>%
  mutate(dep_label = case_when(
    grepl("infl", dep_var_type) ~ "Inflation",
    grepl("lgdp", dep_var_type) ~ "Log GDP per Capita",
    TRUE ~ dep_var_type
  ))

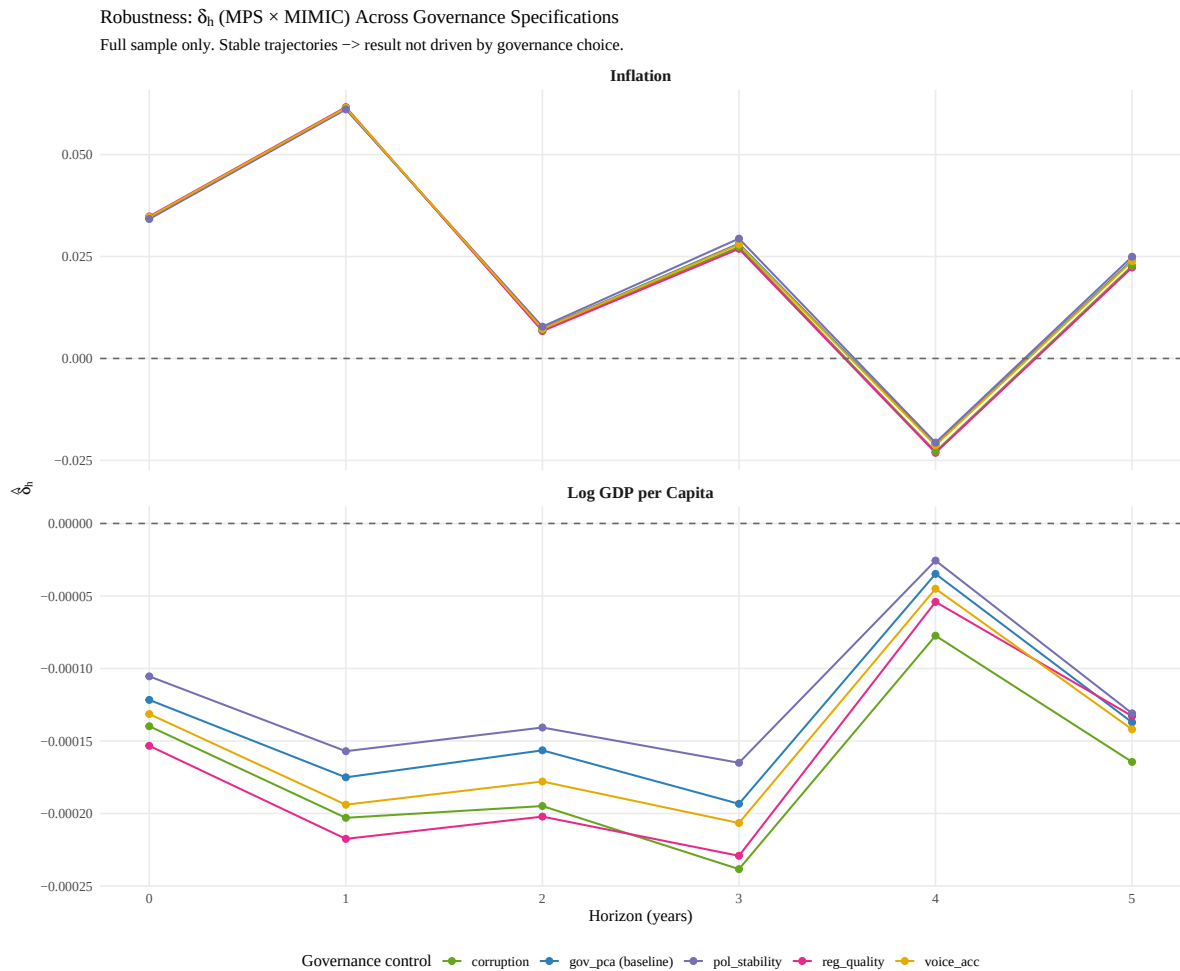
ggplot(irf_robust, aes(x = horizon, y = estimate, color = gov_spec)) +
  geom_line(linewidth = 0.7) +
  geom_point(size = 2) +
  geom_hline(yintercept = 0, linetype = "dashed", color = "grey40") +
  facet_wrap(~ dep_label, ncol = 1, scales = "free_y") +
  scale_x_continuous(breaks = 0:max_horizon) +
  scale_color_manual(
    values = c("gov_pca (baseline)" = "#2C7FB8",
              "reg_quality" = "#E7298A",
              "corruption" = "#66A61E",
              "voice_acc" = "#E6AB02",
              "pol_stability" = "#7570B3"),
    name = "Governance control"
  ) +
  labs(
    title = expression(paste("Robustness: ", delta[h], " (MPS  $\times$  MIMIC) Across Governan
    subtitle = "Full sample only. Stable trajectories -> result not driven by governan
    x = "Horizon (years)",
    y = expression(hat(delta)[h])
  ) +
  theme_minimal(base_size = 12, base_family = "serif") +

```

```

theme(
  panel.grid.minor = element_blank(),
  legend.position = "bottom",
  strip.text = element_text(face = "bold", size = 12)
)

```



```

#
# IRF - DIRECT EFFECT OF MPS ON INFLATION ( _h)
#
# _h captures the direct effect of a monetary policy shock on inflation,
# holding informality constant. This is the "average" transmission effect
# across all levels of informality.
#   _h < 0  $\rightarrow$  contractionary MPS reduces inflation (expected)

```

```

#   _h > 0 → contractionary MPS increases inflation ("price puzzle")
#

irf_infl_beta <- lp_all %>%
  filter(variable == "mps_w",
         dep_var_type == "inflation",
         spec == "baseline")

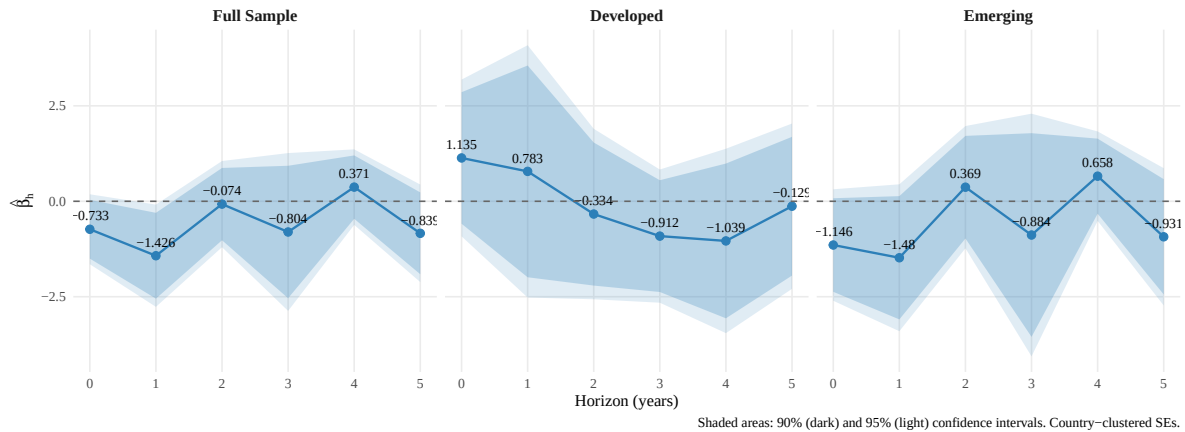
irf_infl_beta$sample <- factor(irf_infl_beta$sample,
                              levels = c("full", "developed", "emerging"),
                              labels = c("Full Sample", "Developed", "Emerging"))

ggplot(irf_infl_beta, aes(x = horizon, y = estimate)) +
  geom_ribbon(aes(ymin = ci_lower_95, ymax = ci_upper_95),
            fill = "#2C7FB8", alpha = 0.15) +
  geom_ribbon(aes(ymin = ci_lower_90, ymax = ci_upper_90),
            fill = "#2C7FB8", alpha = 0.25) +
  geom_line(color = "#2C7FB8", linewidth = 0.8) +
  geom_point(color = "#2C7FB8", size = 2.5) +
  geom_text(aes(label = round(estimate, 3)), nudge_y = 0.35, size = 3.5, family = "serif") +
  geom_hline(yintercept = 0, linetype = "dashed", color = "grey40") +
  facet_wrap(~ sample, ncol = 3) +
  scale_x_continuous(breaks = 0:max_horizon) +
  labs(
    title = "Impulse Response: Direct Effect of MPS on Inflation",
    subtitle = expression(paste(beta[h], " (coefficient on MPS) from the baseline LP s
    x = "Horizon (years)",
    y = expression(hat(beta)[h])),
    caption = "Shaded areas: 90% (dark) and 95% (light) confidence intervals. Country-
  ) +
  theme_minimal(base_size = 12, base_family = "serif") +
  theme(
    panel.grid.minor = element_blank(),
    strip.text = element_text(face = "bold", size = 12),
    plot.subtitle = element_text(size = 10, color = "grey30")
  )

```

Impulse Response: Direct Effect of MPS on Inflation

β_h (coefficient on MPS) from the baseline LP specification



```
#
# IRF - DIRECT EFFECT OF MPS ON LOG GDP PER CAPITA (_h)
#
# _h captures the direct effect of a monetary policy shock on log GDP p.c.
#   _h < 0 → contractionary MPS reduces output (expected)
#   _h > 0 → contractionary MPS increases output (unusual)
#

irf_lgdp_beta <- lp_all %>%
  filter(variable == "mps_w",
         dep_var_type == "lgdp_pc",
         spec == "baseline")

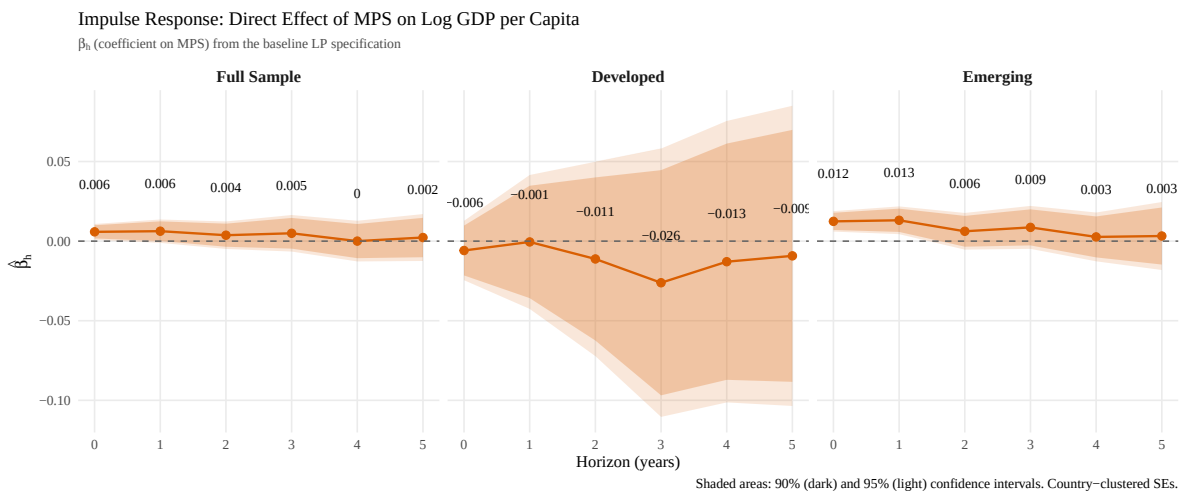
irf_lgdp_beta$sample <- factor(irf_lgdp_beta$sample,
                              levels = c("full", "developed", "emerging"),
                              labels = c("Full Sample", "Developed", "Emerging"))

ggplot(irf_lgdp_beta, aes(x = horizon, y = estimate)) +
  geom_ribbon(aes(ymin = ci_lower_95, ymax = ci_upper_95),
            fill = "#D95F02", alpha = 0.15) +
  geom_ribbon(aes(ymin = ci_lower_90, ymax = ci_upper_90),
            fill = "#D95F02", alpha = 0.25) +
  geom_line(color = "#D95F02", linewidth = 0.8) +
  geom_point(color = "#D95F02", size = 2.5) +
  geom_text(aes(label = round(estimate, 3)), nudge_y = 0.03, size = 3.5, family = "serif") +
  geom_hline(yintercept = 0, linetype = "dashed", color = "grey40") +
  facet_wrap(~ sample, ncol = 3) +
```

```

scale_x_continuous(breaks = 0:max_horizon) +
labs(
  title = "Impulse Response: Direct Effect of MPS on Log GDP per Capita",
  subtitle = expression(paste(beta[h], " (coefficient on MPS) from the baseline LP s
  x = "Horizon (years)",
  y = expression(hat(beta)[h]),
  caption = "Shaded areas: 90% (dark) and 95% (light) confidence intervals. Country-
) +
theme_minimal(base_size = 12, base_family = "serif") +
theme(
  panel.grid.minor = element_blank(),
  strip.text = element_text(face = "bold", size = 12),
  plot.subtitle = element_text(size = 10, color = "grey30")
)

```



```

#
# IRF - DIRECT EFFECT OF MPS ON GDP GROWTH ( _h ) - SIDELINE
#
# _h captures the direct effect of a monetary policy shock on GDP growth.
#   _h < 0 → contractionary MPS reduces growth (expected)
#   _h > 0 → contractionary MPS increases growth (unusual)
#

irf_gdpgr_beta <- lp_all %>%
  filter(variable == "mps_w",
         dep_var_type == "gdp_growth",

```

```

spec == "baseline")

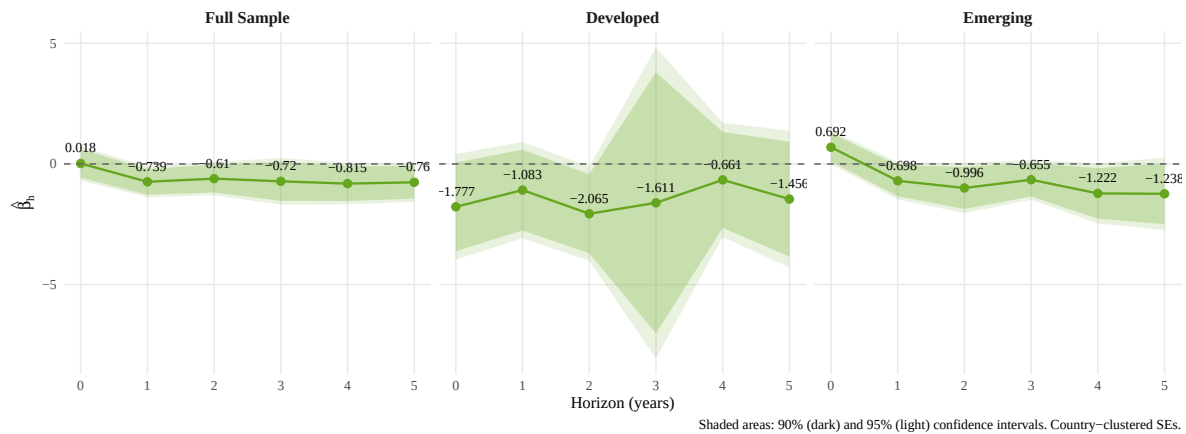
irf_gdpgr_beta$sample <- factor(irf_gdpgr_beta$sample,
                                levels = c("full", "developed", "emerging"),
                                labels = c("Full Sample", "Developed", "Emerging"))

ggplot(irf_gdpgr_beta, aes(x = horizon, y = estimate)) +
  geom_ribbon(aes(ymin = ci_lower_95, ymax = ci_upper_95),
            fill = "#66A61E", alpha = 0.15) +
  geom_ribbon(aes(ymin = ci_lower_90, ymax = ci_upper_90),
            fill = "#66A61E", alpha = 0.25) +
  geom_line(color = "#66A61E", linewidth = 0.8) +
  geom_point(color = "#66A61E", size = 2.5) +
  geom_text(aes(label = round(estimate, 3)), nudge_y = 0.65, size = 3.5, family = "serif") +
  geom_hline(yintercept = 0, linetype = "dashed", color = "grey40") +
  facet_wrap(~ sample, ncol = 3) +
  scale_x_continuous(breaks = 0:max_horizon) +
  labs(
    title = "Impulse Response: Direct Effect of MPS on GDP Growth (Sideline)",
    subtitle = expression(paste(beta[h], " (coefficient on MPS) from the baseline LP s
    x = "Horizon (years)",
    y = expression(hat(beta)[h]),
    caption = "Shaded areas: 90% (dark) and 95% (light) confidence intervals. Country-
  ) +
  theme_minimal(base_size = 12, base_family = "serif") +
  theme(
    panel.grid.minor = element_blank(),
    strip.text = element_text(face = "bold", size = 12),
    plot.subtitle = element_text(size = 10, color = "grey30")
  )

```

Impulse Response: Direct Effect of MPS on GDP Growth (Sideline)

β_h (coefficient on MPS) from the baseline LP specification



```
#
# IRF - DIRECT EFFECT OF MIMIC ON INFLATION ( _h)
#
# _h captures the direct association between lagged informality (MIMIC_{t-1})
# and inflation, holding MPS and the interaction constant. This is the "level"
# effect of the informal economy on price dynamics - separate from how it
# moderates monetary transmission ( _h).
#
#   _h > 0 → higher informality is associated with stronger inflation
#   _h < 0 → higher informality is associated with weaker inflation
#
# CAVEAT: Diagnostic 3 showed MIMIC is 99.6% between-country. Country FE absorb
# most of that variation, so _h is identified off small within-country
# movements and is expected to be imprecise. The interaction _h remains the
# core parameter of interest.
#

irf_infl_gamma <- lp_all %>%
  filter(variable == "mimic_t_minus1",
         dep_var_type == "inflation",
         spec == "baseline")

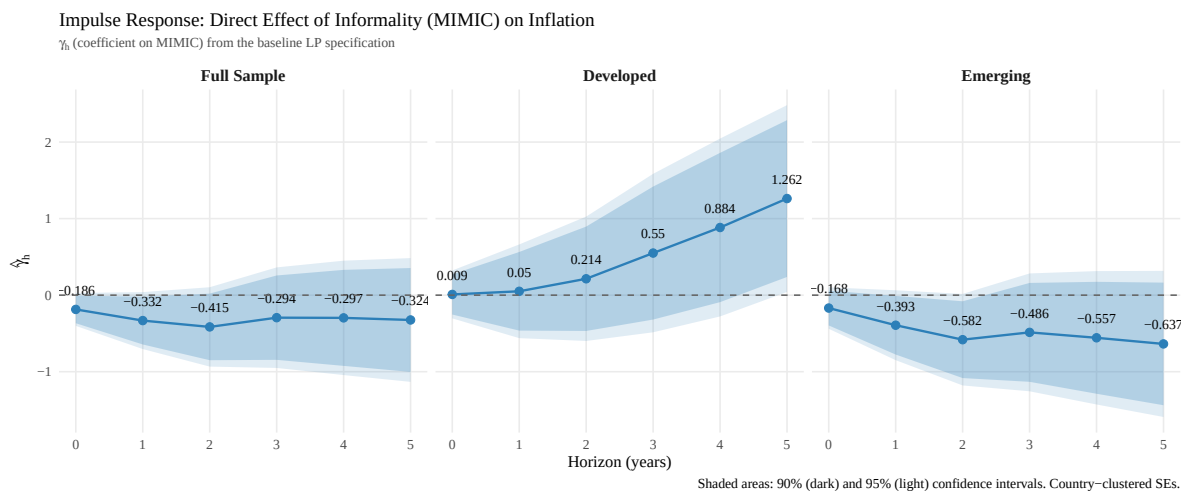
irf_infl_gamma$sample <- factor(irf_infl_gamma$sample,
                               levels = c("full", "developed", "emerging"),
                               labels = c("Full Sample", "Developed", "Emerging"))

ggplot(irf_infl_gamma, aes(x = horizon, y = estimate)) +
```

```

geom_ribbon(aes(ymin = ci_lower_95, ymax = ci_upper_95),
            fill = "#2C7FB8", alpha = 0.15) +
geom_ribbon(aes(ymin = ci_lower_90, ymax = ci_upper_90),
            fill = "#2C7FB8", alpha = 0.25) +
geom_line(color = "#2C7FB8", linewidth = 0.8) +
geom_point(color = "#2C7FB8", size = 2.5) +
geom_text(aes(label = round(estimate, 3)), nudge_y = 0.25, size = 3.5, family = "serif") +
geom_hline(yintercept = 0, linetype = "dashed", color = "grey40") +
facet_wrap(~ sample, ncol = 3) +
scale_x_continuous(breaks = 0:max_horizon) +
labs(
  title = "Impulse Response: Direct Effect of Informality (MIMIC) on Inflation",
  subtitle = expression(paste(gamma[h], " (coefficient on MIMIC) from the baseline LP specification")),
  x = "Horizon (years)",
  y = expression(hat(gamma)[h]),
  caption = "Shaded areas: 90% (dark) and 95% (light) confidence intervals. Country-clustered SEs."
) +
theme_minimal(base_size = 12, base_family = "serif") +
theme(
  panel.grid.minor = element_blank(),
  strip.text = element_text(face = "bold", size = 12),
  plot.subtitle = element_text(size = 10, color = "grey30")
)

```



```

#
# IRF - DIRECT EFFECT OF MIMIC ON LOG GDP PER CAPITA ( _h)
#
# _h captures the direct association between lagged informality (MIMIC_{t-1})
# and log GDP per capita, holding MPS and the interaction constant.
#
#   _h > 0 → higher informality is associated with higher output (unusual)
#   _h < 0 → higher informality is associated with lower output
#           (consistent with the "informality as productivity drag" view)
#
# CAVEAT: Same as Part A - MIMIC is 99.6% between-country (Diagnostic 3), so
# country FE absorb most of its variation. _h is expected to be imprecise;
# the interaction _h is the core parameter.
#

irf_lgdp_gamma <- lp_all %>%
  filter(variable == "mimic_t_minus1",
         dep_var_type == "lgdp_pc",
         spec == "baseline")

irf_lgdp_gamma$sample <- factor(irf_lgdp_gamma$sample,
                               levels = c("full", "developed", "emerging"),
                               labels = c("Full Sample", "Developed", "Emerging"))

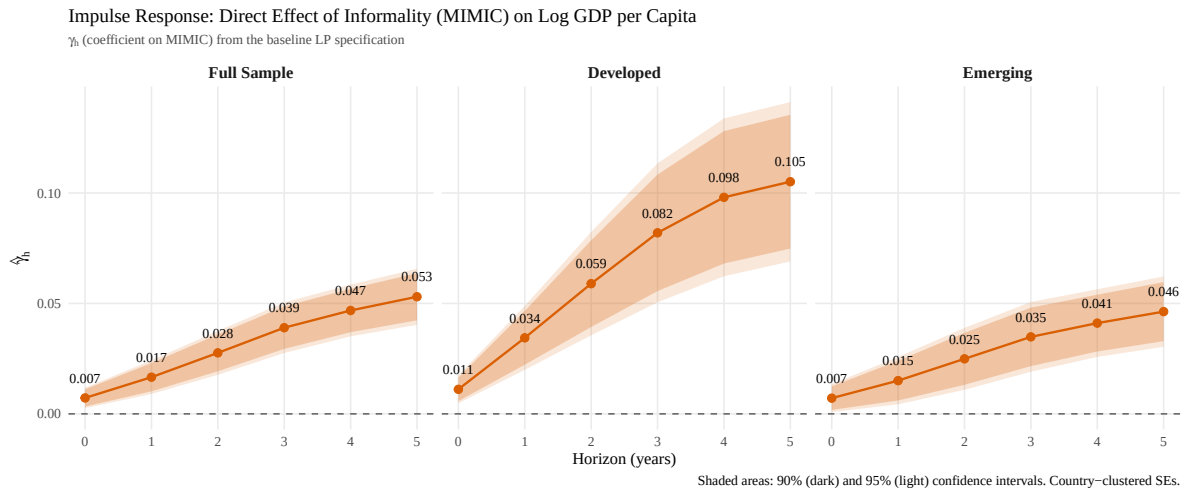
ggplot(irf_lgdp_gamma, aes(x = horizon, y = estimate)) +
  geom_ribbon(aes(ymin = ci_lower_95, ymax = ci_upper_95),
            fill = "#D95F02", alpha = 0.15) +
  geom_ribbon(aes(ymin = ci_lower_90, ymax = ci_upper_90),
            fill = "#D95F02", alpha = 0.25) +
  geom_line(color = "#D95F02", linewidth = 0.8) +
  geom_point(color = "#D95F02", size = 2.5) +
  geom_text(aes(label = round(estimate, 3)), nudge_y = 0.009, size = 3.5, family = "serif") +
  geom_hline(yintercept = 0, linetype = "dashed", color = "grey40") +
  facet_wrap(~ sample, ncol = 3) +
  scale_x_continuous(breaks = 0:max_horizon) +
  labs(
    title = "Impulse Response: Direct Effect of Informality (MIMIC) on Log GDP per Capita",
    subtitle = expression(paste(gamma[h], " (coefficient on MIMIC) from the baseline L")),
    x = "Horizon (years)",
    y = expression(hat(gamma)[h]),
  )

```

```

caption = "Shaded areas: 90% (dark) and 95% (light) confidence intervals. Country-
) +
theme_minimal(base_size = 12, base_family = "serif") +
theme(
  panel.grid.minor = element_blank(),
  strip.text = element_text(face = "bold", size = 12),
  plot.subtitle = element_text(size = 10, color = "grey30")
)

```



```

#
# IRF - DIRECT EFFECT OF MIMIC ON GDP GROWTH ( _h) - SIDELINE
#
# _h captures the direct association between lagged informality (MIMIC_{t-1})
# and the GDP growth rate, holding MPS and the interaction constant.
#
# _h > 0 → higher informality is associated with faster growth
# _h < 0 → higher informality is associated with slower growth
#
# CAVEAT: Same as Parts A and B - _h is expected to be imprecise because
# country FE absorb the overwhelmingly between-country variation in MIMIC.
#

irf_gdpgr_gamma <- lp_all %>%
  filter(variable == "mimic_t_minus1",
    dep_var_type == "gdp_growth",
    spec == "baseline")

```

```

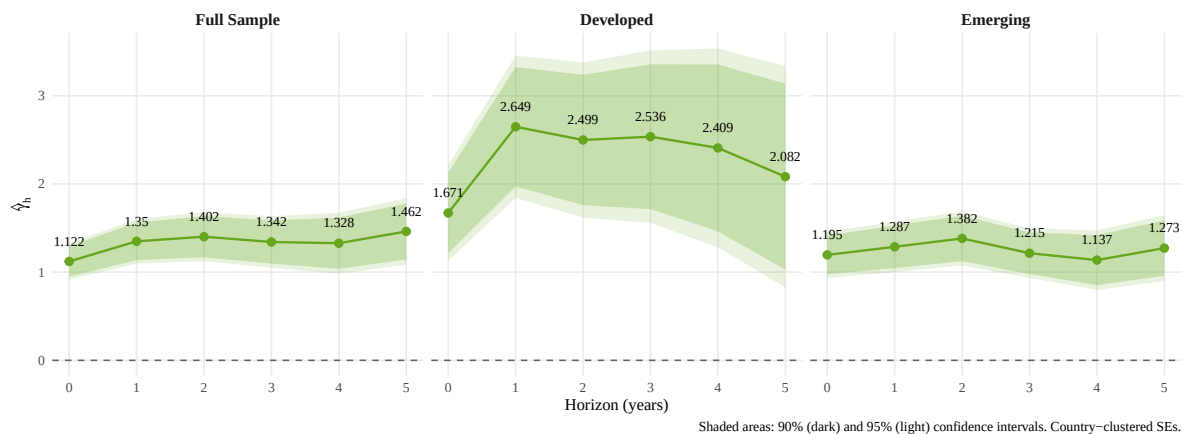
irf_gdpgr_gamma$sample <- factor(irf_gdpgr_gamma$sample,
                                levels = c("full", "developed", "emerging"),
                                labels = c("Full Sample", "Developed", "Emerging"))

ggplot(irf_gdpgr_gamma, aes(x = horizon, y = estimate)) +
  geom_ribbon(aes(ymin = ci_lower_95, ymax = ci_upper_95),
            fill = "#66A61E", alpha = 0.15) +
  geom_ribbon(aes(ymin = ci_lower_90, ymax = ci_upper_90),
            fill = "#66A61E", alpha = 0.25) +
  geom_line(color = "#66A61E", linewidth = 0.8) +
  geom_point(color = "#66A61E", size = 2.5) +
  geom_text(aes(label = round(estimate, 3)), nudge_y = 0.23, size = 3.5, family = "serif") +
  geom_hline(yintercept = 0, linetype = "dashed", color = "grey40") +
  facet_wrap(~ sample, ncol = 3) +
  scale_x_continuous(breaks = 0:max_horizon) +
  labs(
    title = "Impulse Response: Direct Effect of Informality (MIMIC) on GDP Growth (Sideris et al. 2018)",
    subtitle = expression(paste(gamma[h], " (coefficient on MIMIC) from the baseline Ljung-Box test"),
    x = "Horizon (years)",
    y = expression(hat(gamma)[h]),
    caption = "Shaded areas: 90% (dark) and 95% (light) confidence intervals. Country-specific estimates."
  ) +
  theme_minimal(base_size = 12, base_family = "serif") +
  theme(
    panel.grid.minor = element_blank(),
    strip.text = element_text(face = "bold", size = 12),
    plot.subtitle = element_text(size = 10, color = "grey30")
  )

```

Impulse Response: Direct Effect of Informality (MIMIC) on GDP Growth (Sideline)

γ_h (coefficient on MIMIC) from the baseline LP specification



```
#
# SUMMARY TABLE - ALL BASELINE COEFFICIENTS ACROSS HORIZONS
#
# Full regression output for the baseline spec, not just the interaction.
# Useful for the paper's appendix or results tables.
#
```

```
cat("                                \n")
```

```
cat("  FULL COEFFICIENT TABLE - BASELINE SPECIFICATION (Full Sample)\n")
```

```
  FULL COEFFICIENT TABLE - BASELINE SPECIFICATION (Full Sample)
```

```
cat("                                \n")
```

```
cat("\n  Inflation model  \n")
```

```
  Inflation model
```

```

lp_all %>%
  filter(spec == "baseline", sample == "full", dep_var_type == "inflation") %>%
  select(variable, horizon, estimate, std_error, p_value) %>%
  mutate(estimate = round(estimate, 4),
         std_error = round(std_error, 4),
         p_value = round(p_value, 3),
         sig = case_when(p_value < 0.01 ~ "****",
                        p_value < 0.05 ~ "***",
                        p_value < 0.10 ~ "**",
                        TRUE ~ "")) %>%
  as.data.frame() %>%
  print(row.names = FALSE)

```

	variable	horizon	estimate	std_error	p_value	sig
	mps_w	0	-0.7328	0.4677	0.120	
	mimic_t_minus1	0	-0.1864	0.1096	0.092	*
	mps_w_x_mimic	0	0.0346	0.0168	0.042	**
	gdp_growth_w_t_minus1	0	0.0163	0.0364	0.655	
	credit_gdp_t_minus1	0	0.0100	0.0048	0.038	**
	gov_pca	0	-0.1627	0.3598	0.652	
	mps_w	1	-1.4264	0.6832	0.039	**
	mimic_t_minus1	1	-0.3322	0.1898	0.083	*
	mps_w_x_mimic	1	0.0613	0.0248	0.015	**
	gdp_growth_w_t_minus1	1	-0.0001	0.0623	0.999	
	credit_gdp_t_minus1	1	0.0080	0.0069	0.246	
	gov_pca	1	-0.1145	0.6361	0.857	
	mps_w	2	-0.0742	0.5757	0.898	
	mimic_t_minus1	2	-0.4149	0.2645	0.119	
	mps_w_x_mimic	2	0.0074	0.0235	0.755	
	gdp_growth_w_t_minus1	2	0.0304	0.0730	0.677	
	credit_gdp_t_minus1	2	0.0036	0.0091	0.696	
	gov_pca	2	0.4796	0.8004	0.550	
	mps_w	3	-0.8044	1.0547	0.447	
	mimic_t_minus1	3	-0.2941	0.3351	0.382	
	mps_w_x_mimic	3	0.0282	0.0409	0.493	
	gdp_growth_w_t_minus1	3	0.0896	0.0900	0.321	
	credit_gdp_t_minus1	3	0.0062	0.0087	0.477	
	gov_pca	3	1.1538	0.8566	0.180	
	mps_w	4	0.3710	0.5048	0.464	
	mimic_t_minus1	4	-0.2969	0.3813	0.438	

mps_w_x_mimic	4	-0.0210	0.0182	0.251
gdp_growth_w_t_minus1	4	0.0216	0.0853	0.800
credit_gdp_t_minus1	4	0.0012	0.0109	0.914
gov_pca	4	1.5660	0.9722	0.110
mps_w	5	-0.8394	0.6517	0.200
mimic_t_minus1	5	-0.3243	0.4125	0.433
mps_w_x_mimic	5	0.0241	0.0264	0.363
gdp_growth_w_t_minus1	5	-0.0077	0.0816	0.925
credit_gdp_t_minus1	5	0.0029	0.0131	0.826
gov_pca	5	1.6300	1.1175	0.147

```
cat("\n Log GDP p.c. model \n")
```

Log GDP p.c. model

```
lp_all %>%
  filter(spec == "baseline", sample == "full", dep_var_type == "lgdp_pc") %>%
  select(variable, horizon, estimate, std_error, p_value) %>%
  mutate(estimate = round(estimate, 4),
         std_error = round(std_error, 4),
         p_value = round(p_value, 3),
         sig = case_when(p_value < 0.01 ~ "****",
                        p_value < 0.05 ~ "***",
                        p_value < 0.10 ~ "**",
                        TRUE ~ "")) %>%
  as.data.frame() %>%
  print(row.names = FALSE)
```

variable	horizon	estimate	std_error	p_value	sig
mps_w	0	0.0058	0.0025	0.023	**
mimic_t_minus1	0	0.0072	0.0024	0.003	***
mps_w_x_mimic	0	-0.0001	0.0001	0.087	*
infl_w_t_minus1	0	0.0004	0.0006	0.549	
credit_gdp_t_minus1	0	-0.0003	0.0001	0.005	***
gov_pca	0	0.0204	0.0079	0.012	**
mps_w	1	0.0063	0.0038	0.098	*
mimic_t_minus1	1	0.0166	0.0039	0.000	***
mps_w_x_mimic	1	-0.0002	0.0001	0.109	

infl_w_t_minus1	1	0.0003	0.0006	0.652	
credit_gdp_t_minus1	1	-0.0006	0.0002	0.002	***
gov_pca	1	0.0295	0.0093	0.002	***
mps_w	2	0.0037	0.0044	0.400	
mimic_t_minus1	2	0.0276	0.0051	0.000	***
mps_w_x_mimic	2	-0.0002	0.0001	0.241	
infl_w_t_minus1	2	0.0001	0.0006	0.843	
credit_gdp_t_minus1	2	-0.0009	0.0003	0.001	***
gov_pca	2	0.0337	0.0130	0.011	**
mps_w	3	0.0049	0.0059	0.402	
mimic_t_minus1	3	0.0390	0.0058	0.000	***
mps_w_x_mimic	3	-0.0002	0.0002	0.279	
infl_w_t_minus1	3	0.0001	0.0009	0.925	
credit_gdp_t_minus1	3	-0.0010	0.0003	0.001	***
gov_pca	3	0.0353	0.0153	0.023	**
mps_w	4	0.0000	0.0065	0.997	
mimic_t_minus1	4	0.0468	0.0060	0.000	***
mps_w_x_mimic	4	0.0000	0.0002	0.858	
infl_w_t_minus1	4	0.0000	0.0009	0.989	
credit_gdp_t_minus1	4	-0.0009	0.0003	0.009	***
gov_pca	4	0.0270	0.0171	0.117	
mps_w	5	0.0023	0.0075	0.764	
mimic_t_minus1	5	0.0530	0.0065	0.000	***
mps_w_x_mimic	5	-0.0001	0.0002	0.565	
infl_w_t_minus1	5	0.0004	0.0010	0.670	
credit_gdp_t_minus1	5	-0.0006	0.0003	0.060	*
gov_pca	5	0.0207	0.0190	0.279	

```
cat("\n GDP growth model \n")
```

GDP growth model

```
lp_all %>%
  filter(spec == "baseline", sample == "full", dep_var_type == "gdp_growth") %>%
  select(variable, horizon, estimate, std_error, p_value) %>%
  mutate(estimate = round(estimate, 4),
         std_error = round(std_error, 4),
         p_value = round(p_value, 3),
```

```

sig = case_when(p_value < 0.01 ~ "****",
                p_value < 0.05 ~ "***",
                p_value < 0.10 ~ "**",
                TRUE ~ "") %>%
as.data.frame() %>%
print(row.names = FALSE)

```

	variable	horizon	estimate	std_error	p_value	sig
	mps_w	0	0.0179	0.3582	0.960	
	mimic_t_minus1	0	1.1215	0.1047	0.000	***
	mps_w_x_mimic	0	0.0018	0.0109	0.870	
	infl_w_t_minus1	0	0.0469	0.0271	0.086	*
	credit_gdp_t_minus1	0	0.0017	0.0044	0.694	
	gov_pca	0	0.8793	0.3129	0.006	***
	mps_w	1	-0.7393	0.3247	0.025	**
	mimic_t_minus1	1	1.3496	0.1292	0.000	***
	mps_w_x_mimic	1	0.0147	0.0093	0.115	
	infl_w_t_minus1	1	0.0239	0.0444	0.591	
	credit_gdp_t_minus1	1	-0.0007	0.0074	0.927	
	gov_pca	1	0.3708	0.4656	0.427	
	mps_w	2	-0.6096	0.3505	0.084	*
	mimic_t_minus1	2	1.4019	0.1416	0.000	***
	mps_w_x_mimic	2	0.0102	0.0099	0.305	
	infl_w_t_minus1	2	0.0305	0.0368	0.408	
	credit_gdp_t_minus1	2	-0.0016	0.0117	0.892	
	gov_pca	2	-0.4226	0.5276	0.425	
	mps_w	3	-0.7202	0.4932	0.147	
	mimic_t_minus1	3	1.3418	0.1492	0.000	***
	mps_w_x_mimic	3	0.0217	0.0141	0.128	
	infl_w_t_minus1	3	0.0314	0.0317	0.324	
	credit_gdp_t_minus1	3	0.0118	0.0179	0.512	
	gov_pca	3	-1.4180	0.6191	0.024	**
	mps_w	4	-0.8152	0.4371	0.065	*
	mimic_t_minus1	4	1.3279	0.1763	0.000	***
	mps_w_x_mimic	4	0.0260	0.0142	0.069	*
	infl_w_t_minus1	4	0.0062	0.0300	0.837	
	credit_gdp_t_minus1	4	0.0281	0.0211	0.185	
	gov_pca	4	-2.5557	0.8534	0.003	***
	mps_w	5	-0.7603	0.4134	0.068	*
	mimic_t_minus1	5	1.4617	0.1931	0.000	***

mps_w_x_mimic	5	0.0198	0.0117	0.093	*
infl_w_t_minus1	5	0.0153	0.0421	0.716	
credit_gdp_t_minus1	5	0.0485	0.0229	0.036	**
gov_pca	5	-2.5342	0.9743	0.010	**

```
#
# SUMMARY TABLE - ALL BASELINE COEFFICIENTS ACROSS HORIZONS
#
# Full regression output for the baseline spec, not just the interaction.
# Useful for the paper's appendix or results tables.
#
cat("                                \n")
```

```
cat("  FULL COEFFICIENT TABLE - BASELINE SPECIFICATION (Developed)\n")
```

FULL COEFFICIENT TABLE - BASELINE SPECIFICATION (Developed)

```
cat("                                \n")
```

```
cat("\n  Inflation model  \n")
```

Inflation model

```
lp_all %>%
  filter(spec == "baseline", sample == "developed", dep_var_type == "inflation") %>%
  select(variable, horizon, estimate, std_error, p_value) %>%
  mutate(estimate = round(estimate, 4),
         std_error = round(std_error, 4),
         p_value = round(p_value, 3),
         sig = case_when(p_value < 0.01 ~ "****",
                        p_value < 0.05 ~ "**",
```

```

                                p_value < 0.10 ~ "*",
                                TRUE ~ " ")) %>%
as.data.frame() %>%
print(row.names = FALSE)

```

variable	horizon	estimate	std_error	p_value	sig
mps_w	0	1.1346	1.0473	0.286	
mimic_t_minus1	0	0.0093	0.1595	0.954	
mps_w_x_mimic	0	-0.0757	0.0595	0.212	
gdp_growth_w_t_minus1	0	0.0228	0.0493	0.646	
credit_gdp_t_minus1	0	0.0032	0.0040	0.430	
gov_pca	0	-0.1184	0.5577	0.833	
mps_w	1	0.7829	1.6862	0.645	
mimic_t_minus1	1	0.0504	0.3122	0.873	
mps_w_x_mimic	1	-0.0663	0.0933	0.483	
gdp_growth_w_t_minus1	1	0.0769	0.0683	0.268	
credit_gdp_t_minus1	1	0.0051	0.0084	0.550	
gov_pca	1	-0.0199	1.0329	0.985	
mps_w	2	-0.3343	1.1393	0.771	
mimic_t_minus1	2	0.2138	0.4145	0.609	
mps_w_x_mimic	2	-0.0009	0.0677	0.990	
gdp_growth_w_t_minus1	2	0.0608	0.0892	0.500	
credit_gdp_t_minus1	2	0.0035	0.0111	0.755	
gov_pca	2	0.3835	1.2471	0.760	
mps_w	3	-0.9124	0.8900	0.313	
mimic_t_minus1	3	0.5498	0.5281	0.305	
mps_w_x_mimic	3	0.0343	0.0553	0.539	
gdp_growth_w_t_minus1	3	0.0895	0.0969	0.362	
credit_gdp_t_minus1	3	0.0020	0.0067	0.769	
gov_pca	3	1.0135	1.4450	0.488	
mps_w	4	-1.0388	1.2324	0.405	
mimic_t_minus1	4	0.8837	0.5928	0.146	
mps_w_x_mimic	4	0.0489	0.0641	0.450	
gdp_growth_w_t_minus1	4	0.0514	0.1072	0.635	
credit_gdp_t_minus1	4	-0.0009	0.0062	0.887	
gov_pca	4	1.4540	1.4915	0.337	
mps_w	5	-0.1288	1.1047	0.908	
mimic_t_minus1	5	1.2620	0.6227	0.051	*
mps_w_x_mimic	5	-0.0242	0.0551	0.664	
gdp_growth_w_t_minus1	5	0.0448	0.0975	0.649	

credit_gdp_t_minus1	5	-0.0021	0.0089	0.813
gov_pca	5	2.0064	1.3693	0.152

```
cat("\n Log GDP p.c. model \n")
```

Log GDP p.c. model

```
lp_all %>%
  filter(spec == "baseline", sample == "developed", dep_var_type == "lgdp_pc") %>%
  select(variable, horizon, estimate, std_error, p_value) %>%
  mutate(estimate = round(estimate, 4),
         std_error = round(std_error, 4),
         p_value = round(p_value, 3),
         sig = case_when(p_value < 0.01 ~ "****",
                        p_value < 0.05 ~ "***",
                        p_value < 0.10 ~ "**",
                        TRUE ~ "")) %>%
  as.data.frame() %>%
  print(row.names = FALSE)
```

variable	horizon	estimate	std_error	p_value	sig
mps_w	0	-0.0059	0.0095	0.540	
mimic_t_minus1	0	0.0111	0.0032	0.002	***
mps_w_x_mimic	0	0.0003	0.0006	0.588	
infl_w_t_minus1	0	-0.0025	0.0014	0.086	*
credit_gdp_t_minus1	0	-0.0004	0.0002	0.032	**
gov_pca	0	0.0294	0.0131	0.031	**
mps_w	1	-0.0005	0.0215	0.981	
mimic_t_minus1	1	0.0344	0.0074	0.000	***
mps_w_x_mimic	1	-0.0001	0.0012	0.915	
infl_w_t_minus1	1	-0.0031	0.0021	0.146	
credit_gdp_t_minus1	1	-0.0008	0.0003	0.014	**
gov_pca	1	0.0656	0.0271	0.021	**
mps_w	2	-0.0112	0.0311	0.721	
mimic_t_minus1	2	0.0589	0.0119	0.000	***
mps_w_x_mimic	2	0.0005	0.0019	0.783	
infl_w_t_minus1	2	-0.0018	0.0026	0.508	
credit_gdp_t_minus1	2	-0.0011	0.0004	0.004	***

gov_pca	2	0.0872	0.0381	0.028	**
mps_w	3	-0.0261	0.0430	0.547	
mimic_t_minus1	3	0.0820	0.0160	0.000	***
mps_w_x_mimic	3	0.0015	0.0027	0.590	
infl_w_t_minus1	3	0.0006	0.0039	0.871	
credit_gdp_t_minus1	3	-0.0013	0.0004	0.001	***
gov_pca	3	0.0950	0.0459	0.046	**
mps_w	4	-0.0129	0.0451	0.776	
mimic_t_minus1	4	0.0980	0.0182	0.000	***
mps_w_x_mimic	4	0.0006	0.0028	0.841	
infl_w_t_minus1	4	0.0021	0.0052	0.696	
credit_gdp_t_minus1	4	-0.0012	0.0003	0.001	***
gov_pca	4	0.0908	0.0490	0.073	*
mps_w	5	-0.0092	0.0481	0.849	
mimic_t_minus1	5	0.1051	0.0184	0.000	***
mps_w_x_mimic	5	0.0004	0.0031	0.895	
infl_w_t_minus1	5	0.0028	0.0059	0.642	
credit_gdp_t_minus1	5	-0.0009	0.0003	0.001	***
gov_pca	5	0.0651	0.0491	0.194	

```
cat("\n GDP growth model \n")
```

GDP growth model

```
lp_all %>%
  filter(spec == "baseline", sample == "developed", dep_var_type == "gdp_growth") %>%
  select(variable, horizon, estimate, std_error, p_value) %>%
  mutate(estimate = round(estimate, 4),
         std_error = round(std_error, 4),
         p_value = round(p_value, 3),
         sig = case_when(p_value < 0.01 ~ "***",
                        p_value < 0.05 ~ "**",
                        p_value < 0.10 ~ "*",
                        TRUE ~ "")) %>%
  as.data.frame() %>%
  print(row.names = FALSE)
```

variable	horizon	estimate	std_error	p_value	sig
----------	---------	----------	-----------	---------	-----

mps_w	0	-1.7768	1.1158	0.121	
mimic_t_minus1	0	1.6706	0.2770	0.000	***
mps_w_x_mimic	0	0.0745	0.0652	0.261	
infl_w_t_minus1	0	-0.0233	0.0967	0.811	
credit_gdp_t_minus1	0	-0.0094	0.0045	0.044	**
gov_pca	0	1.2836	0.6457	0.055	*
mps_w	1	-1.0832	1.0178	0.295	
mimic_t_minus1	1	2.6486	0.4117	0.000	***
mps_w_x_mimic	1	0.0270	0.0562	0.634	
infl_w_t_minus1	1	0.0433	0.1017	0.673	
credit_gdp_t_minus1	1	-0.0140	0.0063	0.032	**
gov_pca	1	1.1737	1.1350	0.308	
mps_w	2	-2.0645	0.9966	0.046	**
mimic_t_minus1	2	2.4985	0.4491	0.000	***
mps_w_x_mimic	2	0.1073	0.0548	0.058	*
infl_w_t_minus1	2	0.1786	0.1064	0.102	
credit_gdp_t_minus1	2	-0.0139	0.0121	0.259	
gov_pca	2	-0.7261	1.4627	0.623	
mps_w	3	-1.6108	3.2898	0.628	
mimic_t_minus1	3	2.5360	0.4991	0.000	***
mps_w_x_mimic	3	0.0729	0.1694	0.670	
infl_w_t_minus1	3	0.1792	0.0914	0.058	*
credit_gdp_t_minus1	3	-0.0005	0.0209	0.982	
gov_pca	3	-1.8621	1.8438	0.320	
mps_w	4	-0.6610	1.2077	0.588	
mimic_t_minus1	4	2.4086	0.5752	0.000	***
mps_w_x_mimic	4	0.0373	0.0636	0.561	
infl_w_t_minus1	4	0.0439	0.1281	0.734	
credit_gdp_t_minus1	4	0.0168	0.0243	0.494	
gov_pca	4	-3.3970	2.5575	0.193	
mps_w	5	-1.4557	1.4469	0.322	
mimic_t_minus1	5	2.0823	0.6412	0.003	***
mps_w_x_mimic	5	0.0780	0.0825	0.352	
infl_w_t_minus1	5	-0.0266	0.1621	0.871	
credit_gdp_t_minus1	5	0.0385	0.0259	0.147	
gov_pca	5	-5.0558	2.8876	0.089	*

```
#
# SUMMARY TABLE - ALL BASELINE COEFFICIENTS ACROSS HORIZONS
#
# Full regression output for the baseline spec, not just the interaction.
```

```
# Useful for the paper's appendix or results tables.
#
```

```
cat("                                \n")
```

```
cat("  FULL COEFFICIENT TABLE - BASELINE SPECIFICATION (Emerging)\n")
```

```
  FULL COEFFICIENT TABLE - BASELINE SPECIFICATION (Emerging)
```

```
cat("                                \n")
```

```
cat("\n  Inflation model  \n")
```

```
  Inflation model
```

```
lp_all %>%
  filter(spec == "baseline", sample == "emerging", dep_var_type == "inflation") %>%
  select(variable, horizon, estimate, std_error, p_value) %>%
  mutate(estimate = round(estimate, 4),
         std_error = round(std_error, 4),
         p_value = round(p_value, 3),
         sig = case_when(p_value < 0.01 ~ "***",
                        p_value < 0.05 ~ "**",
                        p_value < 0.10 ~ "*",
                        TRUE ~ "")) %>%
  as.data.frame() %>%
  print(row.names = FALSE)
```

	variable	horizon	estimate	std_error	p_value	sig
	mps_w	0	-1.1456	0.7450	0.128	
	mimic_t_minus1	0	-0.1684	0.1382	0.226	

mps_w_x_mimic	0	0.0441	0.0221	0.049	**
gdp_growth_w_t_minus1	0	0.0326	0.0470	0.490	
credit_gdp_t_minus1	0	0.0310	0.0183	0.094	*
gov_pca	0	-0.1690	0.4601	0.714	
mps_w	1	-1.4798	0.9818	0.135	
mimic_t_minus1	1	-0.3928	0.2330	0.095	*
mps_w_x_mimic	1	0.0621	0.0314	0.051	*
gdp_growth_w_t_minus1	1	-0.0207	0.0800	0.796	
credit_gdp_t_minus1	1	0.0221	0.0263	0.404	
gov_pca	1	-0.0187	0.8144	0.982	
mps_w	2	0.3694	0.8172	0.652	
mimic_t_minus1	2	-0.5817	0.3049	0.060	*
mps_w_x_mimic	2	-0.0053	0.0286	0.853	
gdp_growth_w_t_minus1	2	0.0134	0.0913	0.884	
credit_gdp_t_minus1	2	-0.0119	0.0291	0.684	
gov_pca	2	0.8691	1.0436	0.407	
mps_w	3	-0.8841	1.6218	0.587	
mimic_t_minus1	3	-0.4862	0.3928	0.219	
mps_w_x_mimic	3	0.0278	0.0548	0.613	
gdp_growth_w_t_minus1	3	0.0982	0.1147	0.394	
credit_gdp_t_minus1	3	-0.0177	0.0318	0.579	
gov_pca	3	1.7794	1.1296	0.119	
mps_w	4	0.6580	0.5980	0.274	
mimic_t_minus1	4	-0.5570	0.4445	0.214	
mps_w_x_mimic	4	-0.0315	0.0202	0.122	
gdp_growth_w_t_minus1	4	0.0269	0.1078	0.803	
credit_gdp_t_minus1	4	-0.0209	0.0355	0.557	
gov_pca	4	2.1102	1.2750	0.102	
mps_w	5	-0.9306	0.9176	0.313	
mimic_t_minus1	5	-0.6371	0.4869	0.194	
mps_w_x_mimic	5	0.0244	0.0323	0.452	
gdp_growth_w_t_minus1	5	-0.0177	0.1061	0.868	
credit_gdp_t_minus1	5	-0.0065	0.0366	0.860	
gov_pca	5	2.3436	1.4971	0.121	

```
cat("\n Log GDP p.c. model \n")
```

Log GDP p.c. model

```

lp_all %>%
  filter(spec == "baseline", sample == "emerging", dep_var_type == "lgdp_pc") %>%
  select(variable, horizon, estimate, std_error, p_value) %>%
  mutate(estimate = round(estimate, 4),
         std_error = round(std_error, 4),
         p_value = round(p_value, 3),
         sig = case_when(p_value < 0.01 ~ "****",
                        p_value < 0.05 ~ "***",
                        p_value < 0.10 ~ "**",
                        TRUE ~ "")) %>%
  as.data.frame() %>%
  print(row.names = FALSE)

```

	variable	horizon	estimate	std_error	p_value	sig
	mps_w	0	0.0124	0.0033	0.000	***
	mimic_t_minus1	0	0.0071	0.0032	0.031	**
	mps_w_x_mimic	0	-0.0003	0.0001	0.001	***
	infl_w_t_minus1	0	0.0005	0.0006	0.412	
	credit_gdp_t_minus1	0	-0.0002	0.0002	0.439	
	gov_pca	0	0.0195	0.0092	0.037	**
	mps_w	1	0.0131	0.0045	0.004	***
	mimic_t_minus1	1	0.0150	0.0055	0.007	***
	mps_w_x_mimic	1	-0.0003	0.0001	0.005	***
	infl_w_t_minus1	1	0.0006	0.0006	0.344	
	credit_gdp_t_minus1	1	-0.0005	0.0004	0.270	
	gov_pca	1	0.0248	0.0092	0.009	***
	mps_w	2	0.0062	0.0059	0.296	
	mimic_t_minus1	2	0.0249	0.0072	0.001	***
	mps_w_x_mimic	2	-0.0002	0.0002	0.192	
	infl_w_t_minus1	2	0.0004	0.0007	0.523	
	credit_gdp_t_minus1	2	-0.0006	0.0006	0.371	
	gov_pca	2	0.0250	0.0127	0.053	*
	mps_w	3	0.0086	0.0069	0.213	
	mimic_t_minus1	3	0.0349	0.0081	0.000	***
	mps_w_x_mimic	3	-0.0003	0.0002	0.155	
	infl_w_t_minus1	3	0.0002	0.0009	0.852	
	credit_gdp_t_minus1	3	-0.0005	0.0008	0.513	
	gov_pca	3	0.0244	0.0145	0.096	*
	mps_w	4	0.0026	0.0078	0.736	
	mimic_t_minus1	4	0.0411	0.0078	0.000	***

mps_w_x_mimic	4	-0.0001	0.0002	0.631
infl_w_t_minus1	4	0.0000	0.0009	0.993
credit_gdp_t_minus1	4	-0.0004	0.0009	0.640
gov_pca	4	0.0147	0.0153	0.342
mps_w	5	0.0032	0.0109	0.769
mimic_t_minus1	5	0.0463	0.0081	0.000 ***
mps_w_x_mimic	5	-0.0002	0.0003	0.585
infl_w_t_minus1	5	0.0005	0.0010	0.615
credit_gdp_t_minus1	5	-0.0003	0.0010	0.761
gov_pca	5	0.0126	0.0171	0.464

```
cat("\n GDP growth model \n")
```

GDP growth model

```
lp_all %>%
  filter(spec == "baseline", sample == "emerging", dep_var_type == "gdp_growth") %>%
  select(variable, horizon, estimate, std_error, p_value) %>%
  mutate(estimate = round(estimate, 4),
         std_error = round(std_error, 4),
         p_value = round(p_value, 3),
         sig = case_when(p_value < 0.01 ~ "****",
                        p_value < 0.05 ~ "***",
                        p_value < 0.10 ~ "**",
                        TRUE ~ "")) %>%
  as.data.frame() %>%
  print(row.names = FALSE)
```

variable	horizon	estimate	std_error	p_value	sig
mps_w	0	0.6920	0.3604	0.058	*
mimic_t_minus1	0	1.1954	0.1333	0.000	***
mps_w_x_mimic	0	-0.0164	0.0106	0.127	
infl_w_t_minus1	0	0.0525	0.0290	0.073	*
credit_gdp_t_minus1	0	0.0157	0.0139	0.262	
gov_pca	0	0.8778	0.3623	0.018	**
mps_w	1	-0.6981	0.3913	0.078	*
mimic_t_minus1	1	1.2874	0.1463	0.000	***
mps_w_x_mimic	1	0.0128	0.0106	0.228	

infl_w_t_minus1	1	0.0400	0.0495	0.421	
credit_gdp_t_minus1	1	0.0033	0.0175	0.853	
gov_pca	1	0.3327	0.5009	0.508	
mps_w	2	-0.9965	0.5300	0.063	*
mimic_t_minus1	2	1.3822	0.1568	0.000	***
mps_w_x_mimic	2	0.0192	0.0137	0.164	
infl_w_t_minus1	2	0.0318	0.0403	0.432	
credit_gdp_t_minus1	2	0.0084	0.0165	0.613	
gov_pca	2	-0.3854	0.5325	0.471	
mps_w	3	-0.6551	0.4213	0.124	
mimic_t_minus1	3	1.2150	0.1431	0.000	***
mps_w_x_mimic	3	0.0188	0.0120	0.122	
infl_w_t_minus1	3	0.0267	0.0337	0.429	
credit_gdp_t_minus1	3	0.0124	0.0189	0.514	
gov_pca	3	-1.3854	0.5635	0.016	**
mps_w	4	-1.2221	0.6356	0.058	*
mimic_t_minus1	4	1.1369	0.1726	0.000	***
mps_w_x_mimic	4	0.0352	0.0187	0.064	*
infl_w_t_minus1	4	0.0152	0.0306	0.621	
credit_gdp_t_minus1	4	0.0259	0.0246	0.296	
gov_pca	4	-2.4488	0.7814	0.002	***
mps_w	5	-1.2382	0.7640	0.109	
mimic_t_minus1	5	1.2725	0.1910	0.000	***
mps_w_x_mimic	5	0.0298	0.0200	0.140	
infl_w_t_minus1	5	0.0380	0.0420	0.367	
credit_gdp_t_minus1	5	0.0493	0.0286	0.089	*
gov_pca	5	-1.8920	0.8570	0.030	**